MTH107 - Advanced linear algebra

Lecture notes

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Forword

These notes are a work in progress and are far to be complete. This means in particular the following:

- As of today, they start with the content of week 6-7. Part of the content of weeks 1-6 *might* be added in the future, but no guarantees.
- They are still plenty of typos (and hopefully few to no mistakes). If you see some, please let me know by email or directly via learning mall and I will correct them

Things might be presented in a slightly different way that what was done in class, or in a different order.

To go further

In the coming notes, you will see a few red boxes like this one. These are meant for the interested reader that wants to know more and can be skipped without harm. In particular, the content that is written inside a red box will **not** be in the exam.

Theorem 0.1. *Blue boxes are for theorems*

Definition 0.2. Green boxes are for definitions

Standing assumption

Yellow boxes are for standing assumptions for chapters/sections/subsections.

The occasional orange boxes are todo-notes about what need to be changed/added/removed/rewritten. Hopefully you will see few of them.

Finally, these notes are heavily inspired by the book *Linear algebra done right* from Sheldon Axler, as well as by Ye Liu notes (from 2022) and Pietro Sgobba notes.

Contents

1 Reminders about sets, maps, ...

Write the chapter

2 Vector spaces

Write the chapter

These two pages contain a few results for further references. The proofs still need to be written (but you can look at your own notes from the lecture)

Theorem 2.0.1.

Let V be a finite dimensional vector space and let U and W be two subspaces. Then we have

$$\dim(U+W) = \dim(U) + \dim(W) - \dim(U \cap W).$$

Corollary 2.0.2. *Let V be a finite dimensional vector space and let $U_1 \oplus \cdots \oplus U_m$ be a direct sum of subspaces. Then*

$$\dim(U_1 \oplus \cdots \oplus U_m) = \dim(U_1) + \cdots + \dim(U_m).$$

Proof. If $m = 1$, there is nothing to prove. The case $m = 2$ is a simple application of the Theorem, since $U_1 \cap U_2 = \{0\}$ if the sum $U_1 + U_2$ is direct. The case $m \geq 3$ is done by induction. $\square$

Infinite dimensional spaces

For a general vector space V, we still have the formula

$$\dim(U+W) + \dim(U \cap W) = \dim(U) + \dim(W).$$

However, this is not really interesting as for infinite dimensional vector spaces, $\dim(U) + \dim(W) = \max\{\dim(U), \dim(W)\}$. More generally, using bases, one can show

$$\dim(\bigoplus_{\alpha \in I} U_\alpha) = \sum_{\alpha \in I} \dim(U_\alpha).$$

But to understand the above expression we need to make sense of infinite sums, which goes beyond this course.

Lemma 2.0.3. *Let $v_1, \ldots, v_n$ be a linearly dependent list. Then there exists $j \in \{1, \ldots, n\}$ such that $v_j \in \operatorname{span}(v_1, \ldots, v_{j-1})$.*

Theorem 2.0.4.

Let V be a finite dimensional vector space. Then every linearly independent list of length $\dim(V)$ is a basis.

3 Linear maps

This page contains a few results for further references. The proofs still need to be written (but you can look at your own notes from the lecture)

Definition 3.0.1.

Let $\mathcal{E}^n$ be the standard basis of F^n and $\mathcal{E}^m$ be the standard basis of F^m. We define

$$\begin{aligned} \mathrm{L}\colon \mathbf{F}^{m,n} &\longrightarrow \mathcal{L}(\mathbf{F}^n, \mathbf{F}^m) \\ A &\longmapsto \mathrm{L}_A : v \mapsto Av. \end{aligned}$$

Lemma 3.0.2. *Let $\mathcal{E}^n$ be the standard basis of F^n and $\mathcal{E}^m$ be the standard basis of F^m. Then the maps $[\,\cdot\,]_{\mathcal{E}^n}^{\mathcal{E}^m} : \mathcal{L}(\mathbf{F}^n, \mathbf{F}^m) \to \mathbf{F}^{m,n}$ and $\mathrm{L}\colon \mathbf{F}^{m,n} \to \mathcal{L}(\mathbf{F}^n, \mathbf{F}^m)$ are isomorphisms of vector spaces, and one is the inverse of the other.*

Theorem 3.0.3.

Let V and W be vector spaces and let $T\colon V \to W$ be a linear map. If V is finite dimensional, then

$$\dim(V) = \dim(\operatorname{range}(T)) + \dim(\operatorname{null}(T)).$$

3.1 Matrix representation

Carefull: this section need to be rewritten (up to page 14).

Standing assumption

Through this section, V will always be a finite dimensional vector space over $\mathbf{F}$.

3.1.1 More on matrix representation

Let V be a $\mathbf{F}$-vector space and let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis of V. Then we have an isomorphism

$$\begin{aligned} V &\xrightarrow{[\,\cdot\,]_{\mathcal{B}}} \mathbf{F} \\ v = c_1 v_1 + \dots + c_n v_n &\longmapsto [v]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}. \end{aligned}$$

If both V and W are $\mathbf{F}$-vector spaces, with bases $\mathcal{B} = (v_1, \dots, v_n)$ and $\mathcal{D} = (w_1, \dots, w_m)$, then for any linear map $T \in \mathcal{L}(V, W)$ we have constructed a matrix $[T]_{\mathcal{B}}^{\mathcal{D}}$ in $\mathbf{F}^{m,n}$, defined by

$$[T]_{\mathcal{B}}^{\mathcal{D}} = \Big[[Tv_1]_{\mathcal{D}} \cdots [Tv_n]_{\mathcal{D}} \Big]$$

(the i^{th} columns of $[T]_{\mathcal{B}}^{\mathcal{D}}$ is $[Tv_i]_{\mathcal{D}}$). We then checked that for any vector $v \in V$ we have

$$[Tv]_{\mathcal{D}} = [T]_{\mathcal{B}}^{\mathcal{D}} [v]_{\mathcal{B}}.$$

This result is equivalent to say that the following diagram commutes:

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[d, "{[\,\cdot\,]_{\mathcal{B}}}"'] & W \arrow[d, "{[\,\cdot\,]_{\mathcal{D}}}"] \\ \mathbf{F}^n \arrow[r, "{[T]_{\mathcal{B}}^{\mathcal{D}}}"'] & \mathbf{F}^m \end{tikzcd} \qquad \begin{tikzcd} v \arrow[r, mapsto] \arrow[d, mapsto] & Tv \arrow[d, mapsto] \\ {[v]_{\mathcal{B}}} \arrow[r, mapsto] & {[Tv]_{\mathcal{D}} = [T]_{\mathcal{B}}^{\mathcal{D}}[v]_{\mathcal{B}}} \end{tikzcd}$$

Example 3.1.1. Let $V = \mathbf{F}^n$ and let $\mathcal{E} = (e_1 = [1, 0, \dots, 0]^\mathsf{T}, \dots, e_n = [0, \dots, 0, 1]^\mathsf{T})$ be the standard basis. Then $[\,\cdot\,]_{\mathcal{E}} \colon V \to \mathbf{F}^n$ is the identity map $\mathrm{Id}_{\mathbf{F}^n}$. Indeed, for any $x = [x_1, \dots, x_n]^\mathsf{T}$ in V we have $x = \sum_{i=1}^n x_i e_i$, so $[x]_{\mathcal{E}} = [x_1, \dots, x_n]^\mathsf{T}$.

Example 3.1.2. Let V be a vector space and let $\mathcal{B} = (v_1, \dots, v_n)$ be basis for V. Let $\mathrm{Id}_V \colon V \to V$ be the identity map. Then $\mathrm{Id}(v_1) = v_1$, so $[\mathrm{Id}(v_1)]_{\mathcal{B}} = [1, 0, \dots, 0]^\mathsf{T}$. Similarly, $[\mathrm{Id}(v_i)]_{\mathcal{B}}$ is the vector with a 1 in position i and 0 elsewhere. It follows that

$$[\mathrm{Id}_V]_{\mathcal{B}}^{\mathcal{B}} = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix} = \mathrm{Id}_n \in \mathbf{F}^{n,n} \tag{3.1}$$

is the identity matrix of $\mathbf{F}^{n,n}$. Indeed, $[\mathrm{Id}_V v]_{\mathcal{B}} = [\mathrm{Id}_V]_{\mathcal{B}}^{\mathcal{B}} [v]_{\mathcal{B}}$.

Remark 3.1.3. The equation (3.1) does not depend on the choice of the basis $\mathcal{B}$!

The proof of the following lemma is straightforward and let as an exercise.

Lemma 3.1.4. *Let V be a finite dimensional vector space with basis $\mathcal{B}$ and let $T\colon V \xrightarrow{\cong} W$ be an isomorphism. Then, $\mathcal{D} := (Tv_1, \dots, Tv_n)$ is a basis for W.*

Example 3.1.5. Let V be a finite dimensional vector space with basis $\mathcal{B}$ and let $T\colon V \xrightarrow{\cong} W$ be an isomorphism. Then, by the above lemma $\mathcal{D} := (Tv_1, \dots, Tv_n)$ is a basis and

$$[T]_{\mathcal{B}}^{\mathcal{D}} = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix} = \mathrm{Id}_n \in \mathbf{F}^{n,n}$$

is the identity matrix of $\mathbf{F}^{n,n}$.

Remark 3.1.6. The above example shows that any isomorphisms between *finite-dimensional* vector spaces can be represented by the identity matrix if we chose bases appropriately.

Exercise 3.1.7. Let V and W be two vector spaces of dimension n and let $T \in \mathcal{L}(V, W)$. Show that T is an isomorphism if and only if there exists bases $\mathcal{B}$ of V and $\mathcal{D}$ of W such that $[T]_{\mathcal{B}}^{\mathcal{D}}$ is the identity matrix of $\mathbf{F}^{n,n}$.

Before moving to the next result, let us recall the following statement. Let U, V and W be three finite-dimensional vector spaces of dimension p, n and m and with respectives bases $\mathcal{B}$, $\mathcal{C}$ and $\mathcal{D}$. Then for $S \in \mathcal{L}(U, V)$ and $T \in \mathcal{L}(V, W)$ we have $TS \in \mathcal{L}(U, W)$ and

$$[TS]_{\mathcal{B}}^{\mathcal{D}} = [T]_{\mathcal{C}}^{\mathcal{D}}[S]_{\mathcal{B}}^{\mathcal{C}},$$

where $[T]_{\mathcal{C}}^{\mathcal{D}} \in \mathbf{F}^{m,n}$, $[S]_{\mathcal{B}}^{\mathcal{C}} \in \mathbf{F}^{n,p}$ and $[TS]_{\mathcal{B}}^{\mathcal{D}} \in \mathbf{F}^{m,p}$.

Proposition 3.1.8. *Let V and W be two vector spaces, both of dimension n. Let $\mathcal{B}$ be a basis for V and let $\mathcal{D}$ be a basis for W. Let $T \in \mathcal{L}(V, W)$ be a linear map. Then T is invertible (as a linear map) if and only if $[T]_{\mathcal{B}}^{\mathcal{D}} \in \mathbf{F}^{n,n}$ is invertible (as a matrix).*

Moreover, if T is invertible, then $\left([T]_{\mathcal{B}}^{\mathcal{D}}\right)^{-1} = [T^{-1}]_{\mathcal{D}}^{\mathcal{B}}$.

Proof. "$\Rightarrow$" Suppose T is invertible, so $T^{-1} \in \mathcal{L}(W, V)$ exists. We will prove that $[T^{-1}]_{\mathcal{D}}^{\mathcal{B}}$ is the inverse matrix of $[T]_{\mathcal{B}}^{\mathcal{D}}$. We have

$$\begin{aligned} \mathrm{Id}_n &= [\mathrm{Id}_W]_{\mathcal{D}}^{\mathcal{D}} = [TT^{-1}]_{\mathcal{D}}^{\mathcal{D}} = [T]_{\mathcal{B}}^{\mathcal{D}}[T^{-1}]_{\mathcal{D}}^{\mathcal{B}} \\ \mathrm{Id}_n &= [\mathrm{Id}_V]_{\mathcal{B}}^{\mathcal{B}} = [T^{-1}T]_{\mathcal{D}}^{\mathcal{D}} = [T^{-1}]_{\mathcal{D}}^{\mathcal{B}}[T]_{\mathcal{B}}^{\mathcal{D}}. \end{aligned}$$

That is, $[T]_{\mathcal{B}}^{\mathcal{D}}$ is invertible and $\left([T]_{\mathcal{B}}^{\mathcal{D}}\right)^{-1} = [T^{-1}]_{\mathcal{D}}^{\mathcal{B}}$.

"$\Leftarrow$" Suppose $[T]_{\mathcal{B}}^{\mathcal{D}}$ is invertible. We will construct an inverse fot T. So let $\left([T]_{\mathcal{B}}^{\mathcal{D}}\right)^{-1} =: A = [a_{ij}]$ and define $S \in \mathcal{L}(W, V)$ by $[S(w)]_{\mathcal{B}} := A[w]_{\mathcal{D}}$ for all $w \in \mathcal{D}$. That is, S is the unique linear map from W to V such that $S(w_i) = \sum_{j=1}^{n} a_{ij}v_j$. By construction, we have $[S]_{\mathcal{D}}^{\mathcal{B}} = [[S(w_1)]_{\mathcal{B}} \ \dots \ [S(w_n)]_{\mathcal{B}}] = A$. So we conclude that $[ST]_{\mathcal{B}}^{\mathcal{B}} = \mathrm{Id}_V$. In particular, for any $1 \leq i \leq n$, we have $ST(v_i) = \mathrm{Id}_V(v_i)$ and hence $ST = \mathrm{Id}_V$. Similarly, $TS = \mathrm{Id}_W$. $\square$

The case of $V = W$ and $T \in \mathcal{L}(V)$ an operator is particularly interesting, as we can talk about powers. For $k \in \mathbf{N}$ positive we define

$$T^k := \underbrace{T \cdots T}_{k \text{ times}} \in \mathcal{L}(V)$$

while $T^0 := \mathrm{Id}_V$.

Proposition 3.1.9. *Let V be a finite dimensional space with basis $\mathcal{B}$ and let $T \in \mathcal{L}(V)$ be an operator. Then for $k \in \mathbf{N}$ we have*

$$[T^k]^{\mathcal{B}}_{\mathcal{B}} = \left([T]^{\mathcal{B}}_{\mathcal{B}}\right)^k.$$

Proof. The proof is by induction. If $k = 0$ or $k = 1$, this is obvious. Now, suppose that the formula holds for $k \geq 1$ and let us try to prove it for $(k+1) = k+1$. By the product formula

$$[T^{k+1}]^{\mathcal{B}}_{\mathcal{B}} = [T^k]^{\mathcal{B}}_{\mathcal{B}}[T]^{\mathcal{B}}_{\mathcal{B}} = \left([T]^{\mathcal{B}}_{\mathcal{B}}\right)^k [T]^{\mathcal{B}}_{\mathcal{B}} = \left([T]^{\mathcal{B}}_{\mathcal{B}}\right)^{k+1}.$$

□

Remark 3.1.10. By combining Propositions 3.1.8 and 3.1.9, if $T \in \mathcal{L}(V)$ is invertible, then we obtain $[T^k]^{\mathcal{B}}_{\mathcal{B}} = \left([T]^{\mathcal{B}}_{\mathcal{B}}\right)^k$ for any $k \in \mathbf{Z}$.

Exercise 3.1.11. Let $R_\theta \colon \mathbf{R}^2 \to \mathbf{R}^2$ be the map "rotation by angle θ". Let $\mathcal{E} = \left(\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}\right)$ be the standard basis for $\mathbf{R}^2$. Prove that R_θ is a linear map and find $[T_\theta]^{\mathcal{E}}_{\mathcal{E}}$.

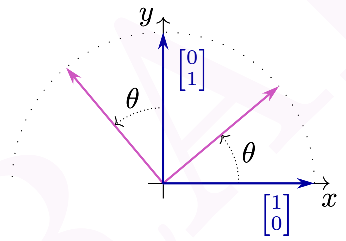

Figure 3.1: Rotation (in purple) of the standard basis vectors (in blue) by an angle θ.

Solution. Let $\begin{bmatrix}x\\y\end{bmatrix}$ be a point in $\mathbf{R}^2$. Define $r := \sqrt{x^2+y^2}$ and $\begin{bmatrix}x'\\y'\end{bmatrix} := T\begin{bmatrix}x\\y\end{bmatrix}$. Finally, let α be the angle between $\begin{bmatrix}x\\y\end{bmatrix}$ and the x-axis. Then we have $\cos(\alpha) = \frac{x}{r}$ and $\sin(\alpha) = \frac{y}{r}$, while $\cos(\alpha+\theta) = \frac{x'}{r}$ and $\sin(\alpha+\theta) = \frac{y'}{r}$. We conclude that

$$R\left(\begin{bmatrix}x\\y\end{bmatrix}\right) = \begin{bmatrix}x'\\y'\end{bmatrix} = \begin{bmatrix}x\cos(\theta) - y\sin(\theta)\\ x\sin(\theta) + y\cos(\theta)\end{bmatrix}.$$

Using the above formula, one easily check that R_θ is linear, and by computing the image of $\begin{bmatrix}1\\0\end{bmatrix}$ and $\begin{bmatrix}0\\1\end{bmatrix}$ we obtain

$$[R_\theta]^{\mathcal{E}}_{\mathcal{E}} = \begin{bmatrix}\cos(\theta) & -\sin(\theta)\\ \sin(\theta) & \cos(\theta)\end{bmatrix}. \tag{3.2}$$

□

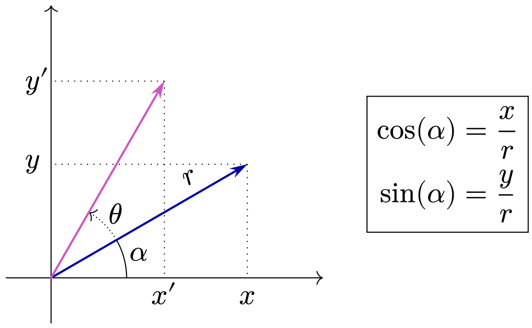

Figure 3.2: A vector (in blue) and its image (in purple) by the rotation of an angle θ.

Remark 3.1.12. The following facts can easily been seen geometrically (see Figure 3.3) but can also be shown using the formula for $[R_\theta]_{\mathcal{E}}^{\mathcal{E}}$.

1. $R_{\theta_1} R_{\theta_2} = R_{\theta_1+\theta_2}$;
2. R_θ is invertible and $(R_\theta)^{-1} = R_{-\theta}$;
3. for any $n \in \mathbf{Z}$ we have $(R_\theta)^n = R_{n\theta}$;
4. for any $k \in \mathbf{Z}$ we have $R_{2k\pi} = \mathrm{Id}_{\mathbf{R}^2}$.

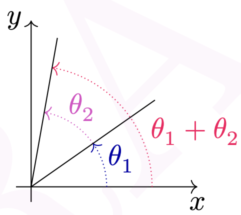

Figure 3.3: Angles addition.

Example 3.1.13. Let $S_1 \colon \mathbf{R}^2 \to \mathbf{R}^2$ be the reflexion along the x-axis, $S_2 \colon \mathbf{R}^2 \to \mathbf{R}^2$ the reflexion along the y-axis and let $\mathcal{E}$ be the standard basis. Find $[S_1]_{\mathcal{E}}^{\mathcal{E}}$ and $[S_2]_{\mathcal{E}}^{\mathcal{E}}$.

Solution. We have $S_1 \left[\begin{smallmatrix} x \\ y \end{smallmatrix}\right] = \left[\begin{smallmatrix} x \\ -y \end{smallmatrix}\right]$ and $S_2 \left[\begin{smallmatrix} x \\ y \end{smallmatrix}\right] = \left[\begin{smallmatrix} -x \\ y \end{smallmatrix}\right]$ for any $\left[\begin{smallmatrix} x \\ y \end{smallmatrix}\right]$ in $\mathbf{R}^2$. By computing the image of the basis vectors, we obtain

$$[S_1]_{\mathcal{E}}^{\mathcal{E}} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$
$$[S_2]_{\mathcal{E}}^{\mathcal{E}} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}.$$

□

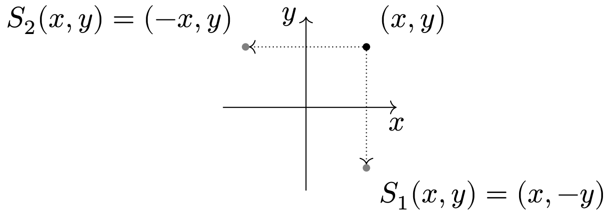

Figure 3.4: Mirror symmetries with respect to the x-axis (S_1) and the y-axis (S_2).

Observe that we the notations from the previous exercises, we have $[S_1S_2]_{\mathcal{E}}^{\mathcal{E}} = [S_2S_1]_{\mathcal{E}}^{\mathcal{E}} = [R_\pi]_{\mathcal{E}}^{\mathcal{E}}$. That is, $S_1S_2 = S_2S_1 = R_\pi$. While $S_1^2 = S_2^2 = (S_1S_2)^2 = \mathrm{Id}_{\mathbf{R}^2}$.

Example 3.1.14. Let $T \in \mathcal{L}(V)$. Suppose that there exists $v \in V$ and $n \in \mathbf{N}_{\geq 1}$ such that $T^n v = 0$, but $T^{n-1}v \neq 0$.

1. Prove that $\mathcal{B} := (v, Tv, \dots, T^{n-1}v)$ is linearly independent.
2. Let $U := \operatorname{span}(\mathcal{B})$ (so $\mathcal{B}$ is a basis for U). Prove that for any $u \in U$, its image Tu is still in U.
3. Let $T|_U \colon U \to U$ be the restriction of T to U. Find $[(T|_U)^k]_{\mathcal{B}}^{\mathcal{B}}$ for $k \in \mathbf{N}_{\geq 1}$.

Solution. 1. Observe that if $m \geq n$, then $T^m v = T^{m-n}T^n v = T^{m-n}0 = 0$. Suppose we can write

$$0_V = \sum_{i=0}^{n-1} \lambda_i T^i v = \lambda_0 v + \lambda_1 Tv + \dots + \lambda_{n-1}T^{n-1}v. \tag{3.3}$$

We want to show that all the λ_i are 0. By applying T^{n-1} to both sides of (3.3), we obtain $0_v = \lambda_0 T^{n-1}v + \lambda_2 0 + \dots + \lambda_{n-1}0$. But this implies that $\lambda_0 = 0$. We can then apply T^{n-2} to both sides of (3.3) to obtain that $\lambda_1 = 0$. By repeating this process, we obtain that all the λ_i are 0 as desired.

2. For an element $u \in U$ we have $u = \lambda_0 v + \dots + \lambda_{n-1}T^{n-1}v$, so

$$T(u) = \lambda_0 Tv + \dots + \lambda_{n-2}T^{n-1}v + \underbrace{\lambda_{n-1}T^n v}_{=0}$$

is also in U.

3. We have $T|_U(T^i v) = T^{i+1}v$, so

$$[T|_U]_{\mathcal{B}}^{\mathcal{B}} = \begin{array}{c} 1\\ 2\\ 3\\ \vdots\\ n \end{array} \begin{bmatrix} 0 & \cdots & \cdots & \cdots & 0\\ 1 & \ddots & & & \vdots\\ 0 & \ddots & \ddots & & \vdots\\ \vdots & \ddots & \ddots & \ddots & \vdots\\ 0 & \cdots & 0 & 1 & 0 \end{bmatrix}.$$

Similarly, $T|_U{}^k(T^i v) = T^{k+i}v$ and

$$[T|_U{}^k]_{\mathcal{B}}^{\mathcal{B}} = \begin{cases} \begin{array}{c} 1 \\ \vdots \\ k \\ k+1 \\ \vdots \\ n \end{array} \begin{bmatrix} 0 & & & & \\ \vdots & \ddots & & & \\ 0 & & \ddots & & \\ 1 & \ddots & & \ddots & \\ & \ddots & & & \\ & & 1 & \underbrace{0 \cdots\cdots 0}_{k} & \end{bmatrix} & \text{if } 1 \leq k \leq n-1, \\ 0 & \text{if } k \geq n. \end{cases}$$

□

Example 3.1.15. Let $D\colon \mathcal{P}_n(\mathbf{R}) \to \mathcal{P}_n(\mathbf{R})$ be the differentiation operator: $D(p) = p'$, and let $\mathcal{B} = (1, x, x^2, \dots, x^n)$ be the standard basis of $\mathcal{P}_n(\mathbf{R})$. Find $[D^k]_{\mathcal{B}}^{\mathcal{B}}$ for $k \in \mathbf{N}_{\geq 1}$

Solution. We have $D(1) = 0$ and $D(x^i) = ix^{i-1}$ for $1 \geq i \geq n$, which gives us the (n,n)-matrix

$$[D]_{\mathcal{B}}^{\mathcal{B}} = \begin{bmatrix} 0 & 1 & 0 & & \\ & 0 & 2 & \ddots & \\ & & 0 & \ddots & 0 \\ & & & \ddots & n \\ & & & & 0 \end{bmatrix}.$$

In general, for $1 \leq k \leq n$, we have

$$D^k(x^i) = \begin{cases} i(i-1)\cdots(i-k+1)x^{i-k} & \text{if } 1 \leq k \leq i, \\ 0 & \text{if } k \geq i+1, \end{cases}$$

which gives us the matrix

$$[D^k]_{\mathcal{B}}^{\mathcal{B}} = \begin{cases} \begin{bmatrix} \overbrace{0 \cdots\cdots 0}^{k} & k! & 0 \cdots\cdots\cdots 0 \\ \ddots & \ddots & \frac{(k+1)!}{1} \quad \vdots \\ & \ddots & \ddots \quad 0 \\ & & \ddots \quad \frac{n!}{(n-k)!} \\ & \ddots & 0 \\ & & \vdots \\ & & 0 \end{bmatrix} & \text{if } 1 \leq k \leq n, \\ 0 & \text{if } k \geq n+1. \end{cases}$$

□

3.1.2 Change of basis

So far, bases have played an important role in studying finite-dimensional vector spaces and linear maps between them. But bases are not unique! So it is natural to ask what happens when we change the basis.

Question 3.1.16. Let $\mathcal{B} = (v_1, \dots, v_n)$ and $\mathcal{D} = (w_1, \dots, w_n)$ be two basis for V. What is the relation between $[v]_{\mathcal{B}}$ and $[v]_{\mathcal{D}}$?

To answer this question, let us first consider the identity map

$$\mathrm{Id}_V \colon \underset{\mathcal{B}}{V} \longrightarrow \underset{\mathcal{D}}{V}.$$

That is, we think of $\mathcal{B}$ as a basis for the V the domain of Id_V, while we think of $\mathcal{D}$ as a basis of the codomain V of Id_V. Then for any $v \in V$ we have:

$$[v]_{\mathcal{D}} = [\mathrm{Id}_V]^{\mathcal{D}}_{\mathcal{B}} [v]_{\mathcal{B}}.$$

In other words, if we write

$$\begin{aligned} v_1 &= a_{11}w_1 + \cdots + a_{n1}w_n \\ &\vdots \\ v_n &= a_{nn}w_1 + \cdots + a_{nn}w_n \end{aligned}$$

we have

$$[\mathrm{Id}_V]^{\mathcal{D}}_{\mathcal{B}} = \begin{bmatrix} a_{11} & a_{12} \cdots\cdots a_{1n} \\ a_{21} & a_{22} \cdots\cdots a_{2n} \\ \vdots & \vdots \qquad\quad \vdots \\ a_{n1} & a_{n2} \cdots\cdots a_{nn} \end{bmatrix} \in \mathbf{F}^{n,n}.$$

We say that $[\mathrm{Id}_V]^{\mathcal{D}}_{\mathcal{B}}$ is the **change of basis matrix** from $\mathcal{B}$ to $\mathcal{D}$.

Example 3.1.17. Let $\mathcal{E} = \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$ be the standard basis of $\mathbf{R}^2$ and let $\mathcal{D} = \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \end{bmatrix} \right)$. Then

$$\begin{aligned} \mathrm{Id}_{\mathbf{R}^2} \begin{bmatrix} 1 \\ 0 \end{bmatrix} &= 1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 0 \begin{bmatrix} 0 \\ -2 \end{bmatrix}, \\ \mathrm{Id}_{\mathbf{R}^2} \begin{bmatrix} 0 \\ 1 \end{bmatrix} &= 0 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \left(-\tfrac{1}{2}\right) \begin{bmatrix} 0 \\ -2 \end{bmatrix}. \end{aligned}$$

Therefore,

$$[\mathrm{Id}_V]^{\mathcal{D}}_{\mathcal{E}} = \begin{bmatrix} 1 & 0 \\ 0 & -\frac{1}{2} \end{bmatrix}.$$

Proposition 3.1.18. *Let $\mathcal{B}$ and $\mathcal{D}$ be two bases for V. Then $[\mathrm{Id}_V]^{\mathcal{D}}_{\mathcal{B}}$ is invertible and $\left([\mathrm{Id}_V]^{\mathcal{D}}_{\mathcal{B}}\right)^{-1} = [\mathrm{Id}_V]^{\mathcal{B}}_{\mathcal{D}}$.*

Proof. This is simply Proposition 3.1.8 applied to the invertible map $\mathrm{Id}_V\colon V \to V$. □

Example 3.1.19. Suppose $\mathcal{D} = \left(\begin{bmatrix}1\\2\end{bmatrix}, \begin{bmatrix}3\\4\end{bmatrix}\right)$. One easily check that $\mathcal{D}$ is linearly independent and so is a basis for $\mathbf{R}^2$. Given $\begin{bmatrix}a\\b\end{bmatrix} \in \mathbf{R}^2$, compute $[v]_{\mathcal{D}}$.

Solution. First we compute $[\mathrm{Id}_V]_{\mathcal{E}}^{\mathcal{D}}$ where $\mathcal{E}$ is the standard basis of $\mathbf{R}^2$. In order to do that we start by solving $\begin{bmatrix}1\\0\end{bmatrix} = c_1\begin{bmatrix}1\\2\end{bmatrix} + c_2\begin{bmatrix}3\\4\end{bmatrix}$. We find $\begin{bmatrix}c_1\\c_2\end{bmatrix} = \begin{bmatrix}-2\\1\end{bmatrix}$. Then we solve $\begin{bmatrix}0\\1\end{bmatrix} = d_1\begin{bmatrix}1\\2\end{bmatrix} + d_2\begin{bmatrix}3\\4\end{bmatrix}$ and find $\begin{bmatrix}d_1\\d_2\end{bmatrix} = \begin{bmatrix}3/2\\-1/2\end{bmatrix}$. Putting this together, we have

$$[\mathrm{Id}_V]_{\mathcal{E}}^{\mathcal{D}} = \begin{bmatrix}-2 & \frac{3}{2}\\ 1 & -\frac{1}{2}\end{bmatrix}.$$

Finally, we have

$$[v]_{\mathcal{D}} = [\mathrm{Id}_V]_{\mathcal{E}}^{\mathcal{D}}[v]_{\mathcal{E}} = \begin{bmatrix}-2 & \frac{3}{2}\\ 1 & -\frac{1}{2}\end{bmatrix}\begin{bmatrix}a\\b\end{bmatrix} = \begin{bmatrix}-2a+\frac{3}{2}b\\ a-\frac{b}{2}\end{bmatrix}.$$

□

Observe that it was also possible to directly solve the system

$$\begin{bmatrix}a\\b\end{bmatrix} = \begin{bmatrix}1 & 3\\ 2 & 4\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix}.$$

Exercise 3.1.20. Let V be a vector space of dimension n and let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis for V. Let $[n] := \{1, 2, \dots, n\}$ and let $f\colon [n] \to [n]$ be an injective map (so f is bijective). Show that $\mathcal{D} := (v_{f(1)}, \dots, v_{f(n)})$ is also a basis for V.

Can you describe $[\mathrm{Id}_V]_{\mathcal{B}}^{\mathcal{D}}$? *Hint:* if $[v]_{\mathcal{B}} = [c_1, \dots, c_n]^{\mathsf{T}}$, then $[v]_{\mathcal{D}} = [c_{f(1)}, \dots, c_{f(n)}]^{\mathsf{T}}$.

We have seen how coordinate vectors change under change of basis. But how about matrix representations of linear maps? Let us consider the simpler case of operators. So let $T \in \mathcal{L}(V)$ and let $\mathcal{B}$ and $\mathcal{D}$ be two bases for V.

Question 3.1.21. What is the relation between $[T]_{\mathcal{B}}^{\mathcal{B}}$ and $[T]_{\mathcal{D}}^{\mathcal{D}}$?

Answer.

$$\begin{tikzcd}
{}^{\mathcal{B}}V \arrow[r, "T"] \arrow[d, "\mathrm{Id}_V"'] & V^{\mathcal{B}} \arrow[d, "\mathrm{Id}_V"] \\
{}_{\mathcal{D}}V \arrow[r, "T"'] & V_{\mathcal{D}}
\end{tikzcd}$$

Using bases as chosen, we have $[\mathrm{Id}_V]_{\mathcal{B}}^{\mathcal{D}}[T]_{\mathcal{B}}^{\mathcal{B}} = [T]_{\mathcal{D}}^{\mathcal{D}}[\mathrm{Id}_V]_{\mathcal{B}}^{\mathcal{D}}$, or

$$[T]_{\mathcal{D}}^{\mathcal{D}} = [\mathrm{Id}_V]_{\mathcal{B}}^{\mathcal{D}}[T]_{\mathcal{B}}^{\mathcal{B}}[\mathrm{Id}_V]_{\mathcal{D}}^{\mathcal{B}} \tag{3.4}$$

where we used the fact that $\left([\mathrm{Id}_V]_{\mathcal{B}}^{\mathcal{D}}\right)^{-1} = [\mathrm{Id}_V]_{\mathcal{D}}^{\mathcal{B}}$. ■

If you tried to compute some example of change of basis matrices, the result you obtained might have seen familiar to you. Indeed, they look like matrix representations of linear transformations. This ressemblance is not a coincidence.

To take a concrete example, let $\mathcal{E} = (e_1 = \left[\begin{smallmatrix}1\\0\end{smallmatrix}\right], e_2 = \left[\begin{smallmatrix}0\\1\end{smallmatrix}\right])$ be the standard basis of $\mathbf{R}^2$ and let R_θ be the rotation of angle θ around the origin. Finally, let $\mathcal{B} := (R_\theta e_1, R_\theta e_2)$. This is a basis of $\mathbf{R}^2$ and

$$[\mathrm{Id}_{\mathbf{R}^2}]^{\mathcal{E}}_{\mathcal{B}} = \begin{bmatrix}\cos(\theta) & -\sin(\theta)\\ \sin(\theta) & \cos(\theta)\end{bmatrix}$$

where you can recognize the matrix $[R_\theta]^{\mathcal{E}}_{\mathcal{E}}$ from Equation (3.2). This phenomenon is more general and we have.

Lemma 3.1.22. *Let V be a vector space with basis $\mathcal{B} = (v_1, \dots, v_n)$ and let $T \in \mathcal{L}(V)$ be an isomorphism. Then $T\mathcal{B} = (Tv_1, \dots, Tv_n)$ is a basis by Lemma 3.1.4 and $[\mathrm{Id}_V]^{\mathcal{B}}_{T\mathcal{B}} = [T]^{\mathcal{B}}_{\mathcal{B}}$.*

Proof. The j^{th} column of $[T]^{\mathcal{B}}_{\mathcal{B}}$ is given by the coefficients of $T(v_i) = a_{1i}v_1 + \dots + a_{ni}v_n$ while the i^{th} column of $[\mathrm{Id}_V]^{\mathcal{B}}_{T\mathcal{B}}$ is given by the coefficients of $\mathrm{Id}_V(Tv_i) = Tv_i = a_{1i}v_1 + \dots + a_{ni}v_n$. □

You can think of $[T]^{\mathcal{B}}_{\mathcal{B}}$ as "the space is fixed, vectors are moved by T" and of $[\mathrm{Id}_V]^{\mathcal{B}}_{T\mathcal{B}} = ([\mathrm{Id}_V]^{T\mathcal{B}}_{\mathcal{B}})^{-1}$ as "vectors are fixed, the underlying space is moved by T^{-1}". See Figure 3.5 for an example with $T = R_\theta$ the rotation of angle θ around the origin.

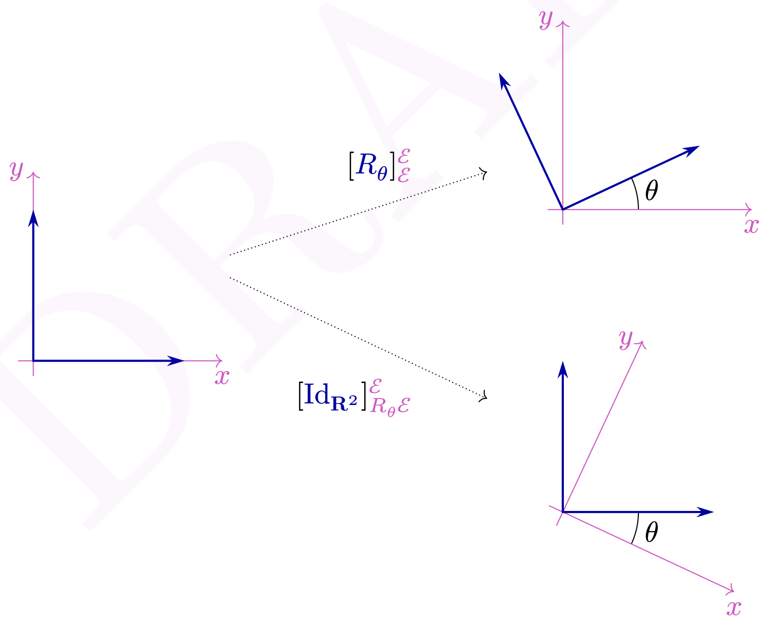

Figure 3.5: Rotation of an angle θ with respect to the standard basis, versus change of basis matrix.

Definition 3.1.23.

Two matrices A and B in $\mathbf{F}^{n,n}$ are **similar** if there exists an invertible matrix P in $\mathbf{F}^{n,n}$ such that $B = PAP^{-1}$.

Lemma 3.1.24. *Similarity is an* equivalence relation*:*

1. *A and A are similar;*
2. *If A and B are similar, then B and A are similar;*
3. *If A is similar to B and B is similar to C, then A is similar to C.*

Proof. 1. $A = \mathrm{Id}_n A \,\mathrm{Id}_n^{-1}$.
2. If $B = PAP^{-1}$, then $A = QBQ^{-1}$ for $Q = P^{-1}$.
3. If $B = PAP^{-1}$ and $C = QBQ^{-1}$, then $C = RAR^{-1}$ for $R = QP$. □

Proposition 3.1.25. *Let V be a vector space and let $T \in \mathcal{L}(V)$. Let $\mathcal{B}$ and $\mathcal{D}$ be two bases for V. Then $[T]_{\mathcal{B}}^{\mathcal{B}}$ and $[T]_{\mathcal{D}}^{\mathcal{D}}$ are similar.*

Proof. This directly follows from Equation (3.4). □

We have seen that any operator on a finite dimensional vector space admits a matrix representation. So a natural question to ask is: does there always exists a nice matrix representation?

Major Question 3.1.26.

Given V and $T \in \mathcal{L}(V)$, find a basis $\mathcal{B}$ such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is "as simple as possible" (for example: diagonal, upper triangular, with many zeroes, ...).

The main idea below this question is that a matrix is "nice" or "simple" if we can do fast computations with it. This is the case as soon as the matrix is sparse enough (has many zeroes).

We will see (Section 5.3 and in particular Theorem 5.3.9) that if V is a finite dimensional complex vector space, then given $T \in \mathcal{L}(V)$ there always exists a basis $\mathcal{B}$ such that

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{bmatrix} \boxed{J_1} & & & \\ & \boxed{J_2} & & \\ & & \ddots & \\ & & & \boxed{J_k} \end{bmatrix}$$

where each J_i is a Jordan block. That is, there exists an integer n_i and a complex number λ_i such that J_i is a $n_i \times n_i$ matrix of the form

$$J_i = \begin{bmatrix} \lambda_i & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & \lambda_i \end{bmatrix}.$$

Since $[T]_{\mathcal{B}}^{\mathcal{B}}$ is a $n \times n$ matrix, we should have $n_1 + \cdots + n_k = n$. But this is the only restriction on the n_i and λ_i. The above expression for $[T]_{\mathcal{B}}^{\mathcal{B}}$ is called the Jordan form of T.

3.2 Projections

Projections are useful tools that allow to go from one "big" space (infinite dimensional) to "smaller" spaces (finite dimensional) to make approximations. We will first discuss general projections, and will then see how we can apply them to make approximation of functions.

3.2.1 Projections: definition and first properties

Definition 3.2.1.

Let $V = U \oplus W$ be a direct sum decomposition. We define the **projection onto** U (along direction W) to be the map

$$P_U = P_{U,W} \colon V \longrightarrow V$$
$$v = u + w \longmapsto u.$$

By direct sum decomposition, for any $v \in V$, there exists a unique $u \in U$ (and a unique $w \in W$) such that $v = u + w$. So the above map P_U is well-defined.

Example 3.2.2. Let $V = \mathbf{R}^2$ and let $U = \{(x,0) \,|\, x \in \mathbf{R}\}$ be the x-axis and $W = \{(y,0) \mid y \in \mathbf{R}\}$ be the y-axis. Then $V = U \oplus W$ and P_U is the first coordinate projection: $P_U \left[\begin{smallmatrix} x \\ y \end{smallmatrix}\right] = x$.

remark 3.2.3. We write P_U, but the projection $P_U \colon U \oplus W \to V$ depends both on U and W! Let us demonstrate this dependence by an example. Let $V = \mathbf{R}^2$ and let $U = \{(x,0) \mid x \in \mathbf{R}\}$ be the x-axis. Let $W_1 = \{(y,0) \mid y \in \mathbf{R}\}$ be the y-axis and $W_2 = \{(y,y) \mid y \in \mathbf{R}\}$ be the line $y = x$. We have $V = U \oplus W_1 = U \oplus W_2$. Now, let $v = \left[\begin{smallmatrix} 2 \\ 1 \end{smallmatrix}\right]$ we have $\left[\begin{smallmatrix} 2 \\ 1 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 2 \\ 0 \end{smallmatrix}\right] + \left[\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right] \in U \oplus W_1$ but also $\left[\begin{smallmatrix} 2 \\ 1 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right] + \left[\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right] \in U \oplus W_2$. Therefore, $P_{U,W_1} \left[\begin{smallmatrix} 2 \\ 1 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 2 \\ 0 \end{smallmatrix}\right]$ is not the same as $P_{U,W_2} \left[\begin{smallmatrix} 2 \\ 1 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right]$. See Figure 3.6 for a picture.

For now, projections were just maps. But it turns out that they are linear maps satisfying many interesting properties.

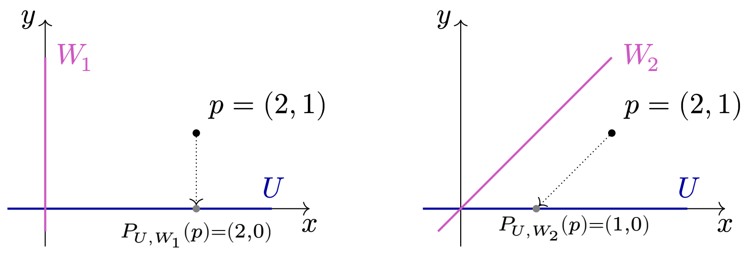

Figure 3.6: Projection of $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ on the x-axis, along direction W_1 (on the right) and along direction W_2 (on the left).

Proposition 3.2.4 (Properties of projection onto P_U)**.** *Let $V = U \oplus W$ be a direct sum decomposition. Then the projection P_U satisfies the following properties:*

1. $P_U \in \mathcal{L}(V)$*;*
2. $P_U(u) = u$ *for all* $u \in U$*;*
3. $P_U(w) = 0_V$ *for all* $w \in W$*;*
4. $\operatorname{range}(P_U) = U$*;*
5. $\operatorname{null}(P_U) = W$*;*
6. $v - P_U(v) \in W$ *for all* $v \in V$*;*
7. $P_U^2 = P_U$*.*

Proof. For v in V, let $v = u + w$ be it's direct sum decomposition: $u \in U$ and $w \in W$.

1. Let v_1 and v_2 be two vectors in V. There exists a unique way to write $v_1 = u_1 + w_1$ and $v_2 = u_2 + w_2$ with $u_1, u_2 \in U$ and $w_1, w_2 \in W$. But then $u_1 + u_2$ is in U and a simple computation gives

$$P_U(v_1 + v_2) = P_U((u_1 + u_2) + (w_1 + w_2)) = u_1 + u_2 = P_U(v_1) + P(v_2).$$

For scalar multiplication, suppose that $\lambda \in \mathbf{F}$ and $v \in V$. Then λu is in U and

$$P_U(\lambda v) = P_U(\lambda u + \lambda w) = \lambda u = \lambda P_U(v).$$

Hence P_U is a linear map from V to V.

2. Suppose u is in U. We have $u = u + 0_V$ with $0_V \in W$. So $P_U(u) = u$.

3. Suppose w is in w. We have $w = 0_V + w$ with $0_V \in U$. So $P_U(w) = 0_V$.

4. By definition of P_U, its range is included in U. Furthermore, 2 implies that U is included in the range of P_U. Therefore, we have the equality $\operatorname{range}(P_U) = U$.

5. It follows from 3 that $W \subseteq \operatorname{null}(P_U)$. For the other inclusion, let v be in the nullspace of P_U. That is $P_U(v) = u = 0$, so $v = w$ is in W.

6. For $v \in V$, we have $v - P_U(v) = u + w - u = w \in W$.

7. We have $P_U^2(v) = P_U(u) = u = P_U(v)$, so $P_U^2 = P_U$. $\square$

Projections are an easy way to test if a vector u belongs to a direct summand. Indeed, if $V = U \oplus W$, then by Proposition 3.2.4, $v \in V \iff P(v) = v$.

We have defined projections with respect to a direct sum decomposition. But properties 4 and 5 in Proposition 3.2.4 say that we can recover the spaces U and W from the linear map P_U. And we can use properties 1 and 7 to define abstract projections.

Definition 3.2.5.

An operator $P \in \mathcal{L}(V)$ is a **projection** if it satisfies $P^2 = P$.

Projections onto give us examples of projections.

Lemma 3.2.6. *If $V = U \oplus W$, then P_u is a projection.*

Proof. This is properties 1 and 7 of Proposition 3.2.4). □

We also have the following two basic examples.

Example 3.2.7. For any vector space V, both the zero map $0 \in \mathcal{L}(V)$ and the identity map Id_V are projections.

The above example should remind you that 0 and 1 are both solution to the real equation $z^2 = z$. But in a vector space of dimension $\dim(V) \geq 2$, the zero map and the identity map are not the only projections as demonstrated by Example 3.2.2. To elaborate a little bit: 0 is a solution of $z^2 = z$ while 1 is the only invertible solution (divide by z on both sides). Since non-zero complex numbers are invertible, these are all the solutions. Similarly, 0 is a solution of $P^2 = P$, while Id_V is the only invertible solution. But if V is a vector space of dimension at least 2, then $\mathcal{L}(V)$ is also a vector space of dimension at least 2 and hence there exists non-zero linear maps that are not invertible.

Below is an example showing the usefulness of matrix representations to find projections.

Example 3.2.8. Let $T \in \mathcal{L}(\mathbf{F}^2)$ be an operator. Suppose that there exists a basis $\mathcal{B}$ such that $[T]_{\mathcal{B}}^{\mathcal{B}} = A = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$. Then T is a projection. Indeed, since $[T^2]_{\mathcal{B}}^{\mathcal{B}} = \left([T]_{\mathcal{B}}^{\mathcal{B}}\right)^2$, the operator P is a projection if and only if $A^2 = A$. It is then easy to check that $\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}^2 = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$.

The same trick works with any matrix A such that $A^2 = A$. For example, if $[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$.

It turns out that *projections* (Definition 3.2.5) and *projections onto* (Definition 3.2.1) are exactly the same things as demonstrated by the following proposition.

Proposition 3.2.9. *Let V be a vector space and let $P \in \mathcal{L}(V)$ be a projection. Then $V = \mathrm{range}(P) \oplus \mathrm{null}(P)$ and $P = P_{\mathrm{range}(P)}$.*

Proof. It is clear that $\mathrm{range}(P) + \mathrm{null}(P) \subseteq V$. So we need to prove that the sum is direct and that it is equal to V. Let $v \in V$. Then one have $v = P(v) + (v - P(v))$ with $P(v) \in \mathrm{range}(P)$ and $v - P(v)$ is in $\mathrm{null}(P)$. Indeed, $P(v - P(v)) = P(v) - P^2(v) = P(v) - P(v) = 0$. Now, for the direct sum part. If $v \in \mathrm{range}(P) \cap \mathrm{null}(P)$ we have $v = P(w)$ and $0 = P(v) = P^2(w) = P(w) = v$, so $\mathrm{range}(P) \cap \mathrm{null}(P) = 0$ which finishes the proof of $\mathrm{range}(P) + \mathrm{null}(P) \subseteq V$.

Finally, let $v \in V$. Then $v = P(v) + (v - P(v))$ is the unique decomposition $v = u + w$ with $u \in \mathrm{range}(P)$ and $w \in \mathrm{null}(P)$. Therefore, $= P_{\mathrm{range}(P)}(v) = P(v)$ and we just demonstrated $P = P_{\mathrm{range}(P)}$. □

Another way to interpret Proposition 3.2.9 is that there is a one-to-one correspondance between projections $P \in \mathcal{L}(V)$ and direct sums decomposition $V = U \oplus W$ where order matter. We conclude this subsection by the following result that is a generalization of Proposition 3.2.9.

Theorem 3.2.10.

Suppose $P_1, \dots, P_k \in \mathcal{L}(V)$ are projections such that $P_1 + \cdots + P_k = \mathrm{Id}_V$ and $P_i P_j = 0_{\mathcal{L}(V)}$ if $i \neq j$. Then $V = \mathrm{range}(P_1) \oplus \cdots \oplus \mathrm{range}(P_k)$.

Proof. Let v be any vector in V. Then

$$v = \mathrm{Id}_V v = (P_1 + \cdots + P_k)v = P_1 v + \cdots + P_k v \in \mathrm{range}(P_1) + \cdots + \mathrm{range}(P_k) \tag{3.5}$$

and thus $V = P_1 v + \cdots + P_k v \in \mathrm{range}(P_1) + \cdots + \mathrm{range}(P_k)$. We now prove that the sum is direct. Suppose $0 = v_1 + \cdots + v_k$ with $v_i \in \mathrm{range}(P_i)$ for $1 \leq i \leq k$. We want to prove that all the v_i are zero. By definition of the range, there exist w_i such that $P_i w_i = v_i$. Using that P_i is a projection we have $P_i v_i = P_i^2 w_i = P_i w_i = v_i$ while $P_i v_j = P_i P_j w_j = 0 w_j = 0$ if $i \neq j$. Now, if we apply P_i to both sides of (3.5) we obtain $0 = v_i$. By doing this for all $1 \leq i \leq k$ we have that all the v_i are zero as desired, and so the sum is direct. □

Now that we have proven the Theorem, let us discuss why it is a generalization of Proposition 3.2.9. Let $P \in \mathcal{L}(V)$ be a projection, $P_1 := P$ and $P_2 := \mathrm{Id}_V - P$. One easily check the following properties:

- P_2 is a projection: $P_2^2 = \mathrm{Id}_V^2 - \mathrm{Id}_V P - P \mathrm{Id}_V + P^2 = \mathrm{Id}_V - P - P + P = P_2$;
- $P1 + P_2 = \mathrm{Id}_V$;
- and $P_1 P_2 = P^2 - P = 0 = P_2 P_1$.

One can hence can apply Theorem 3.2.10 to P_1 and P_2 to obtain $V = \mathrm{range}(P) \oplus \mathrm{range}(\mathrm{Id}_v - P)$. Finally, $\mathrm{range}(\mathrm{Id}_v - P) = \mathrm{null}(P)$ and we recover the Proposition. Indeed, if v is in $\mathrm{null}(P)$, then $v = v - 0 = v - P(v) = (\mathrm{Id}_V - P)(v)$, which proves $\mathrm{null}(P) \subseteq \mathrm{range}\,\mathrm{Id}_V - P$. For the other inclusion, if $v \in \mathrm{range}\,\mathrm{Id}_V - P$, then $v = (\mathrm{Id}_V - P)(w)$ for some w and so $P(v) = P(\mathrm{Id}_V - P)(w) = P(w) - P^2(w) = 0$.

3.2.2 Left/right inverses and projections

An easy way to create projections is to use left and right inverses.

Proposition 3.2.11. *Let V and W be two vector spaces and let $T \in \mathcal{L}(V, W)$ and $S \in \mathcal{L}(W, V)$ be such that $TS = \mathrm{Id}_W$. Then $V = \mathrm{range}(S) \oplus \mathrm{null}(T)$ and $ST \in \mathcal{L}(V)$ is a projection to* $\mathrm{range}(S)$ *with nullspace* $\mathrm{null}(T)$.

Proof. We first show that ST is an abstract projection: $(ST)^2 = S(TS)T = S\,\mathrm{Id}_W T = ST$.

It now remains to show that $\mathrm{range}(ST) = \mathrm{range}(S)$ and $\mathrm{null}(ST) = \mathrm{null}(T)$. If v is in $\mathrm{range}(ST)$, then there exists $u \in V$ such that $v = ST(u) = S(T(u)) \in \mathrm{range}(S)$. So $\mathrm{range}(ST) \subseteq \mathrm{range}(S)$. For the other inclusion, let $v \in \mathrm{range}(S)$. Thus there exists $w \in W$ with $v = S(w) = S\,\mathrm{Id}_W(w) = STS(w) = ST(T(w))$ which shows that $\mathrm{range}(S) \subseteq \mathrm{range}(ST)$.

For the nullspaces, if $v \in \mathrm{null}(T)$, then $ST(v) = S(0) = 0$ and hence $\mathrm{null}(T) \subseteq \mathrm{null}(ST)$. For the other inclusion, let v be an element in $\mathrm{null}(ST)$. We have $0_V = ST(v)$ and by applying T to both sides $0_W = TST(v) = \mathrm{Id}_W T(v) = T(v)$, which shows that $\mathrm{null}(ST) \subseteq \mathrm{null}(T)$. □

Observe that for a given $T \in \mathcal{L}(V, W)$ there exists a right inverse S such that $TS = \mathrm{Id}_W$ if and only if T is surjective. If such an inverse exists it is not unique, unless T is an isomorphism. This is related to the fact that given a general subspace U ($= \mathrm{null}(T)$) of V, there exists more than one direct sum complement X ($= \mathrm{range}(S)$) such that $V = U \oplus X$. Let us exemplify this.

Example 3.2.12. Let $T\colon \mathbf{R}^2 \to \mathbf{R}$ be the map $(x, y) \mapsto x$. This is a linear map with $\mathrm{null}(T) = \{(0, y) \mid y \in \mathbf{R}\}$. Define $S_1, S_2\colon \mathbf{R} \to \mathbf{R}^2$ by

$$\begin{aligned} S_1(x) &= (x, 0) \\ S_2(x) &= (x, x). \end{aligned}$$

These two maps are linear and one easily verify that $TS_1 = TS_2 = \mathrm{Id}_{\mathbf{R}}$. By the above proposition, both S_1T and S_2T are projections. On one hand, S_1T is the projection onto $\mathrm{range}(S_1T) = \mathrm{range}(S_1) = \{(x, 0 \mid x \in \mathbf{R}\}$ and along direction $\mathrm{null}(T)$. On the other hand, S_2T is the projection onto $\mathrm{range}(S_2T) = \mathrm{range}(S_2) = \{(x, x \mid x \in \mathbf{R}\}$ and along direction $\mathrm{null}(T)$. See Figure 3.7.

Many examples of the phenomenon described in Proposition 3.2.11 arise when you want to do approximations. Suppose we had some "complicated" space V (usually a space of functions) where we want to be able to do some approximations. We can consider a linear map $T\colon V \to \mathbf{F}^n$ which we think of as "taking samples/measurements/evaluating". We then construct a right inverse $S\colon \mathbf{F}^n \to V$ which "selects a solution/reconstructs something" using this data. Finally, the map $ST \in \mathcal{L}(V)$ will be a projections onto U a nice subspace of V, for example $U = \mathcal{P}_n(\mathbf{R})$.

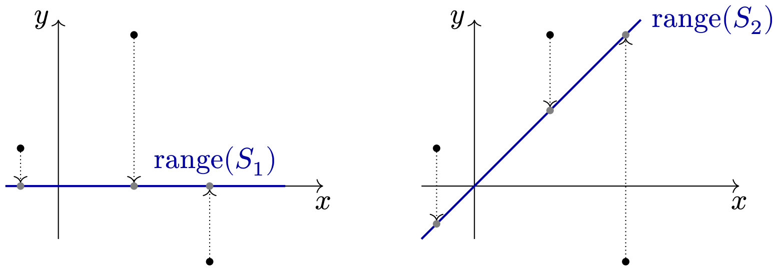

Figure 3.7: Two projections along the y-axis direction. The left one is onto the x-axis, and the right one is onto the diagonal $y = x$.

Example 3.2.13. Let $V = \mathcal{C}^\infty(\mathbf{R}) = \{f\colon \mathbf{R} \to \mathbf{R} \mid f^{(n)} \text{ exists } \forall n\}$ be the space of smooth real functions. This space is infinite dimensional and quite complicated. To better understand it, we want to approximate it by polynomials. That is, we would like to have projections $P_n\colon \mathcal{C}^\infty(\mathbf{R}) \to \mathcal{P}_n(\mathbf{R})$. If $n = 0$, then we will approximate continuous functions by constant functions, if $n = 1$ the approximation will be by lines, by parabola if $n = 2$ and so on. Of course, bigger is n, better is going to be the approximation, but also more complicated. In order to define the projections, we will use Proposition 3.2.11. So we need linear surjective maps $T_n\colon \mathcal{C}^\infty(\mathbf{R}) \to \mathbf{R}^{n+1}$ and right inverses maps $S_n\colon \mathbf{R}^{n+1} \to \mathcal{C}^\infty(\mathbf{R})$. Let us define

$$T_n(f) := (f(a), f'(a), \dots, f^{(n)}(a)) \in \mathbf{R}^{n+1},$$
$$S_n(x_0, \dots, x_n) := x_0 + x_1(x-a) + \frac{x_2}{2!}(x-a)^2 + \dots + \frac{x_n}{n!}(x-a)^n \in \mathcal{P}_n(\mathbf{R}) \subseteq \mathcal{C}^\infty(\mathbf{R}).$$

These maps are linear (exercise) and are left/right inverses. Indeed,

$$\begin{aligned}
(T_nS_n)(1,0,\dots,0) &= T_n(1) = (1,0,\dots,0),\\
(T_nS_n)(0,1,0,\dots,0) &= T_n(x-a) = (0,1,0,\dots,0),\\
(T_nS_n)(0,0,1,0,\dots,0) &= T_n\left(\tfrac{(x-a)^2}{2}\right) = (0,0,1,0,\dots,0),\\
&\vdots\\
(T_nS_n)(0,\dots,0,1) &= T_n\left(\tfrac{(x-a)^n}{n!}\right) = (0,\dots,0,1).
\end{aligned}$$

Therefore, $P_n := S_nT_n \in \mathcal{L}(\mathcal{C}^\infty(\mathbf{R}))$ is a projection to $\operatorname{range}(S_n) = \mathcal{P}_n(\mathbf{R})$, which has dimension $n+1$. In fact,

$$S_nT_n(f) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots + \frac{f^{(}n)(a)}{n!}(x-a)^n$$

is the n^{th} Taylor polynomial of f at $x = a$. Finally, the $\operatorname{null}(S_nT_n) = \operatorname{null}(T_n)$ is the set of all functions g such that $0 = g(a) = g'(a) = \dots = g^{(n)}(a)$.

The sequence of Taylor polynomials forms a better and better *local approximation* of f around a. Local approximation means that for x close to a, the value of $f(x)$ and $((S_nT_n)(f))(x)$ are nearly the same. More formally, since $V = \operatorname{range}(S_nT_n) \oplus \operatorname{null}(S_nT_n)$, every function f can be written in a unique way as $p+g$, where p a polynomial of degree at most n and $g \in \mathrm{null}(T_n)$. In other words, $(f-(S_nT_n)f)^{(i)}(a) = 0$ for all $0 \leq i \leq n$. That is, near a, the function $(f-(S_nT_n)f)$ is close to be the constant function 0.

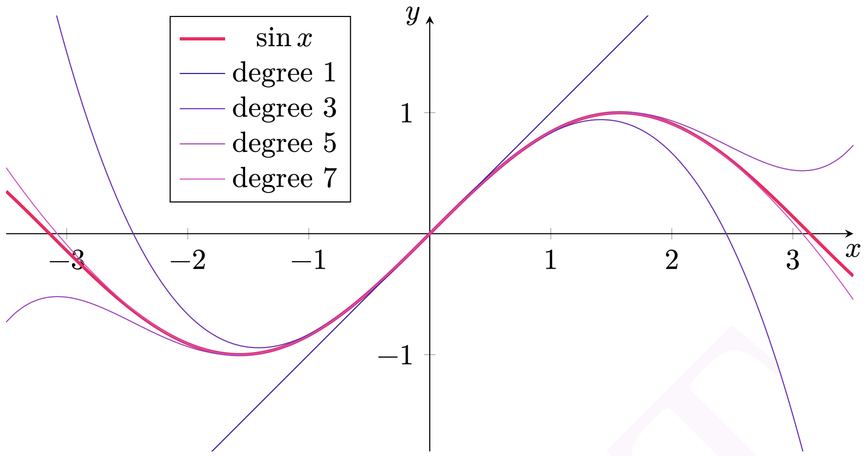

Figure 3.8: Taylor polynomials of degree n of $\sin(x)$ at $x=0$ for $n \in \{1,3,5,7\}$.

Example 3.2.14. In the previous example, we approximated a function f by a polynomial P such that P and f agree on a (that is $P(a) = f(a)$) and also the the i^{th} derivatives of P and f agree on a ($P^{(i)}(a) = f^{(i)}(a)$ for $1 \leq i \leq n$). This is a local approximation of the function f. Another way to approximate f is to find a polynomial G such that $f(x) = G(x)$ for many values of x. This is what we will do now, using Lagrange polynomials. Let us fix a sequence of pairwise distinct real numbers $(x_0, x_1, x_2, \dots)$ and define

$$\begin{aligned} Q_n \colon \mathcal{C}^\infty &\longrightarrow \mathbf{R}^{n+1} \\ f &\longmapsto (f(x_0), f(x_1), \dots, f(x_n)). \end{aligned}$$

For the other direction, we define R_n by its value on the standard basis vectors $(e_1 = (1,0\dots,0), \dots, e_{n+1} = (0,\dots,0,1))$:

$$\begin{aligned} R_n(e_i) &:= \prod_{\substack{0\leq m\leq n\\ m\neq i}} \frac{x-x_m}{x_i-x_m} \\ &= \frac{x-x_0}{x_i-x_0} \cdots \frac{x-x_{i-1}}{x_i-x_{i-1}} \frac{x-x_{i+1}}{x_i-x_{i+1}} \cdots \frac{x-x_n}{x_i-x_n}. \end{aligned}$$

The maps Q_n and R_n are linear and left/right inverses: $Q_nR_n = \mathrm{Id}_{\mathbf{R}^{n+1}}$ (exercise).

The polynomial $R_n(e_i)$ is known as the i^{th} *Lagrange polynomial*. This is the unique polynomial p of degree at most n such that $p(x_j) = 0$ if $j \neq i$ and $p(x_i) = 1$. This property directly implies that $(Q_nR_n)(e_i) = e_i$ and so $Q_nR_n = \mathrm{Id}_{\mathbf{R}^n}$. Therefore, $R_nQ_n \in \mathcal{L}(\mathcal{C}^\infty)$ is a projection to $\mathrm{range}(R_n) = \mathcal{P}_n(\mathbf{R})$, which has dimension $n+1$. In fact, $R_nQ_n(f)$ is a polynomial of degree at most n such that for $0 \leq i \leq n$ we have $(R_nQ_n(f))(x_i) = f(x_i)$.

In this sense, the sequence of Lagrange polynomials forms another series of approximations of f.

This time, $\mathrm{null}(R_n Q_n) = \mathrm{null}(Q_n) = \{g \in \mathcal{C}^\infty \mid 0 = g(x_0) = \cdots = g(x_n)\}$.

4 Inner product spaces

Standing assumption

In this chapter, $\mathbf{F}$ will always be either $\mathbf{R}$ the field of real numbers or $\mathbf{C}$ the field of complex numbers. V will always denotes a vector space over $\mathbf{F}$.

When going from $\mathbf{R}^2$ to abstract vector spaces, we generalized the notions of vector addition and of scalar multiplication. But they are some other features of $\mathbf{R}^2$ that we didn't capture. For example, in $\mathbf{R}^2$, one can talk about the length of a vector, or about the angle between two vectors. For a general vector space V, we do not have (yet) such notions as "length" or "angle".

4.1 Inner products and norms

Recall (from highschool or MTH008) that for $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$ two vectors in $\mathbf{R}^2$ we have the dot product $x \bullet y := x_1 y_1 + x_2 y_2 \in \mathbf{R}$. This is useful as the length of x is equal to $\sqrt{x_1^2 + x_2^2} = \sqrt{x \bullet x}$.

4.1.1 Dot product in $\mathbf{R}^n$

We first generalize scalar product to $\mathbf{R}^n$.

Definition 4.1.1.

Let $x = [x_1, \dots, x_n]^\mathsf{T}$ and $y = [y_1, \dots, y_n]^\mathsf{T}$ be two vectors in $\mathbf{R}^n$. We define their **dot product**[a] to be

$$x \bullet y := x_1 y_1 + \dots + x_n y_n \in \mathbf{R}.$$

[a] The dot product is sometimes also called scalar product; which is not the same thing as the scalar multiplication.

Definition 4.1.2.

Let $x = [x_1, \dots, x_n]^\mathsf{T}$ be a vector in $\mathbf{R}^n$. We define its **norm** to be

$$\|x\| := \sqrt{x \bullet x} = \sqrt{x_1^2 + \dots + x_n^2} \in \mathbf{R}.$$

The norm $\|x\|$ of x is a generalization of the length in $\mathbf{R}$, $\mathbf{R}^2$ or $\mathbf{R}^3$. We may think of it as the distance between the point $(0, \dots, 0)$ and the point $(x_1, \dots, x_n)$.

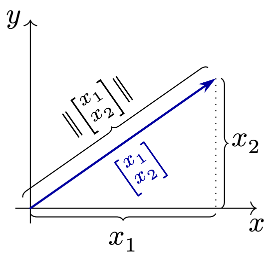

Figure 4.1: The norm of the vector $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in \mathbf{R}^2$ is $\left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right\| = \sqrt{x_1^2 + x_2^2}$.

Proposition 4.1.3. *The dot product in $\mathbf{R}^n$ satisfies the following properties.*

1. *For all $x \in \mathbf{R}^n$ we have $x \bullet x \geq 0$;*
2. *For all $x \in \mathbf{R}^n$ we have $x \bullet x = 0 \iff x = 0$;*
3. *Fix $y \in \mathbf{R}^n$. The map $\bullet y \colon \mathbf{R}^n \to \mathbf{R}^n$, $x \mapsto x \bullet y$ is linear;*
4. *For all $x, y \in \mathbf{R}^n$, we have $x \bullet y = y \bullet x$.*

Proof. 1. For $x = [x_1, \dots, x_n]^\top \in \mathbf{R}^n$, we have $x \bullet x = x_1^2 + \cdots + x_n^2 \geq 0$.
2. $x \bullet x = 0$ if and only if $x_1^2 + \cdots + x_n^2 = 0$, if and only if all the x_i are 0, if and only if $x = 0$.
3. Using the definition, one can easily check that $(x + x') \bullet y = x \bullet y + x' \bullet y$ and $(\lambda x) \bullet y = \lambda(x \bullet y)$ for $\lambda \in \mathbf{R}$.
4. This directly follows from the definition and the commutativity of addition in $\mathbf{R}$. □

The above proposition suggests the right direction to generalize the dot product to other real vector spaces. Before doing that, we will turn our attention to complex vector spaces.

4.1.2 Dot product in $\mathbf{C}^n$

Can we simply mimic the dot product in $\mathbf{R}^n$ for $\mathbf{C}^n$? No! Indeed, if we use the same definition then we are going to have a problem for the length of i. Indeed, $\sqrt{i^2} = \sqrt{-1}$ is not a real number, while we would like $\|z\|$ to be the "length of z" and hence a positive real number. In order to solve this problem we will use the following definitions.

Definition 4.1.4.

Let $z = [z_1, \dots, z_n]^\top$ and $w = [w_1, \dots, w_n]^\top$ be two vectors in $\mathbf{C}^n$. We define their **dot product** to be

$$z \bullet w := z_1 \overline{w}_1 + \cdots + z_n \overline{w}_n,$$

where $\overline{w}_1$ is the complex conjugate of w_1.

Definition 4.1.5.

Let $z = [z_1, \dots, z_n]^\mathsf{T}$ be a vector in $\mathbf{C}^n$. We define its **norm** to be

$$\|z\| := \sqrt{z \bullet z}.$$

If w_i is a real number, then $\overline{w_i} = w_i$, so the above definition are indeed generalizations of the dot product and norm on $\mathbf{R}^n$.

Before going further, we prove one important property of the complex dot product.

Proposition 4.1.6. *For every $z \in \mathbf{C}^n$ the dot product $z \bullet z$ is a non-negative real number.*

Proof. For $w = a + bi \in \mathbf{C}$ with a and b in $\mathbf{R}$, we have $\overline{w} = a - bi$ and so $w\overline{w} = (a+bi)(a-bi) = a^2+b^2 = |w|^2$ is in $\mathbf{R}_{\geq 0}$. So we have $z \bullet z = |z|_1{}^2 + \dots + |z|_n{}^2 \in \mathbf{R}_{\geq 0}$. □

Corollary 4.1.7. *For every $z \in \mathbf{C}^n$ we have $\|z\| \in \mathbf{R}_{\geq 0}$.*

Complex dot product enjoys properties similar to the real dot product.

Proposition 4.1.8. *The dot product in $\mathbf{C}^n$ satisfies the following properties.*

1. *For all $z \in \mathbf{C}^n$ we have $z \bullet z \in \mathbf{R}_{\geq 0}$;*
2. *For all $z \in \mathbf{C}^n$ we have $z \bullet z = 0 \iff z = 0$;*
3. *Fix $w \in \mathbf{C}^n$. The map $\bullet\, w \colon \mathbf{C}^n \to \mathbf{C}^n$, $z \mapsto z \bullet w$ is linear;*
4. *For all $w, z \in \mathbf{C}^n$, we have $z \bullet c = \overline{w \bullet z}$.*

Proof. Exercise. □

4.1.3 Inner product spaces

We can use the dot products in $\mathbf{R}^n$ and $\mathbf{C}^n$ as inspiration to define a generalization of "inner product" on abstract real or complexe vector spaces.

Definition 4.1.9.

Let $\mathbf{F}$ be either $\mathbf{R}$ or $\mathbf{C}$ and let V be a $\mathbf{F}$-vector space. An **inner product** on V is a function

$$\begin{aligned} \langle \cdot, \cdot \rangle \colon V \times V &\longrightarrow \mathbf{F} \\ (u, v) &\longmapsto \langle u, v \rangle \end{aligned}$$

satisfying the following properties:

1. *(Positivity)* $\langle v, v \rangle \in \mathbf{R}_{\geq 0}$ for all $v \in V$;
2. *(Definiteness)* $\langle v, v \rangle = 0 \iff v = 0$;

3. *(Linearity in 1^{st} variable)*[a] For all $u, v, w \in V$ and all $\lambda \in \mathbf{F}$ we have

$$\langle u+v, w\rangle = \langle u, w\rangle + \langle v, w\rangle,$$
$$\langle \lambda v, w\rangle = \lambda\langle v, w\rangle;$$

4. *(Conjugate symmetry)* $\langle v, w\rangle = \overline{\langle w, v\rangle}$ for all $v, w \in V$.

[a] For us, an inner product is linear in the first variable. Some people prefer to define inner product to be linear in the second variable. Both definitions are fine, but once you have chosen one, you should stick with it.

Remark 4.1.10. If V is an $\mathbf{R}$-vector space, then condition $\langle v, w\rangle = \overline{\langle w, v\rangle}$ is equivalent to $\langle v, w\rangle = \langle w, v\rangle$. Therefore, if $\langle\,\cdot\,,\cdot\,\rangle$ is an inner-product on a real vector space, it is also linear in the second variable. That is, for all $u, v, w \in V$ and all $\lambda \in \mathbf{R}$ we have

$$\langle u, v+w\rangle = \langle u, v\rangle + \langle u, w\rangle,$$
$$\langle v, \lambda w\rangle = \lambda\langle v, w\rangle.$$

Since $\langle\,\cdot\,,\cdot\,\rangle$ is linear on both variable, it is says to be **bilinear**. For real vector spaces, we can replace conditions 3 and 4 in the definition of inner product by the condition that $\langle\,\cdot\,,\cdot\,\rangle$ is bilinear.

Remark 4.1.11. If $\langle\,\cdot\,,\cdot\,\rangle$ is an inner-product on a $\mathbf{C}$ vector space, then it is not linear in the second coordinate, but **semilinear**. That is, for all $u, v, w \in V$ and all $\lambda \in \mathbf{C}$ we have

$$\langle u, v+w\rangle = \langle u, v\rangle + \langle u, w\rangle,$$
$$\langle v, \lambda w\rangle = \overline{\lambda}\langle v, w\rangle.$$

In this case, we say that $\langle\,\cdot\,,\cdot\,\rangle$ is **sesquilinear**, which means 1.5-*linear*. For complex vector spaces, we can replace conditions 3 and 4 in the definition of inner product by the condition that $\langle\,\cdot\,,\cdot\,\rangle$ is sesquilinear.

The following result is an easy consequence of the definition.

Lemma 4.1.12. *Let $\langle\,\cdot\,,\cdot\,\rangle$ be an inner product on a vector space V. Then for every $v \in V$ we have*

$$\langle v, 0\rangle = 0 = \langle 0, v\rangle.$$

Definition 4.1.13.

Let $\mathbf{F}$ be either $\mathbf{R}$ or $\mathbf{C}$. An **inner product space** over $\mathbf{F}$ is an $\mathbf{F}$ vector space equipped with an inner product $\langle\,\cdot\,,\cdot\,\rangle$.

We will often simply write V instead of $(V, \langle\,\cdot\,,\cdot\,\rangle)$ for an inner product space.

Example 4.1.14. $\mathbf{R}^n$ with the dot product is an inner product real vector space, sometimes called an **Euclidean space**. $\mathbf{C}^n$ with the dot product is an inner product complex vector space.

Example 4.1.15. Let $c_1, \ldots, c_n > 0$. Define $\langle \cdot , \cdot \rangle \colon \mathbf{F}^n \to \mathbf{F}^n$ by

$$\langle (z_1, \ldots, z_n), (w_1, \ldots, w_n) \rangle := c_1 z_1 \overline{w}_1 + \cdots + c_n z_n \overline{w}_n.$$

Then $\langle \cdot , \cdot \rangle$ is an inner product.

Proof. We have $\langle z, z \rangle = c_1 |z_1|^2 + \cdots + c_n |z_n|^2$. This implies both that $\langle z, z \rangle$ is a positive real number and that $\langle z, z \rangle = 0 \iff z = 0$.

The other two defining properties of an inner product are easily checked. For linearity in 1^{st} coordinate:

$$\begin{aligned} \langle \lambda z, w \rangle &= c_1 \lambda z_1 \overline{w}_1 + \cdots + c_n \lambda z_n \overline{w}_n \\ &= \lambda (c_1 z_1 \overline{w}_1 + \cdots + c_n z_n \overline{w}_n) = \lambda \langle z, w \rangle. \end{aligned}$$

For conjugate symmetry:

$$\begin{aligned} \overline{\langle w, z \rangle} &= \overline{c_1 w_1 \overline{z}_1 + \cdots + c_n w_n \overline{z}_n} \\ &= \overline{c}_1 z_1 \overline{w}_1 + \cdots + \overline{c}_n z_n \overline{w}_n \\ &= c_1 z_1 \overline{w}_1 + \cdots + \overline{c}_n z_n \overline{w}_n \\ &= \langle z, w \rangle, \end{aligned}$$

where we used both that $\overline{\overline{z_i}} = z_i$ and that $\overline{c}_i = c_i$ since the c_i are real numbers. □

Remark 4.1.16. The above example shows that there can be more than one inner product structure on the same vector space.

Example 4.1.17. Let $\mathcal{C}^0([-1,1]) = \{f \colon [-1,1] \to \mathbf{R} \mid f \text{ is continuous}\}$ be the real vector space of continuous functions from $[-1,1]$ to $\mathbf{R}$. Define $\langle \cdot, \cdot \rangle : \mathcal{C}^0([-1,1]) \times \mathcal{C}^0([-1,1]) \to \mathbf{R}$ by

$$\langle f, g \rangle := \int_{-1}^{1} f(x) g(x) \, \mathrm{d}x.$$

Then $\langle \cdot , \cdot \rangle$ is an inner product.

Proof. If both f and g are a continuous functions defined on $[-1,1]$, then $f \cdot g$ is also continuous and hence integrable. This implies that $\langle \cdot , \cdot \rangle$ is well-defined. Moreover,

$$\langle f, f \rangle = \int_{-1}^{1} f(x) f(x) \, \mathrm{d}x \geq 0.$$

This proves positivity.

It is clear that $\langle 0, 0 \rangle = 0$. Definiteness is a little more difficult to show and we will do it later.

Linearity of $\langle \cdot, \cdot \rangle$ is the well-known linearity of definite integral.

The definition is symmetric in 1st and 2nd variable: $\langle f, g\rangle = \langle g, f\rangle$ and we have conjugate symmetry since $\langle f, g\rangle$ is a real number.

In order to prove definiteness, we need to show that if $\int_{-1}^{1} f^2(x)\,\mathrm{d}x = 0$, then $f^2 = 0$ and therefore $f = 0$. The proof of this claim requires analysis. We will actually show the converse, namely: *if $g\colon [-1, 1]$ is a non-negative continuous function such that $g(x_0) \neq 0$ for some $x_0 \in [-1, 1]$, then $\int_{-1}^{1} g(x)\,\mathrm{d}x \neq 0$.*

Let $\varepsilon := {}^1\!/\!{}_2 g(x_0) > 0$. By continuity, there exists $\delta > 0$ such that if $|x - x_0| < \delta$, then $|g(x) - g(x_0)| < \varepsilon$. In particular, if x is in $[x_0 - \delta, x_0 + \delta]$, then $g(x) > \varepsilon$.

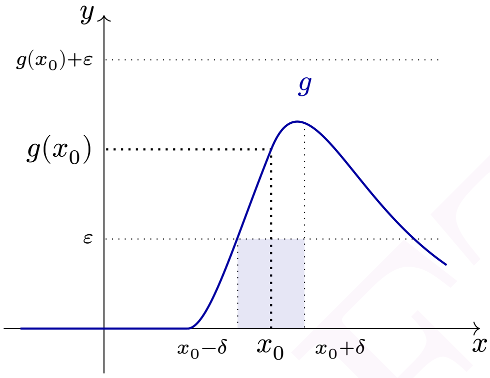

Figure 4.2: A non-zero positive continuous function. Its integral is at least the area of the blue rectangle: $\int_{x_0-\delta}^{x_0+\delta} g(x)\,\mathrm{d}x \geq 2\delta\varepsilon$.

We conclude that

$$\begin{aligned}\int_{-1}^{1} g(x)\,\mathrm{d}x &= \int_{-1}^{x_0-\delta} g(x)\,\mathrm{d}x + \int_{x_0-\delta}^{x_0+\delta} g(x)\,\mathrm{d} + \int_{x_0+\delta}^{1} g(x)\,\mathrm{d}x > 2\\ &= \int_{-1}^{x_0-\delta} g(x)\,\mathrm{d}x + 2\delta\varepsilon + \int_{x_0+\delta}^{1} g(x)\,\mathrm{d}x \geq 0 + 2\delta\varepsilon + 0 > 0.\end{aligned}$$

This proves the claim, and finishes the proof of definiteness. $\square$

Example 4.1.18. $\langle p, q\rangle := \int_0^\infty p(x)q(x)e^{-x}\,\mathrm{d}x$ defines an inner product on $\mathcal{P}(\mathbf{R})$. The proof is left as an exercise.

Definition 4.1.19.

Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space over $\mathbf{F}$. For a vector $v \in V$, we define it's **norm** to be $\|v\| := \sqrt{\langle v, v\rangle}$.

If V is an inner product space, we can regard the norm as a function

$$\begin{aligned} \|v\|: V &\longrightarrow \mathbf{R}_{\geq 0} \\ v &\longmapsto \sqrt{\langle v, v\rangle}. \end{aligned}$$

Remark 4.1.20. The norm depends on the choice of an inner product. Different inner products will give different norms.

Example 4.1.21. For the dot product on $\mathbf{F}^n$, the corresponding norm is the standard norm: $\|(z_1, \dots, z_n)\| = \sqrt{z_1^2 + \cdots + z_n^2}$.

Example 4.1.22. For $\mathcal{C}^0([-1,1])$ with inner product $\langle f, g\rangle = \int_{-1}^1 f(x)g(x)\,\mathrm{d}x$, the norm is $\|f\| = \sqrt{\int_{-1}^1 f^2(x)\,\mathrm{d}x}$.

The norm is not linear, but behaves well with respect to scalar multiplication.

Proposition 4.1.23. *Let V be an inner product space over $\mathbf{F}$ with norm $\|\cdot\|$. Then*

1. *For all $v \in V$, we have $\|v\| = 0$ if and only if $v = 0$;*
2. *For all $v \in V$ and all $\lambda \in \mathbf{F}$ we have $\|\lambda v\| = |\lambda| \cdot \|v\|$.*[1]

Proof. For the first statement, $\|v\| = 0 \iff \langle v, v\rangle = 0 \iff v = 0$.

For the second statement,

$$\begin{aligned} \|\lambda v\| &= \sqrt{\langle \lambda v, \lambda v\rangle} = \sqrt{\lambda\bar{\lambda}\langle v, v\rangle} \\ &= \sqrt{|\lambda|^2\langle v, v\rangle} = |\lambda| \cdot \|v\|. \end{aligned}$$

□

In practice, it is often easier to compute $\|v\|^2 = \langle v, v\rangle$. If you do that, don't forget to take the square-root after computing $\|v\|^2$.

4.2 Orthogonality and its consequences

Now that we have generalized the notion of length, we would like to generalize the notion of angle. In general, this is not possible, but one can at least generalize the notion of perpendicularity. Recall that in $\mathbf{R}^2$ if θ is the angle between two vectors u and v we have

$$u \bullet v = \|u\|\|v\| \cos(\theta). \tag{4.1}$$

We conclude that u and v are perpendicular if and only if $\langle u, v\rangle = 0$. One will use this notion to generalize perpendicularity. In fact, we will see later that Equation (4.1) can also be used to generalize the notion of angles for real inner product spaces.

4.2.1 Orthogonality

The following fundamental definition is inspired by Equation (4.1).

[1] $|\lambda| \in \mathbf{R}_{\geq 0}$ is the modulus (absolute value if $\mathbf{F} = \mathbf{R}$) of λ.

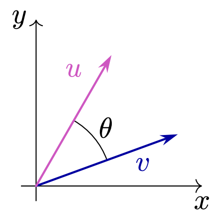

Figure 4.3: An angle between two vectors in $\mathbf{R}^2$. The angle satisfies the relation $u \bullet v = \|u\|\|v\|\cos(\theta)$.

Definition 4.2.1.

Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. Two vectors u and v in V are **orthogonal**, written $u \perp v$ if $\langle u, v \rangle = 0$.

The notion of orthogonality depends on the inner product we have chosen!

Observe that $u \perp v$ if and only if $v \perp u$. Orthogonality possesses other interesting properties, notably in relation to the 0 vector.

Proposition 4.2.2 (Orthogonality and 0_V). *Let V be an inner product space. Then*

1. *0_V is orthogonal to every vector $v \in V$;*
2. *A vector $v \in V$ is orthogonal to itself if and only if $v = 0_V$.*

Proof. 1 This follows from $\langle 0_V, v \rangle = 0_v$.
2 If v is orthogonal to itself, then $0_V = \langle v, v \rangle$ which implies $v = 0_V$. □

We now recover a classical theorem from Euclidean geometry that holds true in any inner product vector space over $\mathbf{R}$ or $\mathbf{C}$.

Theorem 4.2.3 (Pythagorean Theorem).

Suppose u and v are two orthogonal vectors in an inner product space. Then

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2.$$

Proof. We have

$$\begin{aligned} \|u + v\|^2 &= \langle u + v, u + v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle \\ &= \|u\|^2 + 0 + 0 + \|v\|^2 = \|u\|^2 + \|v\|^2. \end{aligned}$$

□

If $\mathbf{F} = \mathbf{R}$, then the converse also holds:

Proposition 4.2.4. *Let V be an inner product space over $\mathbf{R}$ and let u and v be two vectors. Then u and v are orthogonal if and only if $\|u + v\|^2 = \|u\|^2 + \|v\|^2$.*

Proof. By the proof of the previous proposition, we have

$$\begin{aligned}\|u+v\|^2 &= \|u\|^2 + \|v\|^2 + \langle u, v\rangle + \langle v, u\rangle \\ &= \|u\|^2 + \|v\|^2 + 2\langle u, v\rangle.\end{aligned}$$

So $\|u+v\|^2 = \|u\|^2 + \|v\|^2$ if and only if $\langle u, v\rangle = 0$, that is if and only if u and v are orthogonal. □

Observe that the proof of Proposition 4.2.4 does not work for inner product spaces over $\mathbf{C}$. Indeed, in this case we have $\|u+v\|^2 = \|u\|^2 + \|v\|^2 + \langle u, v\rangle + \overline{\langle u, v\rangle}$. But in general, $\langle u, v\rangle + \overline{\langle u, v\rangle} = 0$ does not imply that $\langle u, v\rangle = 0$. Let us see that in details with an easy example.

Example 4.2.5. Let $\mathbf{C}$ be the complex vector space of dimension 1, equipped with the standard dot product $\langle w, z\rangle = w \bullet z = w\bar{z}$. So the norm of a complex number is the same as its modulus: $\|z\| = |z|$ for all $z \in \mathbf{C}$. Let $u = 1$ and $v = i$. Then $\|1+i\|^2 = 2 = \|1\| + \|i\|$. But $\langle 1, i\rangle = 1\bar{i} = -i \neq 0$ so 1 and i are not orthogonal.

As demonstrated by the above example, Proposition 4.2.4 is not necessarily true for a complex vector space. But be careful: a same space V can be endowed with multiple inner products, and Proposition 4.2.4 might be wrong for some of them, but true for some other ones.

Example 4.2.6. Let $\mathbf{C}$ be the complex vector space of dimension 1, equipped with the non-standard inner product $\langle a+bi, c+di\rangle_{\mathbf{R}} := ac + bd$. One can check that $\langle \cdot, \cdot \rangle_{\mathbf{R}}$ is indeed a $\mathbf{C}$ inner product.[2] As in the previous example, the $\langle \cdot, \cdot \rangle_{\mathbf{R}}$-norm of a complex number is the same as its modulus: $\|z\| = |z|$ for all $z \in \mathbf{C}$. However, since $\langle \cdot, \cdot \rangle_{\mathbf{R}}$ has values in $\mathbf{R}$, it satisfies Proposition 4.2.4. In particular, $\langle 1, i\rangle = 1 \cdot 0 + 0 \cdot 1 = 0$. So 1 and i are $\langle \cdot, \cdot \rangle_{\mathbf{R}}$-orthogonal.

We know that in general, direct sum complements are not unique. However, in an inner product space any subspace U admits a unique *orthogonal complement.* Let us first show that when U is of dimension 1. We will show it in more generality later.

Proposition 4.2.7. *Let V be an inner product space and u be any non-zero vector. Then, for any $v \in V$ there exists a unique decomposition $v = \lambda u + w$ with $\lambda \in \mathbf{F}$ and such w orthogonal to u. In fact, we have*

$$v = \frac{\langle v, u\rangle}{\|u\|^2} u + w.$$

Proof. The idea is to first find the *orthogonal projection* of v onto $\operatorname{span}(u)$. This will both give λ and $w = v - \lambda u$. See Figure 4.4. Let $w := v - \lambda u$ for some $\lambda \in \mathbf{F}$ to be determined. We want w to be orthogonal to u, that is $\langle w, u\rangle = 0$. So let us compute:

$$0 = \langle w, u\rangle = \langle v - \lambda u, u\rangle = \langle v, u\rangle - \lambda\langle u, u\rangle.$$

[2] Actually, $\langle w, z\rangle_{\mathbf{R}} = \operatorname{Re}(w \bullet z)$ where $\bullet$ is the standard complex dot product. So $\langle \cdot, \cdot \rangle_{\mathbf{R}}$ has values in $\mathbf{R}$ and is simply the standard real product if we view $\mathbf{C} \cong \mathbf{R}^2$ as a real vector space.

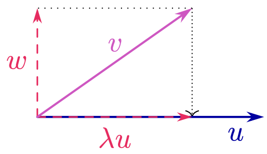

Figure 4.4: An orthogonal decomposition of v with respect to u.

So w and u are orthogonal if and only if $\lambda = \frac{\langle v,u\rangle}{\|u\|^2}$. This proves both the existence and the unicity. $\square$

Here is a rewriting of the previous proposition in terms of direct sum complements.

Proposition 4.2.8. *Let V be an inner product space and U be a one dimensional subspace. Then there exists a unique* orthogonal complement *W of U. That is: W is a direct sum complement of U ($V = U \oplus W$) and $\langle u, w\rangle = 0$ for all $u \in U$ and $w \in W$.*

Proof. Since U is one dimensional, we have $U = \operatorname{span}(u)$ for every non-zero vector $u \in U$. Let us fix such a vector i. Let $W := \{w \in V \mid \forall u \in U, w \perp u = 0\}$. We will see later that W is a subspace of V. By Proposition 4.2.7, any vector $v \in V$ admits a unique decomposition of the form $v = \lambda u + w$ for $\lambda \in \mathbf{F}$ and $w \in W$. Since $U = \operatorname{span}(u)$, this is equivalent to the uniqueness of the decomposition $v = u' + w$ for $u' \in U$ and $w \in W$. The unicity of this decomposition is exactly the fact that $V = U \oplus W$.

Finally, if X is another orthogonal complement, then $X \subset W$ by definition of W. If $X \neq W$, then any $w \in W \setminus X$ cannot have a decomposition $w = u + x \in U + X$. So $X = W$ and W is unique. $\square$

4.2.2 Cauchy-Schwartz inequality

Cauchy-Schwary inequalities exist for sums, series, integrals, … They are really useful both in algebra and in analysis. All these inequalities follow from a general inequality for inner product spaces, which we now prove.

Theorem 4.2.9 (Cauchy-Schwarz inequality)**.**

Let V be an inner product space. Then for all u and v in V we have

$$|\langle u, v\rangle| \leq \|u\| \cdot \|v\|. \tag{4.2}$$

Moreover, we have $|\langle u, v\rangle| = \|u\| \cdot \|v\|$ if and only ifu and v are linearly dependent.

Proof. If $u = 0_V$, then u and v are linearly dependent and Equation (4.2) is $0 = 0 \cdot \|v\|$.

Let us now suppose that $u \neq 0_V$ and consider the decomposition $v = \frac{\langle v,u\rangle}{\|u\|^2}u + w$ where w is orthogonal to u given by Proposition 4.2.7. By the Pythagorean Theorem

$$\|v\|^2 = \left\| \frac{\langle v,u\rangle}{\|u\|^2}u \right\|^2 + \|w\|^2 = \left| \frac{\langle v,u\rangle}{\|u\|^2} \right|^2 \|u\|^2 + \|w\|^2 \geq \frac{|\langle v,u\rangle^2|}{\|u\|^2}.$$

That is $|\langle u,v\rangle|^2 = |\langle v,u\rangle^2| \leq \|v\|^2\|u\|^2$.

Finally, we have equality in (4.2) if and only if $\|w\|^2 = 0$, if and only if $w = 0$. But w is zero if and only if u and v are linearly dependent. □

Some special cases of the Cauchy-Schwartz inequality may be familiar to you. Observe that in application, we usually use the squared version of the inequality: $|\langle u,v\rangle|^2 \leq \|u\|^2\|v\|^2$.

Example 4.2.10. Let $V = \mathbf{R}^n$ be the Euclidean space with standard dot product. The Cauchy-Schwartz inequality applied to $(x_1, \dots, x_n)$ and $(y_1, \dots, y_n)$ gives

$$\Big|\sum_{i=1}^n x_i y_i\Big|^2 \leq \Big(\sum_{i=1}^n x_i^2\Big)\Big(\sum_{i=1}^n y_i^2\Big).$$

Example 4.2.11. Let $V = \mathcal{C}^0([-1,1])$ be the space of continuous functions from $[-1,1]$ to $\mathbf{R}$ with inner product $\langle f,g\rangle = \int_{-1}^1 f(x)g(x)\,\mathrm{d}x$. Then for f and g in $V = \mathcal{C}^0([-1,1])$ the Cauchy-Schwartz inequality is

$$\Big(\int_{-1}^1 f(x)g(x)\,\mathrm{d}x\Big)^2 \leq \Big(\int_{-1}^1 f(x)^2\,\mathrm{d}x\Big)\Big(\int_{-1}^1 g(x)^2\,\mathrm{d}x\Big).$$

It is an interesting exercise to try to give an analysis proof of this inequality.

4.2.3 Angles in inner product spaces over $\mathbf{R}$

When $\mathbf{F} = \mathbf{R}$ we can define angles using the Cauchy-Schwartz inequality and Equation (4.1). Let V be an inner product space over $\mathbf{R}$. By the Cauchy-Schwartz inequality, for non-zero u and v in V we have

$$-1 \leq \frac{\langle u,v\rangle}{\|u\|\|v\|} \leq 1.$$

Definition 4.2.12.

Let V be an inner product space over $\mathbf{R}$ and let u and v be two non-zero vectors. We define the **angle between** u **and** v by

$$\angle(u,v) := \arccos \frac{\langle u,v\rangle}{\|u\|\|v\|} \in [0,\pi].$$

If one of u or v is 0_V, we say that the angle between u and v is arbitrary.

Using this definition of angles, we have that u and v are orthogonal if and only if $\langle u, v\rangle = 0$, if and only if $\angle(u, v) = \frac{\pi}{2}$.

Remark 4.2.13. If $\mathbf{F} = \mathbf{C}$, the above definition does not make sense! Indeed, in this case $\langle u, v\rangle$ can be a non-real complex number.

4.2.4 Classical geometry results in inner product spaces

Using the notion of norm and orthogonality, it is possible to recover some of the classical geometry results in a more general context.

Theorem 4.2.14 (Triangle inequality)**.**

Let V be an inner product space. Then for any u and v in V we have

$$\big|\|u\| - \|v\|\big| \leq \|u + v\| \leq \|u\| + \|v\|.$$

Moreover, we have

- $\|u + v\| = \|u\| + \|v\|$ *if and only if there exists* $\lambda \in \mathbf{R}_{\geq 0}$ *such that* $u = \lambda v$ *or* $v = \lambda u$*;*
- $|\|u\| - \|v\|| \leq \|u + v\|$ *if and only if there exists* $\mu \in \mathbf{R}_{\leq 0}$ *such that* $u = \mu v$ *or* $v = \mu u$*.*

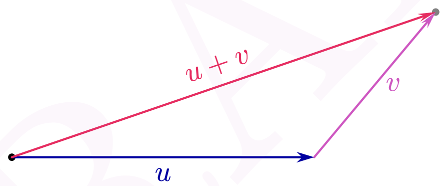

Figure 4.5: The shortest path between the black dot and the graydot is given by the vector $u + v$.

Proof. We have

$$\begin{aligned}\|u + v\|^2 &= \langle u + v, u + v\rangle \\ &= \langle u, u\rangle + \langle v, v\rangle + \langle u, v\rangle + \overline{\langle u, v\rangle} \\ &= \|u\|^2 + \|v\|^2 + 2\operatorname{Re}\langle u, v\rangle.\end{aligned}$$

Since $-|\langle u, v\rangle| \leq \operatorname{Re}\langle u, v\rangle \leq |\langle u, v\rangle|$ and $|\langle u, v\rangle| \leq \|u\| \cdot \|v\|$ by the Cauchy-Schwarz inequality, we have

$$\big|\|u\| - \|v\|\big| \leq \|u + v\| \leq \|u\| + \|v\|.$$

Now, the right inequality is an equality if and only if $\mathrm{Re}\langle u, v\rangle = |\langle u, v\rangle| = \|u\|\|v\|$, if and only if $\langle u, v\rangle = \|u\|\|v\|$. Indeed, $\mathrm{Re}\langle u, v\rangle = |\langle u, v\rangle|$ is equivalent to $|\langle u, v\rangle| = \langle u, v\rangle$. Finally, $\langle u, v\rangle = \|u\|\|v\|$ if and only if there exists $\lambda \in \mathbf{R}_{\geq 0}$ such that $u = \lambda v$ or $v = \lambda u$.

The proof for the left equality is similar. □

Figure 4.5 explains the name *triangle inequality*. This inequality is really important as it shows that $\|u - v\|$ is a **distance**. Meaning that:

1. $\|u - v\| \in \mathbf{R}_{\geq 0}$ (the distance is always a positive number);
2. $\|u - u\| = 0$ (if you don't move, the distance is 0);
3. If $u \neq v$, then $\|u - v\| > 0$ (if you move, the distance is positive);
4. $\|u - v\| = \|v - u\|$ (the distance from u to v is the same as the distance from v to u);
5. For all $w \in V$, $\|u - v\| \leq \|u - w\| + \|w - u\|$ (if on the way between u to v we path at w, this cannot shorten the distance).

We now recover another result from classical geometry.

Theorem 4.2.15 (Parallelogram equality)**.**

Let V be an inner product space. Then for any u and v in V we have

$$\|u + v\|^2 + \|u - v\|^2 = 2(\|u\|^2 + \|v\|^2).$$

Proof. The proof is just direct computations. Figure 4.6 is here to explain the name of the theorem.

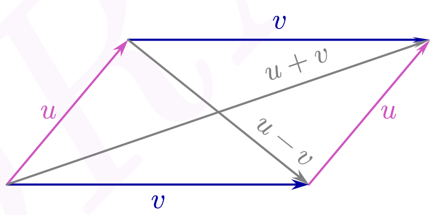

Figure 4.6: A parallelogram and its two diagonals.

$$\begin{aligned}
\|u + v\|^2 + \|u - v\|^2 &= \langle u + v, u + v\rangle + \langle u - v, u - v\rangle \\
&= \langle u, u\rangle + \langle u, v\rangle + \langle v, u\rangle + \langle v, v\rangle \\
&\quad + \langle u, u\rangle - \langle u, v\rangle - \langle v, u\rangle + \langle v, v\rangle \\
&= 2\langle u, u\rangle + 2\langle v, v\rangle \\
&= 2(\|u\|^2 + \|v\|^2).
\end{aligned}$$

□

Corollary 4.2.16 (Appolonius's identity)**.** *Let ABC be a triangle in $\mathbf{F}^n$, and let D be the midpoint of AB. Then*

$$\|BC\|^2 + \|AC\|^2 = \frac{1}{2}\|AB\|^2 + 2\|CD\|^2.$$

Proof. Let $u := \overrightarrow{DA}$ and $v := \overrightarrow{DC}$. Then $\overrightarrow{AC} = v - u$, $\overrightarrow{DB} = -u$ and $\overrightarrow{BC} = v + u$. See Figure 4.7. By the parallelogram equality

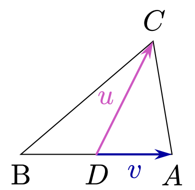

Figure 4.7: A triangle ABC, where D is the midpoint of AB.

$$\|\overrightarrow{BC}\|^2 + \|\overrightarrow{AC}\|^2 = 2\left(\left\|\frac{1}{2}\overrightarrow{AB}\right\|^2 + \|\overrightarrow{CD}\|^2\right) = \frac{1}{2}\|\overrightarrow{AB}\|^2 + 2\|\overrightarrow{CD}\|^2.$$

□

4.3 Orthonormal bases

In this section we will use inner product and orthogonality to define nice bases with specifically good properties.

4.3.1 Orthonormality

We start with a definition.

Definition 4.3.1.

Let V be an inner product space over $\mathbf{F}$. A list of vectors $(v_1, \dots, v_n)$ is **orthogonal** if its elements are pairwise orthogonal. That is, if for all $i, j \in \{1, \dots, n\}$ with $i \neq j$ the vectors v_i and v_j orthogonal.
The list $(v_1, \dots, v_n)$ is **orthonormal** if it is orthogonal and all the vectors have norm 1.

Infinite dimensional spaces

It is possible to define orthogonality and orthonormality in a similar way for an infinite family $(v_\alpha)_{\alpha \in I}$.

Observe that a list of vectors $(v_1, \dots, v_n)$ is orthonormal if and only if[3]

$$\langle v_i, v_j \rangle = \delta_{i,j} := \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

Observe that the notion of orthogonal/orthonormal list depends on the choice of an inner product!

We already have encountered examples of orthonormal lists.

Example 4.3.2. The standard basis of $\mathbf{F}^n$ is orthonormal for the dot product.

But other examples exist as well.

Example 4.3.3. The list $\left(u = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}, v = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} \right)$ in $\mathbf{F}^3$ is orthonormal for the dot product.

Indeed, we have

$$\begin{aligned} u \bullet u &= \tfrac{1}{3} + \tfrac{1}{3} + \tfrac{1}{3} = 1, \\ v \bullet v &= \tfrac{1}{2} + \tfrac{1}{2} + 0 = 0, \\ u \bullet v &= \tfrac{-1}{\sqrt{6}} + \tfrac{1}{\sqrt{6}} + 0 = 0. \end{aligned}$$

Proposition 4.3.4. *Let V be an inner product space and let $(v_1, \dots, v_n)$ be an orthonormal list of vectors. Then for every $\lambda_1, \dots, \lambda_n \in \mathbf{F}$ we have*

$$\|\lambda_1 v_1 + \dots + \lambda_n v_n\|^2 = |\lambda_1|^2 + \dots + |\lambda_1|^2.$$

Proof. This is a direct application of the Pythagorean Theorem:

$$\begin{aligned} \|\lambda_1 v_1 + \dots + \lambda_n v_n\|^2 &= \|\lambda_1 v_1\|^2 + \dots + \|\lambda_n v_n\|^2 \\ &= |\lambda_1|^2 \|v_1\|^2 + \dots + |\lambda_n|^2 \|v_n\|^2 \\ &= |\lambda_1|^2 + \dots + |\lambda_n|^2. \end{aligned}$$

□

Corollary 4.3.5. *An orthonormal list $(v_1, \dots, v_n)$ is linearly independent.*

Proof. Let $(v_1, \dots, v_n)$ be an orthonormal list. Then , if $\lambda_1 v_1 + \dots + \lambda_n v_n = 0$, the previous proposition tells us that $0 = |\lambda_1|^2 + \dots + |\lambda_n|^2$ and therefore all the λ_i are 0. □

Infinite dimensional spaces

Corollary 4.3.5 is true even for infinite lists. In fact, we have that any *orthogonal* list (finite or infinite) of non-zero vectors is linearly independent! Can you prove it for finite lists?

It is natural to ask which orthonormal lists are also spanning. Since an orthonormal lists is linearly independent, it is spanning if and only if it is a basis.

[3] $\delta_{i,j}$ is called the *Kronecker delta*.

Definition 4.3.6.

Let V be an inner product space over $\mathbf{F}$. A list of vectors $(e_1, \dots, e_n)$ is an **orthonormal basis** if it is both an orthonormal list and a basis.

Infinite dimensional spaces

Similarly, one say that an infinite family $(e_\alpha)_{\alpha\in I}$ is an orthonormal family if it is both an orthonormal list and a basis.

It follows from Example 4.3.2 that the standard basis of $\mathbf{F}^n$ is an orthonormal basis, but it is not the only one.

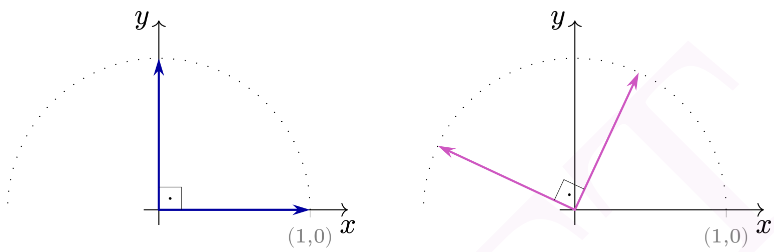

Figure 4.8: Two orthonormal bases of $\mathbf{R}^2$.

Proposition 4.3.7. *If* $\dim(V) = n$*, any orthonormal list with of length* n *is an orthonormal basis.*

Proof. Any orthonormal list is linearly independent. But any linearly independent list of length $\dim(V)$ is a basis by Theorem 2.0.4. □

Example 4.3.8. $u_1 = [1/2, 1/2, 1/2, 1/2]^\intercal$, $u_2 = [1/2, 1/2, -1/2, -1/2]^\intercal$, $u_3 = [1/2, -1/2, -1/2, 1/2]^\intercal$, $u_4 = [-1/2, 1/2, -1/2, 1/2]^\intercal$ is an orthonormal basis of $\mathbf{F}^4$.

Since $\dim(\mathbf{F}^4) = 4$, it is enough to check that the family is orthonormal. It is easy to show that $\|u_i\| = 4 \cdot 1/4 = 1$ for $1 \le i \le 4$. The fact that $u_i \bullet u_j = 0$ if $i \neq j$ is left as an exercice.

The main advantage of orthonormal basis is the following result.

Proposition 4.3.9 (Orthonormal decomposition)**.** *Let* V *be a finite dimensional inner product space and let* $(e_1, \dots, e_n)$ *be an orthonormal basis. Then for any vector* $v \in V$ *we have*

1. $v = \langle v, e_1\rangle e_1 + \cdots + \langle v, e_n\rangle e_n$;
2. $\|v\|^2 = |\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_1\rangle|^2$;
3. $\langle u, v\rangle = \langle u, e_1\rangle\overline{\langle v, e_1\rangle} + \cdots + \langle u, e_n\rangle\overline{\langle v, e_n\rangle}$ *for every* $u \in V$.

Proof. 1. Since $(e_1, \dots, e_n)$ is a basis, there exists $\lambda_i \in \mathbf{F}$ such that $v = \lambda_1 e_1 + \dots + \lambda_n e_n$. Then for $1 \leq i \leq n$ we have

$$\begin{aligned}\langle v, e_i \rangle &= \langle \lambda_1 e_1 + \dots + \lambda_n e_n, e_i \rangle \\ &= \lambda_1 \langle e_1, e_i \rangle + \dots + \lambda_n \langle e_n, e_i \rangle \\ &= \lambda_i 1 = \lambda_i.\end{aligned}$$

2. The formula for $\|v\|^2$ follows by using the Pythagorean Theorem on 1 and the fact that $\|\langle v, e_i\rangle e_i\| = |\langle v, e_i \rangle| \|e_i\| = |\langle v, e_i \rangle|$.
3. This follows from taking the inner product $\langle u, \cdot \rangle$ on both sides of 1 and using the conjugate symmetry $\langle e_i, v \rangle = \overline{\langle v, e_i \rangle}$. $\square$

Observe that $\langle v, e_1 \rangle e_1$ is nothing else than the projection of v onto $\operatorname{span} e_1$ along direction $\operatorname{span}(e_1)^\perp := \{w \in V \mid \langle w, e_1 \rangle = 0\}$. We will make this explicit in the next subsection.

Infinite dimensional spaces

Proposition 4.3.9 is also true for infinite dimensional spaces. The trick is that if $(e_\alpha)_{\alpha \in I}$ is an infinite orthonormal basis, then for a given v the inner product $\langle v, e_\alpha \rangle$ is non-zero only for finitely many α.

Proposition 4.3.9.1 is particularly useful for the matrix representation of a vector.

Corollary 4.3.10. *Let V be a finite dimensional inner product space and let $\mathcal{B} = (e_1, \dots, e_n)$ be an orthonormal basis. Then for any vector $v \in V$ we have*

$$[v]_{\mathcal{B}} = \begin{bmatrix} \langle v, e_1 \rangle \\ \vdots \\ \langle v, e_n \rangle \end{bmatrix}.$$

Let us end this subsection with the following announcement.

Theorem 4.3.11.

Let V be a finite dimensional inner product space. Then V admits an orthonormal basis.

We will prove this result later in Section 4.3.3, but we still state it as it is a good motivation for is coming next.

Infinite dimensional spaces

In Theorem 4.3.11, we do not ask V to be finite dimensional, only U. However, the theorem is not true for a general U. That is, infinite dimensional subspaces might not have orthonormal basis. An example of such subspace if given in Example 4.3.17.

4.3.2 Orthogonal projections

In Proposition 4.2.8 we saw that if U is a 1 dimensional subspace of V, then it has a unique *orthogonal complement*. Our aim is now to formally define orthogonal complements and generalize Proposition 4.2.8 to subspaces of dimension bigger than 1.

Definition 4.3.12.

Let V be an inner product space. For a subset $U \subset V$ we define its **orthogonal complement** to be

$$U^{\perp} := \{v \in V \mid \forall u \in U, \langle u, v\rangle = 0\}.$$

That is, $U^{\perp}$ is the set of vectors that are orthogonal to all vectors of U.

Example 4.3.13. Let $V = \mathbf{R}^2$ and $U_1 = \{\left[\begin{smallmatrix}1\\2\end{smallmatrix}\right]\}$. Then $U_1^{\perp} = \{\left[\begin{smallmatrix}x\\y\end{smallmatrix}\right] \in \mathbf{R}^2 \mid x + 2y = 0\}$; see Figure 4.9.

For $U_2 = \{\left[\begin{smallmatrix}t\\t\end{smallmatrix}\right] \mid t \in \mathbf{R}\}$ we have $U_2^{\perp} = \{\left[\begin{smallmatrix}x\\y\end{smallmatrix}\right] \in \mathbf{R}^2 \mid \forall t \in \mathbf{R}, tx + ty = 0\} = \{\left[\begin{smallmatrix}x\\-x\end{smallmatrix}\right] \in \mathbf{R}^2 \mid x \in \mathbf{R}\}$.

For $U_3 = \{\left[\begin{smallmatrix}t\\1\end{smallmatrix}\right] \in \mathbf{R}^2 \mid t \in \mathbf{R}\}$ we have $U_3 = \{\left[\begin{smallmatrix}x\\y\end{smallmatrix}\right] \in \mathbf{R}^2 \mid \forall t \in \mathbf{R}, tx + y = 0\} = \{\left[\begin{smallmatrix}0\\0\end{smallmatrix}\right]\}$.

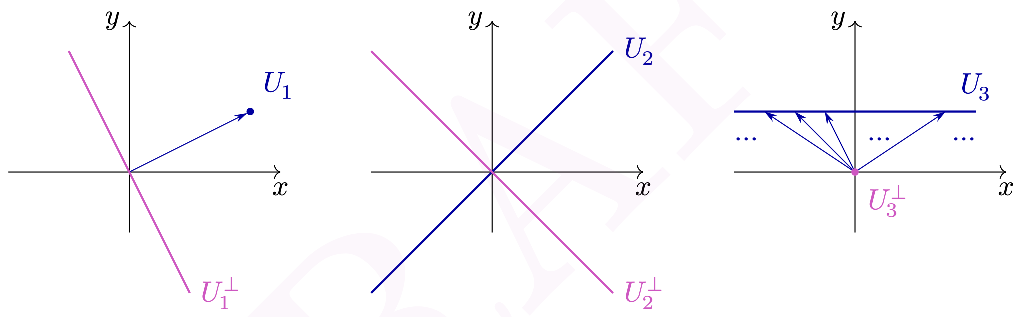

Figure 4.9: Orthogonal complements in $\mathbf{R}^2$.

One of the advantage of orthogonal complements is that they automatically are subspaces. In fact, we have

Proposition 4.3.14. *Let V be an inner product space. Then the following holds.*

1. *If U is a subset of V, then $U^{\perp}$ is a subspace of V;*
2. *$\{0\}^{\perp} = V$ and $V^{\perp} = \{0\}$;*
3. *If U is a subset of V, then $U \cap U^{\perp} \subseteq \{0\}$ (with equality if and only if U contains 0);*
4. *If $U \subseteq W$ are two subsets of V, then $W^{\perp} \subset U^{\perp}$.*

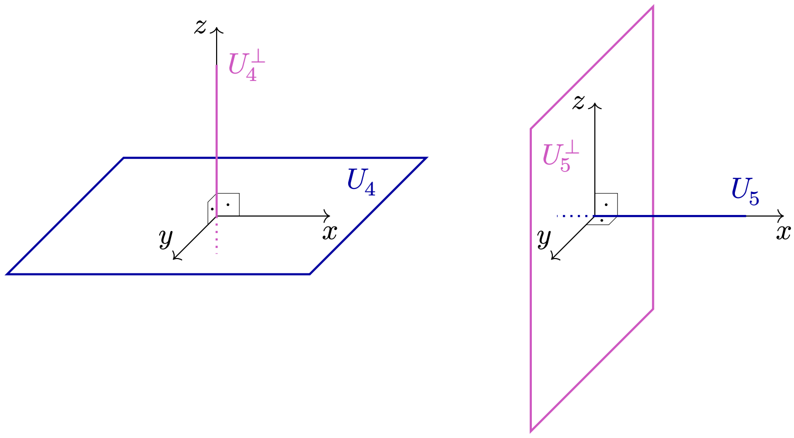

Figure 4.10: Orthogonal complements in $\mathbf{R}^3$.

Proof. 1. Firstly, since 0 is orthogonal to every vector in V, it belongs to $U^\perp$. Then, if both v_1 and v_2 belong to $U^\perp$, for all $u \in U$ we have $\langle v_1 + v_2, u\rangle = \langle v_1, u\rangle + \langle v_2, u\rangle = 0 + 0 = 0$ and therefore $v_1 + v_2$ belongs to $U^\perp$. Similarly, for $v \in U^\perp$ and $\lambda \in \mathbf{F}$ we have $\langle \lambda v, u\rangle = \lambda\langle v, u\rangle = \lambda 0 = 0$ for every $u \in U$ and so λv belongs to $U^\perp$.

2. On one hand, the vector 0 is orthogonal to every other vector, which implies $\{0\}^\perp = V$. On the other hand, if v is in $V^\perp$ then it is orthogonal to itself and therefore equal to 0. This shows that $V^\perp \subset \{0\}$ and we have equality since $V^\perp$ is a subspace.

3. Suppose that $U \cap U^\perp$ is not empty and let v be any element inside the intersection. Then, as above, we have $\langle v, v\rangle = 0$ and so $v = 0$. This shows that $U \cap U^\perp \subseteq \{0\}$. We have equality if and only if both U and $U^\perp$ contains 0, but $U^\perp$ is a subspace and hence always contains 0.

4. Let v be in $W^\perp$. Then $\langle v, w\rangle = 0$ for every $w \in W$ and therefore in particular for every $w \in U$. So v is in $U^\perp$ and $W^\perp \subset U^\perp$. □

If U is a subspace of V, in order to describe $U^\perp$, it is enough to check orthogonality with respect to a spanning family.

Lemma 4.3.15. *Let V be an inner product space and let U be a finite dimensional subspace with a spanning family $(u_1, \dots, u_n)$. We have*

$$U^\perp = \{v \in V \mid \forall j \in \{1, \dots, n\}, \langle v, u_j\rangle = 0\}.$$

Proof. The left-to-right inclusion is obvious, so we only need to check the right to left inclusion. Let V be vector such that $\langle v, u\rangle_j = 0$ for all $j \in \{1, \dots, n\}$. For every $u \in U$, we have $u = \lambda_1 u_1 + \dots + \lambda_n u_n$ for some $\lambda_j \in \mathbf{F}$ (not necessarily unique). By linearity, $\langle v, u\rangle = \lambda_1\langle v, u_1\rangle + \dots + \lambda_n\langle v, u_n\rangle = 0$ and therefore v is in $U^\perp$. □

Infinite dimensional spaces

The lemma is also true (with essentially the same proof) for an infinite dimensional subspace U.

Another nice property of orthogonal complements is that they can be use for direct sum decompositions. In order to prove the forthcoming theorem, we will assume that Theorem 4.3.11 is true, even if we postpone its proof to the next subsection. Actually, our proof of Theorem 4.3.16, as well as of its consequences, does not need Theorem 4.3.11 if we replace the hypothesis *U finite dimensional* by *U admits an orthonormal basis.*

Theorem 4.3.16.

Let V be an inner product space and let U be a finite dimensional subspace with orthonormal basis $(u_1, \dots, u_n)$. Finally, let $(e_1, \dots, e_n)$ be the standard basis of $\mathbf{F}^n$. Consider the linear maps $T\colon V \to \mathbf{F}^n$ and $S\colon \mathbf{F}^n \to V$ given by

$$T(v) := \begin{bmatrix} \langle v, e_1 \rangle \\ \vdots \\ \langle v, e_n \rangle \end{bmatrix},$$
$$S(e_i) := u_i,$$

where S is extended linearly to $\mathbf{F}^n$. Then we have $TS = \mathrm{Id}_{\mathbf{F}^n}$, $V = U \oplus U^\perp$, and $ST \in \mathcal{L}(V)$ is the projection onto U along the direction $U^\perp$.

Proof.

$$\begin{aligned} TS(e_i) = T(u_i) &= [\langle u_i, u_1 \rangle, \dots, \langle u_i, u_n \rangle]^\mathsf{T} \\ &= [0, \dots, 0, 1, 0, \dots, 0]^\mathsf{T} = e_i. \end{aligned}$$

Therefore, $TS(e_i) = \mathrm{Id}_{\mathbf{F}^n}(e_i)$ for all basis vectors e_i, and since both TS and $\mathrm{Id}_{\mathbf{F}^n}$ are linear we have $TS = \mathrm{Id}_{\mathbf{F}^n}$.

By Proposition 3.2.11, the map ST is a projection onto $\mathrm{range}(S)$ along the direction $\mathrm{null}(T)$ and $V = \mathrm{range}(S) \oplus \mathrm{null}(T)$.

Since $S(e_i)$ belongs to U for basis vectors, we have $\mathrm{range}(S) \subseteq U$. Moreover, for any $u \in U$, there exists a unique decomposition $u = \lambda_1 u_1 + \dots + \lambda_n u_n$ and so $u = S(\lambda_1 e_1 + \dots + \lambda_n e_n)$. This shows that $\mathrm{range}(S) = U$. Finally, a vector $v \in V$ is in $\mathrm{null}(T)$ if and only if $\langle v, e_i \rangle = 0$ for all i, if and only if v is in $U^\perp$. So $\mathrm{null}(T) = U^\perp$. □

Infinite dimensional spaces

For the Theorem to be true, we do not need V to be finite dimensional, only U. In fact, we do not even need U to be finite dimensional, the existence of an orthonormal basis is enough to conclude, with a similar proof. However, an arbitrary subspace U of V does not necessarily admit an orthonormal basis and so the Theorem might fails for such subspaces. This is demonstrated in the next example.

Example 4.3.17. Let $V = \mathcal{C}^0([-1,1])$ be the space of continuous functions from $[-1,1]$ to $\mathbf{R}$ equipped with the inner product $\langle f, g\rangle = \int_{-1}^{1} f(x)g(x)\,\mathrm{d}x$. Let $U := \{f \in V \mid f(0) = 0\}$. This is an infinite dimensional proper subspace of V (the constant function $f = 1$ is in $V \setminus U$). Using analysis, one can show that $U^\perp = \{0\}$ (exercise). Therefore, we have $U + U^\perp = U + \{0\} = U \subsetneq V$. As a consequence, we conclude that U does not admit an orthonormal basis (and not even an orthogonal basis).

As a direct corollary of Theorem 4.3.16, we can compute the dimension of $U^\perp$.

Corollary 4.3.18. *Let V be a finite dimensional inner product space and let U be a subspace. Then $\dim(U^\perp) = \dim(V) - \dim(U)$.*

Another important consequence of Theorem 4.3.16 is that taking orthogonal complement twice is the same as doing nothing.

Corollary 4.3.19. *Let V be an inner product space and let U be a finite dimensional subspace. Then $(U^\perp)^\perp = U$.*

Proof. "$\subseteq$" Let $u \in U$. Then for any $v \in U^\perp$, $\langle u, v\rangle = 0$ which show that u is in $(U^\perp)^\perp$.

"$\supseteq$" Let $v \in (U^\perp)^\perp \subseteq V = U \oplus U^\perp$. So there exists a unique decomposition $v = u + w$ with $u \in U$ and $w \in U^\perp$. We want to show that v is in U, or equivalently that $w = 0$. We have $u \in U \subseteq (U^\perp)^\perp$. So $w = v - u$ belongs both to $(U^\perp)^\perp$ and to $U^\perp$. But then $w \in (U^\perp)^\perp \cap U^\perp = \{0\}$ which is what we wanted to prove. □

Using Corollary 4.3.19 and Proposition 4.3.14 we can prove the following result.

Lemma 4.3.20. *Let V be an inner product space and let U be a finite dimensional subspace of V. Then $U^\perp = \{0\} \iff U = V$.*

Proof. If $U^\perp = \{0\}$, then $U = (U^\perp)^\perp = \{0\}^\perp = V$. If $U = V$, then $U^\perp = V^\perp = \{0\}$. □

We can use Theorem 4.3.16 to define specific projections.

Definition 4.3.21.

Let V be an inner product space and let U be a finite dimensional space. So $V = U \oplus U^\perp$. The **orthogonal projection** of V onto U is the projection P_U onto U along direction $U^\perp$. That is, this is the map

$$\begin{aligned} P_U \colon V = U \oplus U^\perp &\to U \\ v = u + w &\longmapsto u. \end{aligned}$$

Example 4.3.22. Let V be an inner product space. For u a non-zero vector of V, define $U := \operatorname{span} u$. Then for any $v \in V$, by Proposition 4.2.7 we have $v = \frac{\langle v,u\rangle}{\langle u,u\rangle}u + w$ with $w \in U^\perp$. So $P_U(v) = \frac{\langle v,u\rangle}{\langle u,u\rangle}u$.

Example 4.3.23. Let V be an inner product space and let U be a finite dimensional subspace. Then the map ST from Theorem 4.3.16 is an orthogonal projection. Observe that while we used an orthonormal basis $\mathcal{B}$ of U to construct T and S, the map ST does not depend on $\mathcal{B}$, but only on U! Indeed, ST only depends on $\text{range}(S) = U$ and $\text{null}(T) = U^\perp$, and none of this subspaces depend on the choice of an orthonormal basis for U.

Orthogonal projections are projections. They hence satisfies all properties of projections of Proposition 3.2.4. They also satisfy two additional properties.

Proposition 4.3.24. *Let V be an inner product space and let U be a finite dimensional subspace. Then the orthogonal projection onto U satisfies*

8. $\|P_U v\| \leq \|v\|$ *for every* $v \in V$;

9. *For every orthonormal basis* $(e_1, \dots, e_n)$ *of* U *we have* $P_U v = \langle v, e_1\rangle e_1 + \dots + \langle v, e_n\rangle e_n$.

Proof. 8 Let $v = u + w$ with $u \in U$ and $w \in U^\perp$. By the Pythagorean Theorem, $\|P_U v\|^2 = \|u\|^2 \leq \|u\|^2 + \|w\|^2 = \|v\|^2$. See Figure 4.11.

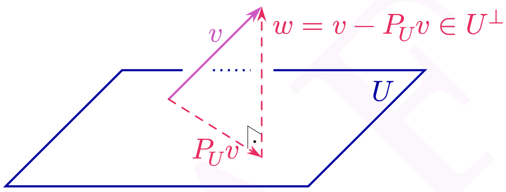

Figure 4.11: An orthogonal projection of v onto U and the decomposition $v = P_U v + w$ with $w \in U^\perp$.

9 This is (the proof of) Theorem 4.3.16. □

Exercise 4.3.25. Let V be an inner product space and let $P \in \mathcal{L}(V)$ be such that $P^2 = P$. Suppose that for all $u \in \text{range}(P)$ and $w \in \text{null}(P)$ we have $\langle u, w\rangle = 0$. Prove that there exists a subspace U of V such that P is the orthogonal projection onto U.

Solution. By Proposition 3.2.9, P is the projection onto $U = \text{range}(P)$ along direction $W = \text{null}(P)$. So we only need to prove that $W = U^\perp$. By hypothesis, $W \subset U^\perp$. Now, let v be an element of $U^\perp$. We have the decomposition $v = u + w$ with $u \in U$ and $w \in W$. Since both v and w are in $U^\perp$ we have $0 = \langle v, u\rangle = \langle u, u\rangle + \langle w, u\rangle = \langle u, u\rangle$. □

Infinite dimensional spaces

Most of the results of this subsection are corollaries of Theorem 4.3.16. That means that they remain true for any infinite dimensional subspace U satisfying $V = U \oplus U^\perp$. For such subspaces U, Corollary 4.3.19 and Lemma 4.3.20 are still true. However, Corollary 4.3.19 and Lemma 4.3.20 fail for general U, as demonstrated by Example 4.3.17.

It is also possible to define orthogonal projections onto U (in the sense of Definition 4.3.21) as soon as $V = U \oplus U^{\perp}$. Proposition 4.3.24 remains true in this context.

4.3.3 The Gram-Schmidt procedure

In the last subsection, we defined orthogonal complements. We then assumed Theorem 4.3.11 (every finite dimensional vector space admits an orthonormal basis) to prove many results about orthogonal complements and orthogonal projections. We now finally prove this theorem.

First of all, observe that if $\dim(V) = 0$, then the empty set is an orthonormal basis! If $\dim(V) = 1$, for any non-zero vector $v \in V$ the vector $\frac{v}{\|v\|}$) is an orthonormal basis of V.

If $\dim V = 2$, let $v_1 \in V$ be any non-zero vector and define $U := \operatorname{span}(v_1)$, so $\frac{v_1}{\|v_1\|}$ is an orthonormal basis of U. Using Proposition 4.2.8, $V = U \oplus U^{\perp}$. Moreover, the same proposition tells us that the orthogonal projection P_U onto U exists. Now, let $v_2 \in V \setminus U$ and consider

$$u_2 := v_2 - P_U v_2 = \frac{\langle v_2, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1.$$

By Proposition 4.2.7, u_2 is orthogonal to v_1. Moreover, $u_2 \neq 0$ as this would implies $v_2 \in U$. Hence, $(\frac{v_1}{\|v_1\|}, \frac{u_2}{\|u_2\|})$ is an orthonormal basis for V.

Let us have a closer look at the construction of our orthonormal basis for an inner product space of dimension 2. We can start with any basis (v_1, v_2). Then we obtain and orthogonal basis (u_1, u_2) where $u_1 = v_1$ and $u_2 = v_2 - P_U v_2$ is v_2 minus its "v_1-component". Finally, we obtain an orthonormal basis (e_1, e_2) by dividing each of the vector u_i by its norm.

The above procedure can be extended to any finite dimensional inner product space.

Add a picture of the procedure?

Theorem 4.3.26 (Gram-Schmidt procedure).

Let V be an inner product space and let $(v_1, \ldots, v_n)$ be a linearly independent list. Define inductively

$$\begin{aligned}
u_1 &:= v_1, \\
u_2 &:= v_2 - \frac{\langle v_2, u_1 \rangle}{\|u_1\|^2} u_1, \\
u_3 &:= v_3 - \frac{\langle v_3, u_1 \rangle}{\|u_1\|^2} u_1 - \frac{\langle v_3, u_2 \rangle}{\|u_2\|^2} u_2, \\
&\vdots \\
u_n &:= v_n - \frac{\langle v_n, u_1 \rangle}{\|u_1\|^2} u_1 - \cdots - \frac{\langle v_n, u_{n-1} \rangle}{\|u_{n-1}\|^2} u_{n-1}.
\end{aligned}$$

Finally, for $j \in \{1, \ldots, n\}$ define $e_k := \frac{u_k}{\|u_k\|}$.

Then $(u_1, \dots, u_n)$ is an orthogonal list and $(e_1, \dots, e_n)$ is an orthonormal list. Moreover, for $j \in \{1, \dots, n\}$ we have

$$\operatorname{span}(e_1, \dots, e_j) = \operatorname{span}(u_1, \dots, u_j) = \operatorname{span}(v_1, \dots, v_j).$$

Proof. The proof is done by induction on n. If $n = 1$, then $u_1 = v_1$ is orthogonal and $\operatorname{span}(u_1) = \operatorname{span}(v_1)$. Moreover, by linear independence, $v_1 \neq 0$, so $\|u_1\| \neq 0$ which implies that e_1 is well-defined and is a non-zero multiple of u_1. This directly gives $\operatorname{span}(e_1) = \operatorname{span}(u_1)$. Finally, $\|e_1\| = \left\| \frac{u_1}{\|u_1\|} \right\| = 1$.

Suppose that the Theorem holds for $n-1 \geq 1$, so $(e_1, \dots, e_{n-1})$ is an orthonormal basis for $U := \operatorname{span}(e_1, \dots, e_{n-1}) = \operatorname{span}(u_1, \dots, u_{n-1}) = \operatorname{span}(v_1, \dots, v_{n-1})$. Since U has an orthonormal basis, we can apply Example 4.3.17 to it to obtain $V = U \oplus U^\perp$. Moreover, v_n does not belong to U as the list $(v_1, \dots, v_n)$ is linearly independent. Considering $P_U \in \mathcal{L}(V)$ the orthogonal projection onto U, by Proposition 4.3.24.9 we have

$$\begin{aligned} P_U(v_n) &= \langle v_n, e_1 \rangle e_1 + \dots + \langle v_n, e_{n-1} \rangle e_{n-1} \\ &= \frac{\langle v_n, u_1 \rangle}{\|u_1\|^2} u_1 - \dots - \frac{\langle v_n, u_{n-1} \rangle}{\|u_{n-1}\|^2} u_{n-1}. \end{aligned}$$

Hence $u_n = v_n - P_U(v_n)$ belongs to $U^\perp$ and $(u_1, \dots, u_n)$ is an orthogonal list of non-zero vectors. Since it has length $n = \dim(V)$, this implies that it is an orthogonal basis and that $(e_1, \dots, e_n)$ is an orthonormal basis of V.

We have proved that $\operatorname{span}(e_1, \dots, e_j) = \operatorname{span}(u_1, \dots, u_j) = \operatorname{span}(v_1, \dots, v_j)$ whenever $j = n$ (in which case they are equal to V). For $j \leq n-1$ this follows from the induction hypothesis. ☐

Exercise 4.3.27. Find an orthonormal basis for $\mathcal{P}_2(\mathbf{R})$, where the inner product is given by $\langle p, q \rangle := \int_{-1}^1 p(x)q(x)\,\mathrm{d}x$.

Solution. Start with the standard basis $(v_1 = 1, v_2 = x, v_3 = x^2)$ and apply Gram-Schmidt to it. We have

$$\begin{aligned} u_1 &= v_1 = 1, \\ u_2 &= v_2 - \frac{\langle v_2, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1, \\ u_3 &= v_3 - \frac{\langle v_3, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1 - \frac{\langle v_3, u_2 \rangle}{\langle u_2, u_2 \rangle} u_2. \end{aligned}$$

Let us compute $\langle u_1, u_1 \rangle = \int_{-1}^1 1\,\mathrm{d}x = 2$ and $\langle v_2, u_1 \rangle = \int_{-1}^1 x\,\mathrm{d}0$. We also have $\langle v_3, u_1 \rangle =$

$\int_{-1}^{1} x^2 \cdot 1 \,\mathrm{d}x = 2/3$, $\langle u_2, u_2\rangle = \int_{-1}^{1} x \cdot x \,\mathrm{d}x = 2/3$, and $\langle v_3, u_2\rangle = \int_{-1}^{1} x^2 \cdot x \,\mathrm{d}x = 0$. So

$$\begin{aligned}
u_1 &= v_1 = 1, \\
u_2 &= x - \frac{0}{2}1 = x, \\
u_3 &= x^2 - \frac{2/3}{2}1 - \frac{0}{2/3}x = x^2 - \frac{1}{3}.
\end{aligned}$$

Finally, we compute $\langle u_3, u_3\rangle = \int_{-1}^{1}(x^2 - 1/3)\,\mathrm{d}x = 8/45$ and we normalize the u_i to obtain an orthonormal basis for $\mathcal{P}_2(\mathbf{R})$:

$$\begin{aligned}
e_1 &= \frac{u_1}{\|u_1\|} = \frac{1}{\sqrt{2}}, \\
e_2 &= \frac{u_2}{\|u_2\|} = \sqrt{\frac{3}{2}}x, \\
e_3 &= \frac{u_3}{\|u_3\|} = \sqrt{\frac{45}{8}}\left(x^2 - \frac{1}{3}\right).
\end{aligned}$$

□

As a corollary of the Gram-Schmidt procedure, we obtain

Theorem 4.3.28.

Every finite dimensional inner product space has an orthonormal basis.

Proof. Since V is finite dimensional, it has a basis $(v_1, \dots, v_n)$. Applying the Gram-Schmidt procedure to this basis gives an orthonormal spanning list, that is an orthonormal basis. □

Infinite dimensional spaces

The same idea gives us that every inner product space of countable dimension (as for example $\mathcal{P}(\mathbf{F})$) has an orthonormal basis. However, this does not work for inner product spaces of uncountable dimension, as for example $\mathbf{F}^\infty$ or $\mathcal{C}^0([-1,1])$.

4.4 Orthogonality, matrices and minimizations problems

In finite dimensional spaces, orthogonal projections can conveniently be described using matrices. This can in turn be used to solve minimization problems, to approximate functions or to find best-fit curves.

4.4.1 Conjugate transpose matrices

Let $\mathcal{E}$ be the standard basis of $\mathbf{F}^n$ and let $T \in \mathcal{L}(\mathbf{F}^n)$ be a linear map. One can associate to T a matrix $[T]^{\mathcal{E}}_{\mathcal{E}} \in \mathbf{F}^{n,n}$. We have seen that T is a projection if and only if $\left([T]^{\mathcal{E}}_{\mathcal{E}}\right)^2 = 1$. At this point, it is natural to ask: can we detect on $[T]^{\mathcal{E}}_{\mathcal{E}}$ if T is an orthogonal projection? The answer is yes!

Before going further, let us recall a classical definition from high school.

Definition 4.4.1.

Let $A \in \mathbf{R}^{m,n}$ be a matrix. The **transpose matrix** of A is the matrix $A^{\perp}$ given by taking the symmetric of A along the diagonal:

$$A = \begin{bmatrix} a_{11} & a_{12} \cdots\cdots & a_{1n} \\ a_{21} & a_{22} \cdots\cdots & a_{2n} \\ \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} \cdots\cdots & a_{mn} \end{bmatrix} \longmapsto A^{\perp} := \begin{bmatrix} a_{11} & a_{21} \cdots\cdots & a_{m1} \\ a_{12} & a_{22} \cdots\cdots & a_{n} \\ \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots \\ a_{1n} & a_{2n} \cdots\cdots & a_{mn} \end{bmatrix}.$$

When we generalized the dot product from real vector spaces to complex vector spaces (Section 4.1.2), we had to introduce complex conjugation in the mix order to keep the nice properties of the real dot products. A similar phenomenon happens for matrices.

Definition 4.4.2.

Let $A \in \mathbf{F}^{m,n}$ be a matrix. The **conjugate transpose matrix** (also called **adjoint matrix** of A is the matrix $A^{*} := \overline{A^{\intercal}} = \overline{A}^{\intercal} \in \mathbf{F}^{n,m}$.

Observe that if $\mathbf{F} = \mathbf{R}$, then $A^* = A^{\intercal}$ is simply the transpose matrix. Here are some properties of the conjugate transpose of a matrix. For real matrices, they translate to properties of the transpose matrix.

Proposition 4.4.3. *The conjugate transpose of a matrix satisfies the following properties.*

1. *For all vectors u, v in $\mathbf{F}^n$ we have $\langle u, v\rangle = u \bullet v = v^* u$;*
2. *For all $A \in \mathbf{F}^{m,n}$ we have $(A^*)^* = A$;*
3. *For all $A, B \in \mathbf{F}^{m,n}$ we have $(A+B)^* = A^* + B^*$;*
4. *For all $A \in \mathbf{F}^{m,n}$ and $B \in \mathbf{F}^{n,k}$, we have $(AB)^* = B^* A^*$;*
5. *For all $A \in \mathbf{F}^{m,n}$ we have $\operatorname{null}(A^*) = \operatorname{col}(A)^{\perp}$ (with respect to the standard dot product).*

Proof. Items 1 to 3 directly follow from the definitions and their proof is an easy exercise.
4. We have $(AB)^* = \overline{(AB)^{\intercal}} = \overline{B^{\intercal} A^{\intercal}} = \overline{B^{\intercal}} \cdot \overline{A^{\intercal}} = B^* A^*$.

5. Recall that $\operatorname{col}(A) = \{AV \mid v \in \mathbf{F}^n\}$. Now, let u be any vector in $\mathbf{F}^n$. We have

$$\begin{aligned} u \in \operatorname{col}(A)^\perp &\iff \forall v \in \mathbf{F}^n, u \bullet Av = 0 \\ &\iff \forall v \in \mathbf{F}^n, 0 = (Av)^* u = v^* A^* u = A^* u \bullet v \\ &\iff A^* u = 0 \\ &\iff u \in \operatorname{null}(A^*). \end{aligned}$$

□

The above proposition implies the following fundamental properties of adjoint matrices.

Lemma 4.4.4. *Let $A \in \mathbf{F}^{m,n}$ be matrix. Then for every vectors $v \in \mathbf{F}^n$ and $w \in \mathbf{F}^m$ we have*

$$Av \bullet w = v \bullet A^* w.$$

Proof. We have

$$\begin{aligned} Av \bullet w &= w^* Av \\ &= (A^* w)^* v \\ &= v \bullet A^* w. \end{aligned}$$

□

Infinite dimensional spaces

Lemma 4.4.4 can be used to define the adjoint of an operator in the following way. Let V and W be possibly infinite dimensional inner product spaces, and let $T\colon V \to W$ be a linear map. We say that $S\colon W \to V$ is an **adjoint** of T and write $S = T^*$ if for all $v \in V$ and $w \in W$ we have $\langle Tv, w\rangle_W = \langle v, Sw\rangle_V$. If V and W are finite dimensional, then the adjoint always exists and is unique. Moreover, for $T \in \mathcal{L}(\mathbf{F}^n, \mathbf{F}^m)$ we have $[T^*]_{\mathcal{E}_n}^{\mathcal{E}_m} = \left([T]_{\mathcal{E}_m}^{\mathcal{E}_n}\right)^*$.
If V and W are infinite dimensional, but "nice enough" (that is, are *Hilbert spaces*) and if T is continuous, then the adjoint of T always exists, is unique and enjoys properties similar to the finite dimensional case.

Lemma 4.4.5. *Let $A \in \mathbf{F}^{m,n}$ be a matrix. Then the column of A are linearly independent if and only if the matrix $A^*A \in \mathbf{F}^{n,n}$ is invertible.*

Proof. We will prove the equivalence: the columns of A are linearly dependent if and only if A^*A is not invertible. Before doing that, we recall that the columns of a matrix $B \in \mathbf{F}^{m,n}$ are linearly dependent if and only if linear map represented by A is non-injective. That is, if and only if there exists a non-zero $u \in \mathbf{F}^n$ such that $Bu = 0$. Moreover, if B is a square matrix, this is also equivalent to the invertibility of B. Indeed, in this case the linear map represented by B is injective if and only if it is bijective, if and only if it is invertible.

"$\Rightarrow$" If the columns of A are linearly dependent, then there exists a non-zero $u \in \mathbf{F}^n$ such that $Au = 0$. But for this non-zero vector we have $(A^*A)u = A^*(Au) = A^*0 = 0$, which implies that A^*A is not invertible.

"⇐" If A^*A is not invertible, then there exists a non-zero vector $v \in \mathbf{F}^n$ with $A^*Av = 0$. But then we have $0 = v^*A^*Av = (Av)^*(Av) = \langle Av, Av \rangle = \|Av\|^2$, which implies $Av = 0$. Since $v \neq 0$, the columns of A are linearly independent. □

By the above Lemma, if the columns of A are linearly independent, then the matrix $A(A^*A)^{-1}A^*$ is well-defined. This is not any matrix, but a matrix representing an orthogonal projection.

Theorem 4.4.6.

Let $A \in \mathbf{F}^{m,n}$ be a matrix with linearity independent columns. Let $U := \mathrm{col}(A)$ be the subspace of $\mathbf{F}^m$ spanned by the columns of A. Then $P := A(A^*A)^{-1}A^* \in \mathbf{F}^{m,m}$ represents the orthogonal projection onto U (with respect to the standard dot product).

Proof. We will use left/right inverses. Let $S := A(A^*A)^{-1}$ and $T := A^*$. It is immediate that $TS = \mathrm{Id}_n$, and therefore by Proposition 3.2.11, $ST = A(A^*A)^{-1}A$ represents a projection onto $\mathrm{col}(S)$ along direction $\mathrm{null}(T)$.

On one hand, since A^*A is invertible, we have $\mathrm{col}(S) = \mathrm{col}(A) = U$. On the other hand, $\mathrm{null}(T) = \mathrm{null}(A^*) = \mathrm{col}(A)^\perp$ by Proposition 4.4.3. □

We will see later (in Example 4.4.14) that for applications it is enough to compute $(A^*A)^{-1}A^*$.

Theorem 4.4.7.

A $n \times n$ matrix A represents an orthogonal projection (with respect to the standard dot product) if and only if $A^2 = A = A^*$.

Proof. Write P for the linear map represented by A, that is $[Pv]_\mathcal{E} = A[v]_\mathcal{E}$.

"⇐" Let $U := \mathrm{col}(A) = \mathrm{range}(P)$ and $W := \mathrm{null}(A) = \mathrm{null}(P)$. Since $A^2 = A$, we have $P^2 = P$ and the map P is a projection onto $\mathrm{col}(A)$ along direction (A). Using Proposition 4.4.3 we have $\mathrm{col}(A)^\perp = \mathrm{null}(A^*) = \mathrm{null}(A)$.Therefore, P is an orthogonal projection (onto $\mathrm{col}(A)$).

"⇒" Suppose P is an orthogonal projection (onto $\mathrm{col}(A)$ along direction $\mathrm{null}(A) = \mathrm{col}(A)^\perp$). Let us write $P*$ for the linear map represented by A^*. We will show that $P*$ is also projection onto $\mathrm{col}(A)$ along direction $\mathrm{null}(A)$. This will imply that $P^* = P$ and therefore that $A^* = A$.

Since P is a projection, we have $A^2 = A$. But then $(A^*)^2 = A^*A^* = (AA)^* = A^*$. Therefore, P^* is a projection onto $\mathrm{col}(A^*)$ along direction $\mathrm{null}(A^*)$. By Proposition 4.4.3 we have $\mathrm{null}(A^*) = \mathrm{col}(A)^\perp$ which is itself equal to $\mathrm{null}(A)$. Applying the same proposition to A^* we obtain $\mathrm{null}(A^*) = \mathrm{null}(A) = \mathrm{col}(A^*)^\perp$. That is, P^* is the orthogonal projection onto $\mathrm{null}(A^*) = \mathrm{null}(A)$ and hence equal to P. □

4.4.2 QR decomposition

One can use the Gram-Schmidt procedure to find nice decomposition of invertible matrices. Let V be a finite dimensional inner product space over $\mathbf{F}$ and let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis for V. By applying Gram-Schmidt to $\mathcal{B}$ we obtain an orthogonal basis $\mathcal{D} = (u_1, \dots, u_n)$ and an orthonormal basis $\mathcal{E} = (e_1, \dots, e_n)$ of V.

Question 4.4.8. What does the change of basis matrices $[\mathrm{Id}]_{\mathcal{E}}^{\mathcal{B}}$ and $[\mathrm{Id}]_{\mathcal{B}}^{\mathcal{E}}$ look like?

Answer. We have $[\mathrm{Id}]_{\mathcal{E}}^{\mathcal{B}} = [\mathrm{Id}]_{\mathcal{D}}^{\mathcal{B}}[\mathrm{Id}]_{\mathcal{E}}^{\mathcal{D}}$. From the Gram-Schmidt procedure, we have

$$[\mathrm{Id}]_{\mathcal{D}}^{\mathcal{B}} = \begin{bmatrix} 1 & * & \cdots & * \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & * \\ 0 & \cdots & 0 & 1 \end{bmatrix} \qquad [\mathrm{Id}]_{\mathcal{E}}^{\mathcal{D}} = \begin{bmatrix} \frac{1}{\|u_1\|} & & \\ & \ddots & \\ & & \frac{1}{\|u_n\|} \end{bmatrix}$$

for some coefficients $*$ in $\mathbf{F}$. So we have just proven that

$$[\mathrm{Id}]_{\mathcal{E}}^{\mathcal{B}} = \begin{bmatrix} \frac{1}{\|u_1\|} & * & \cdots & * \\ & \ddots & \ddots & \vdots \\ & & \ddots & * \\ & & & \frac{1}{\|u_n\|} \end{bmatrix}$$

is an invertible upper triangular matrix, and so is

$$[\mathrm{Id}]_{\mathcal{E}}^{\mathcal{B}} = \begin{bmatrix} \|u_1\| & * & \cdots & * \\ & \ddots & \ddots & \vdots \\ & & \ddots & * \\ & & & \|u_n\| \end{bmatrix}.$$

■

Example 4.4.9. Let $V = \mathbf{R}^n$ be the Euclidean space with standard dot product. And let $A = [v_1 \ \dots \ v_n]$ be an invertible $n \times n$ matrix (v_i is the i^{th} column of A). Then $\mathcal{B} = (v_1, \dots, v_n)$ is a basis for $\mathbf{R}^n$ and applying Gram-Schmidt to $\mathcal{B}$ one obtain an orthonormal basis $\mathcal{E} = (e_1, \dots, e_n)$ of $\mathbf{R}^n$. Now, write $Q := [e_1 \ \dots \ e_n]$ and $R := [\mathrm{Id}]_{\mathcal{B}}^{\mathcal{E}}$. By the above, R is an invertible upper triangular matrix, while Q is an *orthogonal matrix*[4] ($QQ^\top = \mathrm{Id}_n = Q^\top Q$) and

$$A = QR.$$

This is called the QR **decomposition** of A.

If $V = \mathbf{C}^n$, then we similarly obtain an UR **decomposition** of $A = UR$, where R is an invertible triangular matrix and U is an *unitary matrix* ($UU^* = \mathrm{Id}_n = U^*U$).

[4] Be careful: despite a similar name, an orthogonal matrix is not the matrix of an orthogonal projections! Indeed, by Theorem 4.4.7, if a real matrix M represents a projection then $M^2 = M$. If the matrix M is also orthogonal in the above sense, then it is invertible an hence $M = \mathrm{Id}_n$.

4.4.3 Minimization problems

In this subsection we will see how to use linear algebra to solve minimization problems of the following form.

Problem 4.4.10. Let V be an inner product space and let U be a subspace. Given a vector $v \in V$, we would like to find $u \in U$ such that $\|v - u\|$ is as small as possible.

A typical situation where we want to solve this problem is when V is a complicated inner product space, U is a nicer subspace and given $v \in V$ we are looking at an approximation $u \in U$ of v. The approximation is good if the "error" $\|v - u\|$ is as small as possible. A typical example of a pair $U \subseteq V$ is $V = \mathcal{C}^\infty(\mathbf{R})$ the space of smooth functions and $U = \mathcal{P}(\mathbf{R})$ the subspace of polynomials.

The following result shows that Problem 4.4.10 admits a unique solution.

Theorem 4.4.11.

Let V be an inner product space and let U be a finite dimensional subspace of V. Write P_U for the orthogonal projection onto U. Then for any $v \in V$ we have

$$\|v - P_U v\| \leq \|v - u\|$$

for all $u \in U$. Moreover, $\|v - P_U v\| = \|v - u\|$ if and only if $u = P_U v$.

Proof. We have

$$\begin{aligned} \|v - P_U v\|^2 &\leq \|v - P_U v\|^2 + \|P_U v - u\|^2 \\ &= \|v - u\|^2, \end{aligned}$$

where the last equality is by the Pythagorean Theorem since $P_U v - u \in U$ and $v - P_U v \in U^\perp$ are orthogonal.

We have equality if and only if $\|P_U v - u\| = 0$, that is if and only if $P_U v - u = 0$. $\square$

See Figure 4.12 for an example of the inequality from the theorem.

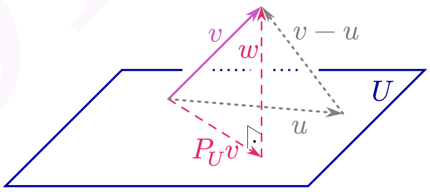

Figure 4.12: The orthogonal projection of v onto U, with "error" $\|w\| = \|v - P_U(v)\|$ and a worst approximation u with error $\|v - u\| > \|v - P_U(v)\|$.

Observe that in Theorem 4.4.11, U has to be finite dimensional[5], but V can be infinite dimensional, which will be the case in many applications.

[5] To be more precise, we only need $U \oplus U^\perp = V$, which is true for finite dimensional subspaces.

Theorem 4.4.11 is often combined with Proposition 4.3.24.9 to compute explicit solutions to minimization problems. Now, let us see some application of Theorem 4.4.11, starting with an easy example.

Example 4.4.12. Let $V = \mathbf{R}^4$ with the standard dot product. Le $v := [1,2,3,4]^\mathsf{T}$, $v_1 := [1,1,0,0]^\mathsf{T}$ and $v_2 := [1,1,1,2]^\mathsf{T}$ be three vectors in V. Find $u \in U := \operatorname{span}(v_1, v_2)$ minimizing $\|v - u\|$.

Solution. By the theorem, we know that $u = P_U v$. To find $P_U v$ we need an orthonormal basis for U. To obtain such a basis, we can apply Gram-Schmidt to (v_1, v_2). So we have $u_1 = v_1 = [1,1,0,0]^\mathsf{T}$ and $\|u_1\|^2 = \langle u_1, u_1\rangle = 2$. Therefore

$$u_2 = v_2 - \frac{\langle v_2, u_1\rangle}{\langle u_1, u_1\rangle} u_1 = \begin{bmatrix} 1\\1\\1\\2 \end{bmatrix} - \frac{2}{2}\begin{bmatrix} 1\\1\\0\\0 \end{bmatrix} = \begin{bmatrix} 0\\0\\1\\2 \end{bmatrix}$$

and $\|u_2\|^2 = 1 + 2^2 = 5$. Normalizing the u_i by their norm we obtain an orthonormal basis $(e_1 = \frac{1}{\sqrt{2}}[1,1,0,0]^\mathsf{T}, e_2 = \frac{1}{\sqrt{5}}[0,0,1,2]^\mathsf{T})$ for U. Finally we compute the projection

$$\begin{aligned} u = P_U v &= \langle v, e_1\rangle e_1 + \langle v, e_2\rangle e_2 \\ &= \frac{3}{\sqrt{2}}\frac{1}{\sqrt{2}}\begin{bmatrix} 1\\1\\0\\0 \end{bmatrix} + \frac{11}{\sqrt{5}}\frac{1}{\sqrt{5}}\begin{bmatrix} 0\\0\\1\\2 \end{bmatrix} \\ &= \begin{bmatrix} 1.5\\1.5\\2.2\\4.4 \end{bmatrix}. \end{aligned}$$

□

We will now do a more realistic example and use linear algebra to compute an approximation of the sine function.

Example 4.4.13. Suppose we want to find a polynomial u of degree at most 5 that approximates the sine function as well as possible on the interval $[-\pi, \pi]$.

Solution. The sine function is continuous and hence belongs to $\mathcal{C}^0[-\pi, \pi]$. The space V has a natural inner product:

$$\langle f, g\rangle = \int_{-\pi}^{\pi} f(x)g(x)\,\mathrm{d}x.$$

Polynomials of degree at most 5 form a subspace $\mathcal{P}_5(\mathbf{R}) \subseteq \mathcal{C}^0[-\pi, \pi]$ of finite dimension. So we are looking at $u \in \mathcal{P}_5(\mathbf{R})$ minimizing $\int_{-\pi}^{\pi} |\sin(x) - u(x)|^2\,\mathrm{d}x = \|\sin(x) - u(x)\|^2$. To find such a polynomial, we start from the standards basis $1, x, x^2, x^3, x^4, x^5$ of $\mathcal{P}_5(\mathbf{R})$ and apply Gram-Schmidt to it to obtain an orthornormal basis $(e_1, \dots, e_6)$. We can then use Proposition 4.3.24.9 to compute $P_U \sin(x)$. Using a computer, one can find

$$P_U \sin(x) \approx 0.987\,862x - 0.155\,271x^3 + 0.005\,643\,12x^5,$$

where π has been replaced by a decimal approximation.

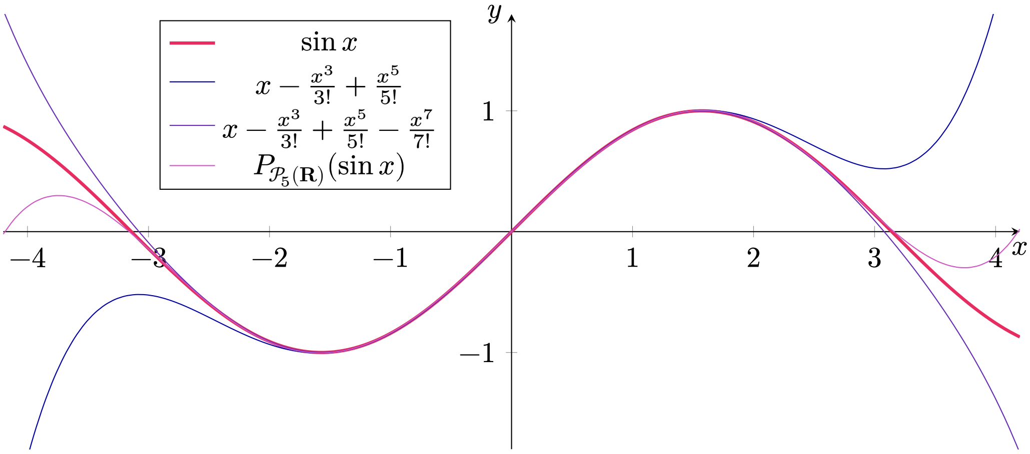

Figure 4.13: Three polynomial approximations of the sine function. By the Taylor polynomials of degree 5 and 7, and a better one by the projection on $\mathcal{P}_5(\mathbf{R})$.

As one can see on Figure 4.13, the polynomial approximation given by the projection of $\sin x$ onto $\mathcal{P}_5(\mathbf{R})$ is much better than the Taylor approximation of the same degree. In fact, $P_{\mathcal{P}_5(\mathbf{R})} \sin(x)$ is even a better approximation that the degree 7 Taylor polynomial of $\sin x$! □

In many real world experiments, we obtain a cloud of points $\{(x_i, y_i) \mid 1 \leq i \leq n\}$ in $\mathbf{R}^2$ that we would like to approximate by an easy function like a line or a parabola. One can use orthogonal projections to find such an approximation. First we need to explicit what we mean by "an approximation". One usual approximation is the *least square regression* where we want to find a function minimizing $(y_1 - f(x_1))^2 + \dots + (y_n - f(x_n))^2$. This corresponds to look at the standard dot product on $\mathbf{R}^n$ and try to minimize $\|y - f(x)\|$, where $y = [y_1, \dots, y_n]^\top$ and $f(x) = [f(x_1), \dots, f(x_n)]^\top$ is the value of the approximating function at the points x_i. Now that we have fixed an inner product on $\mathbf{R}^n$, we need to decide on which subspace we project.

For example, if we want to approximate our datas by a line, we will project on $U_1 = \{[f(x_1), \dots, f(x_n)]^\top \mid f \in \mathcal{P}_1(\mathbf{R})\}$ which is a subspace of dimension 2 of $\mathbf{R}^n$. A basis of this subspace is given by $([x_1, \dots, x_n]^\top, [1, \dots, 1]^\top,)$ (the first vector correspond to the polynomial x, and the second to the constant polynomial 1). In more concrete words, we are looking for two real numbers a and b minimizing

$$\left\| \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} - \left(a \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + b \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \right) \right\|.$$

One can try to minimize the above system by hand, but this is a priori not an easy task. It is easier to use Theorem 4.4.11. So we need to find the orthogonal projection P_U. To do so, we can use Gram-Schmidt. But a more efficient method is to use Theorem 4.4.6

to obtain $\left[\begin{smallmatrix} a \\ b \end{smallmatrix}\right] = A(A^*A)^{-1}y$, where A is the $n \times 2$ matrix:

$$A = \begin{bmatrix} x_1 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{bmatrix}.$$

Let us demonstrate that on a concrete example.

Example 4.4.14. Imagine we measured the following values in an experiment: $(0,6)$, $(1,0)$ and $(2,0)$. It is obvious that no line in $\mathbf{R}^2$ go through these 3 points. However, our model/intuition for this experiment say that these points should lie on line. Maybe our measurements were not precise enough? So we try to find the best-fit line. In order to do that, we need to minimize

$$\left\| \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} - \left(a \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + b \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right) \right\|.$$

To solve this minimization problem, we will use Theorem 4.4.6 for

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix}.$$

So the projection of $[6,0,0]^\mathsf{T}$ onto U is given by

$$\begin{aligned} A \cdot (A^*A)^{-1} A^* \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} &= A \left(\begin{bmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} \\ &= A \begin{bmatrix} 5 & 3 \\ 3 & 3 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 6 \end{bmatrix} \\ &= A \frac{1}{6} \begin{bmatrix} 3 & -3 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 6 \end{bmatrix} \\ &= A \begin{bmatrix} -3 \\ 5 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix}. \end{aligned}$$

As we can see above, the best-fit line is given by the equation $y = -3x + 5$, while the projection of of $[6,0,0]^\mathsf{T}$ onto U is $[5,2,-1]^\mathsf{T} = -3[x_1, x_2, x_3]^\mathsf{T} + 5[1,1,1]^\mathsf{T}$. We can also compute the error:

$$\left\| \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix} \right\| = \sqrt{1^2 + (-2)^2 + (-1)^2} = \sqrt{6}.$$

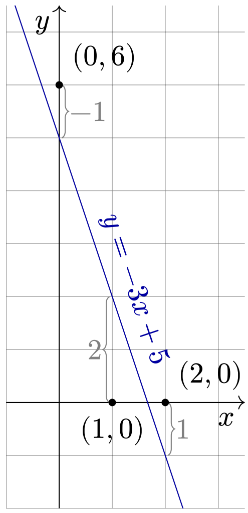

Figure 4.14: Best-fit line for the three black dots. The error is $\sqrt{6}$.

The above example can be adapted to any kind of best-fit curve. For example, if we are looking for a best-fit parabola for our 3 points, our matrix A will be:

$$A = \begin{bmatrix} (x_1)^2 & x_1 & 1 \\ (x_2)^2 & x_2 & 1 \\ (x_3)^2 & x_3 & 1 \end{bmatrix}.$$

because $(1, x, x^2)$ is a basis of $\mathscr{P}_3(\mathbf{R})$. Then we should compute

$$(A^*A)^{-1}A \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} =: \begin{bmatrix} a \\ b \\ c \end{bmatrix}.$$

This gives us that the best fit parabola for $((x_1, y_1), (x_2, y_2), (x_2, y_2))$ is given by the equation $y = ax^2 + bx + c$. Observe that for this specific easy example, our approximation will actually passes through all three points if they are not colinear.

5 Eigenvalues and eigenvectors

In this chapter we will try to better understand operators $T \in \mathcal{L}(V)$. The main idea is to decompose V into smaller subspaces on which T acts, and use this to describe T. This will ultimately (see Section 5.3 and in particular Theorem 5.3.9) allow us to answer

Major Question 3.1.26.

Given V and $T \in \mathcal{L}(V)$, find a basis $\mathcal{B}$ such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is "as simple as possible" (for example: diagonal, upper triangular, with many zeroes, ...).

Before going further into the details, let us recall an easy example we already saw.

Example 5.0.1. Let $V = \mathbf{R}^2$, and let S_θ be the orthogonal reflection across the line L that makes an angle θ with the x-axis, see Figure 5.1. We would like to find a basis $\mathcal{B}$ for which $[T]_{\mathcal{B}}^{\mathcal{B}}$ is "nice", meaning has many zeroes. First, we observe that for any v in L, we have $S_\theta v = v$. So it might be a good idea to take $v_1 = \begin{bmatrix}\cos\theta \\ \sin\theta\end{bmatrix} \in \mathbf{R}^2$ as our first basis vector. Indeed, this vector is of norm 1 and $L = \operatorname{span}(v_1)$.

We can also observe that for any u in $L^\perp$ we have $S_\theta u = -u$, which is also an easy formula. So one take $v_2 = \begin{bmatrix}-\sin\theta \\ \cos\theta\end{bmatrix}$ as our second basis vector. One easily verify that $\operatorname{span}(v_2) = L^\perp$, $v_1 \bullet v_2 = 0$ and $\|v_1\| = \|v_2\| = 1$, so our basis $\mathcal{B} = (v_1, v_2)$ is an orthonormal basis and $\mathbf{R}^2 = L \oplus L^\perp$. Altogether, we have

$$[S_\theta]_{\mathcal{B}}^{\mathcal{B}} = \begin{bmatrix}1 & 0 \\ 0 & -1\end{bmatrix}.$$

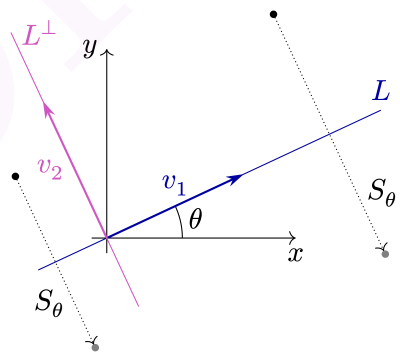

Figure 5.1: Orthogonal reflection S_θ across the line $L = \operatorname{span}\left(\begin{bmatrix}\cos\theta \\ \sin\theta\end{bmatrix}\right)$.

We can see many interesting things happening in this example. First of all, the existence an orthonormal basis $\mathcal{B}$ for which $[S_\theta]_{\mathcal{B}}^{\mathcal{B}}$ is diagonal. Secondly, for any $u \in L$ we have $S_\theta u = u \in L$. Therefore, if we restrict S_θ to L we have an operator $S_\theta|_L \in \mathcal{L}(L)$. Similarly, for $w \in L^\perp$ we have $S_\theta w = -w \in L^\perp$ and we hence have an operator $S_\theta|_{L^\perp} \in \mathcal{L}(L^\perp)$. Altogether, we obtain the following description of S_θ:

$$\begin{aligned} S_\theta \colon \mathbf{R}^2 = L \oplus L^\perp &\to L \oplus L^\perp \\ v = u + w &\longmapsto u - w. \end{aligned}$$

For a general $T \in \mathcal{L}(V)$, we will not obtain a decomposition of V as nice as the decomposition of Example 5.0.1. However, we will still be able to decompose V onto smaller subspaces (Theorem 5.1.46) on which the action of T will not be too complicated (Theorem 5.3.9).

5.1 Eigen-theory

We start by generalizing the idea of "nice vectors for T" from Example 5.0.1. This will give us a first answer to Question 3.1.26: if $T\mathcal{L}(V)$ is an operator on a finite dimensional complex vector space, then there exists a basis for which T is a diagonal by blocks matrix, see Proposition 5.1.50. We will then develop further tool in order to show that such a T admits a Jordan form, see Section 5.3 and in particular Theorem 5.3.9.

5.1.1 Invariant subspaces

One of the important feature of Example 5.0.1 was the existence of subspaces U (L and $L^\perp$) for which the restriction $S_\theta|_U$ was still an operator. Let us generalize this notion.

Definition 5.1.1.

Let V be a vector space over $\mathbf{F}$, let $T \in \mathcal{L}(V)$ be an operator and let $U \subseteq V$ be a subspace. We say that U is **invariant under** T or T**-invariant** if for every $u \in U$ we have $Tu \in U$.

If we denote $TU := \{Tu \mid u \in U\}$, then U is T-invariant if and only if $TU \subseteq U$, if and only if $T|_U \in \mathcal{L}(U)$.

Below are some examples of T-invariant subspaces for various operators T.

Example 5.1.2. For any $T \in \mathcal{L}(V)$ the following four (not necessarily distinct) subspaces of V are all T-invariant: $\{0\}$, V, $\mathrm{null}(T)$ and $\mathrm{range}(T)$.

Indeed, $T0 = 0$ and $TV \subseteq V$. Moreover, for every $u \in \mathrm{null}(T)$, $Tu = 0$ is still in $\mathrm{null}(T)$ while for $v \in \mathrm{range}(T)$, Tv is in $\mathrm{range}(T)$.

Since $\{0\}$ and V are always T-invariant subspaces, we are interested to find the other invariant subspaces of T. However these do not always exist when $\mathbf{F} = \mathbf{R}$.

Example 5.1.3. Let S_θ be the orthogonal reflection across line L from Example 5.0.1. Then both L and $L^\perp$ are S_θ-invariant. In fact, one can easily show that: L and $L^\perp$ are the only invariant subspaces of S_θ — beside the trivial ones $\mathrm{null}(S_\theta) = \{0\}$, $\mathrm{range}(S_\theta) = \mathbf{R}^2$.

Example 5.1.4. Let $V = \mathcal{P}(\mathbf{R})$ and let $D \in \mathcal{L}(\mathcal{P}(\mathbf{R}))$ be the differentiation operator $D(p) = p'$. Then for every n, the subspace $\mathcal{P}_n(\mathbf{R})$ is D-invariant.

Example 5.1.5. Let $V = \mathbf{R}^2$ and let $\theta \in [0, 2\pi)$. Let R_θ be the rotation of angle θ around the origin. Then if $\theta \notin \{0, \pi\}$, the only invariant subspaces under R_θ are $\{0\}$ and V. Indeed, suppose that $U \neq \{0\}$ is a R_θ-invariant subspace and take $0 \neq u \in U$. Since $\theta \neq \{0, \pi\}$, R_θ is not in $\mathrm{span}(u)$. By consequences, $U \supseteq \mathrm{span}(u, R_\theta u) = V$.

On the opposite, every subspace U is an invariant subspace of $R_0 = \mathrm{Id}$ and of R_π.

Example 5.1.6. Let V be any vector space, let Id_V be the identity operator and let $\lambda \in \mathbf{F}$ be any scalar number. Then any subspace U of V is $\lambda\,\mathrm{Id}_V$-invariant. Indeed, for every $u \in U$ we have $\lambda\,\mathrm{Id}_V u = \lambda u \in U$.

As the above examples show, when we say that U is an invariant subspace, it is important to precise for which T it is invariant.

5.1.2 Eigenvalues and eigenvectors

In Example 5.0.1, not only do we have L and $L^\perp$ as non-trivial invariant subspaces, but the action of S_θ on it is particularly easy to describe. Indeed, for $v \in L$ we have $S_\theta v = v$, while for or $u \in L$ we have $S_\theta u = -u$. In both cases, the linear operator S_θ acts as the scalar multiplication by some λ.

Definition 5.1.7.

Let V be a vector space and let $T \in \mathcal{L}(V)$ be an operator. We say that $\lambda \in F$ is an **eigenvalue** of T if there exists a non-zero vector $v \in V$ such that $Tv = \lambda v$.
Such a non-zero vector v satisfying $Tv = \lambda v$ is called an **eigenvector** of T, corresponding to eigenvalue λ.
The **eigenspace** of T corresponding to $\lambda \in \mathbf{F}$ is the set

$$\mathrm{E}(\lambda, T) := \{v \in V \mid Tv = \lambda v\}.$$

Observe that while by definition 0_V is never an eigenvector, 0 might be an eigenvalue. In fact, one always have $\mathrm{E}(0, T) = \mathrm{null}(T)$. So 0 is an eigenvalue of T if and only if T is not injective. We do not allow 0 to be an eigenvector as $T0 = \lambda 0$ is true for all λ (and all T), so this would give a rather uninteresting notion of eigenvalues.

Be careful that the eigenspace of T corresponding to λ always contain 0_V. In fact, $\mathrm{E}(\lambda, T) = \{0_V\} \sqcup \{v \in V \mid v \text{ is an eigenvector corresponding to } \lambda\}$. We add 0 to $\mathrm{E}(\lambda, T)$, because we want it to be a subspace, as suggested by its name.

Lemma 5.1.8. *Let V be a vector space and let $T \in \mathcal{L}(V)$ be an operator. Then for any $\lambda \in \mathbf{F}$ we have*

$$\mathrm{E}(\lambda, T) = \mathrm{null}(T - \lambda\,\mathrm{Id}_V),$$

and this is a T-invariant subspace.

Example 5.1.9. Let S_θ be the orthogonal reflection across the line L, as in Example 5.0.1. Then as discuted above, both 1 and -1 are eigenvalue of S_θ. Moreover, for every u in L, we have $S_\theta v = v$, so $\mathrm{E}(1, S_\theta) \supseteq \mathrm{span}(v_1)$. But if u is not in L, then $S_\theta u \neq u$ and thus $\mathrm{E}(1, S_\theta) = L$. Similarly, one can show that $\mathrm{E}(-1, S_\theta) = L^\perp$.

So we have proved that $\mathbf{R}^2 = \mathrm{E}(1, S_\theta) \oplus \mathrm{E}(-1, S_\theta)$. This is not a coincidence, and we will come back to this later.

One useful properties of eigenspaces is that they contain the 1-dimensional T-invariant subspaces.

Lemma 5.1.10. *Let V be a vector space and let $T \in \mathcal{L}(V)$ be an operator. Then if U is an 1-dimensional T-invariant subspace, there exists an eigenvalue λ such that $U \subseteq \mathrm{E}(\lambda, T)$.*

Proof. Since U is 1-dimensional, for all $0 \neq v \in U$ we have $U = \mathrm{span}(v)$. Fix such a v. Then by T-invariance, $Tv \in U$ and so $Tv = \lambda v$ for some $\lambda \in \mathbf{F}$. Now, for every $w \in U$ there exists μ with $w = \mu v$, and hence $Tw = \mu T v = \mu\lambda v = \lambda w$. This proves that U is contained in $\mathrm{E}(\lambda, T)$. □

Observe that the definition of $\mathrm{E}(\lambda, T)$ makes sense even if λ is not an eigenvalue, and that Lemma 5.1.8 is true for all $\lambda \in \mathbf{F}$, not only for eigenvalues of T. Actually, we can use $\mathrm{E}(\lambda, T)$ to characterize eigenvalues.

Proposition 5.1.11. *Let $V \neq \{0\}$ be a vector space and let $T \in \mathcal{L}(V)$ be an operator. Then for $\lambda \in \mathbf{F}$ the following are equivalent.*

1. *λ is an eigenvalue of T;*
2. *$(T - \lambda \,\mathrm{Id}_V)$ is not injective;*
3. *$\dim(\mathrm{E}(\lambda, T)) \geq \{1\}$.*[1]

If V is finite dimensional, then the above conditions are also equivalent to

4. *$(T - \lambda \,\mathrm{Id}_V)$ is not surjective;*
5. *$(T - \lambda \,\mathrm{Id}_V)$ is not invertible.*

Proof. Let $\lambda \in \mathbf{F}$. Then

$$\begin{aligned} \lambda \text{is an eigenvalue of } T &\iff \exists v \neq 0, Tv = \lambda v \\ &\iff \exists v \neq 0, (T - \lambda \,\mathrm{Id}_V)v = 0 \\ &\iff \mathrm{null}(T - \lambda \,\mathrm{Id}_V) \neq \{0\}. \end{aligned}$$

The last condition is equivalent both to the non-injectivity of $T - \lambda \,\mathrm{Id}_V$ and to the fact that $\{0\} \neq \mathrm{null}(T - \lambda \,\mathrm{Id}_V) = \mathrm{E}(\lambda, T)$. But $U \neq \{0\}$ if and only if $\dim(U) \geq 1$.

If V is finite dimensional, then conditions 2, 4 and 5 are all equivalent by the fundamental theorem of linear algebra (Theorem 3.0.3). □

[1] By $\dim(U) \geq \{1\}$ we mean that either U is finite dimensional with $\dim(U) \geq 1$, or U is infinite dimensional.

Example 5.1.12. Let $T \in \mathcal{L}(\mathbf{F}^2)$ be the linear map $T\left[\begin{smallmatrix} w \\ z \end{smallmatrix}\right] = \left[\begin{smallmatrix} z \\ -w \end{smallmatrix}\right]$. Find the eigenvalues and eigenvectors of T for $\mathbf{F} = \mathbf{R}$ and for $\mathbf{F} = \mathbf{C}$.

Solution. We are looking at $\lambda \in \mathbf{F}$ such that the equation $\lambda\left[\begin{smallmatrix} w \\ z \end{smallmatrix}\right] = T\left[\begin{smallmatrix} w \\ z \end{smallmatrix}\right] = \left[\begin{smallmatrix} z \\ -w \end{smallmatrix}\right])$ admits a non-zero solution. But this equation is equivalent to the system

$$\begin{cases} \lambda w = z, \\ \lambda z = -w. \end{cases}$$

This system implies that $(1 + \lambda^2)w = 0$.

If $\mathbf{F} = \mathbf{R}$ the only possibility to solve $(1 + \lambda^2)w = 0$ is to have $w = 0$ and hence $z = 0$. That is, if $\mathbf{F} = \mathbf{R}$, then T has no eigenvalues (and no eigenvectors).

Now, if $\mathbf{F} = \mathbf{C}$, both $\lambda = i$ and $\lambda = -i$ are eigenvalues of T and we have $\mathrm{E}(i, T) = \{\left[\begin{smallmatrix} w \\ iw \end{smallmatrix}\right] \mid w \in \mathbf{C}\}$ while $\mathrm{E}(-i, T) = \{\left[\begin{smallmatrix} w \\ -iw \end{smallmatrix}\right] \mid w \in \mathbf{C}\}$. $\square$

As we have just seen, operators on real vector spaces do not necessarily have eigenvalues. It turns out that operators on complex vector spaces do always have eigenvalues (and eigenvectors). This makes complex linear algebra easier than real linear algebra.

Theorem 5.1.13.

Let $V \neq \{0\}$ be a non-trivial finite dimensional vector space over $\mathbf{C}$ and let $T \in \mathcal{L}(V)$ be an operator. Then T has at least one eigenvalue.

Proof. Let $n > 0$ be the dimension of V and choose any non-zero vector v. Then the list

$$v, Tv, T^2v, \dots, T^nv$$

has $n + 1$ elements and hence is not linearly independent. Therefore, some linear combination (with not all coefficients equal to 0) of the above vectors equals 0. Equivalently, there exists a non-zero polynomial $p(t)$ such that $p(T)v = 0$. Take such a polynomial $p(t)$ of minimal degree. Observe that since $v \neq 0$, the polynomial $p(t)$ is non-constant.

By the fundamental theorem of algebra, $p(t)$ has at least one complex root $\lambda \in \mathbf{C}$. In other words, there exists a polynomial $q \in \mathcal{P}(\mathbf{C})$ (non-zero, but possibly constant) such that $p(t) = (t - \lambda)q(t)$. This implies that $0 = p(T)v = (T - \lambda\,\mathrm{Id}_V)(q(T)v)$. We have $\deg(q) = \deg(p) - 1$, which by minimality implies that $q(T)v \neq 0$. We have just proved that $T(q(T)v) = \lambda(q(T)v)$ with $q(T)v \neq 0$. So λ is an eigenvalue of T with eigenvector $q(T)v$. $\square$

Corollary 5.1.14. *Let V be a finite dimensional vector space over $\mathbf{C}$ and let $T \in \mathcal{L}(V)$ be an operator. If $\dim(V) \geq 2$, then T has an invariant subspace which is neither $\{0\}$ nor V.*

Proof. Let λ be an eigenvalue and let v be any eigenvector. Then $\mathrm{span}(v)$ is a T-invariant subspace of dimension 1. $\square$

To go further

Theorem 5.1.13 and Corollary 5.1.14 are true for all fields $\mathbf{F}$ such that every polynomial $p(t) \in \mathcal{P}(\mathbf{F})$ has at least one root in $\mathbf{F}$. Such $\mathbf{F}$ are called **algebraically closed fields**. Here is an example of such a field. Start with $\mathbf{Q}$ and add all roots of polynomials to obtain a new field $\overline{\mathbf{Q}}$ with $\mathbf{Q} \subsetneq \overline{\mathbf{Q}} \subsetneq \mathbf{C}$. This new field $\overline{\mathbf{Q}}$ contains $\sqrt{2}$ (root of $x^2 - 2$), i (root of $x^2 + 1$), but does not contains π.

Lemma 5.1.15. *Let $T \in \mathcal{L}(V)$ be an operator. Suppose that v_1 and v_2 are two eigenvectors, corresponding to eigenvalues λ_1 and λ_2 respectively. If $\lambda_1 \neq \lambda_2$ the vectors v_1 and v_2 are linearly independent.*

Proof. Suppose that

$$0 = \mu_1 v_1 + \mu_2 v_2 \tag{5.1}$$

for some $\mu_1, \mu_2 \in \mathbf{F}$. On one hand, if we multiply both sides of Equation (5.1) by λ_1 we have

$$0 = \mu_1 \lambda_1 v_1 + \mu_2 \lambda_1 v_2. \tag{5.2}$$

On the other hand, applying T to both sides of Equation (5.1) gives us

$$\begin{aligned} 0 = T0 &= T(\mu_1 v_1 + \mu_2 v_2) \\ &= \mu_1 \lambda_1 v_1 + \mu_2 \lambda_2 v_2. \end{aligned}$$

Substracting this to Equation (5.2) we obtain

$$\mu_2(\lambda_1 - \lambda_2) v_2 = 0.$$

Since $v_2 \neq 0$ and $\lambda_1 - \lambda_2 \neq 0$ we conclude that $\mu_2 = 0$. A similar argument shows that $\mu_1 = 0$. We conclude that v_1 and v_2 are linearly independent. $\square$

As an easy but important consequence of the above lemma we obtain the following generalization.

Theorem 5.1.16.

Let $T \in \mathcal{L}(V)$ be an operator. Suppose that $v_1, \dots, v_n$ are n eigenvectors, corresponding to eigenvalues $\lambda_1, \dots, \lambda_n$ respectively. If the λ_i are pairwise distinct, then $v_1, \dots, v_n$ is a linearly independent list.

Proof. The proof is by induction. If $n = 1$, then any eigenvector v is a non-zero vector and therefore (v) is a linearly independent list. If $n = 2$, this is the lemma.

Now suppose that $n \geq 3$ and that the statement is proven for all lists of $m < n$ eigenvectors corresponding to distinct eigenvalues. Suppose by contradiction that $v_1, \dots, v_n$ is not linearly independent. Therefore, by Lemma 2.0.3 there exists $j \in \{1, \dots, n\}$ such that $v_j \in \operatorname{span}(v_1, \dots, v_{j-1})$. By induction hypothesis, j cannot be smaller than n and therefore we have

$$v_n = \mu_1 v_1 + \dots + \mu_{n-1} v_{n-1} \tag{5.3}$$

for some $\mu_1, \dots, \mu_{n-1} \in \mathbf{F}$. By applying T to both sides of Equation (5.3) by λ_n and comparing with Equation (5.3) multiplied by λ_n we obtain

$$\begin{aligned}\mu_1\lambda_1 v_1 + \dots + \mu_{n-1}\lambda_{n-1}v_{n-1} &= Tv_n = \lambda_n v_n \\ &= \mu_1\lambda_n v_1 + \dots + \mu_{n-1}\lambda_n v_{n-1}.\end{aligned}$$

Therefore,

$$0 = \mu_1(\lambda_1 - \lambda_n)v_1 + \dots + \mu_{n-1}(\lambda_{n-1} - \lambda_n)v_{n-1}$$

and we conclude that all the μ_j are 0 and hence that $v_n = 0$ which is absurd. □

We can use Theorem 5.1.16 to better understand the sum of the eigenspaces.

Corollary 5.1.17. *Let $T \in \mathcal{L}(V)$ be an operator and let $\lambda_1, \dots, \lambda_m$ be pairwise distinct eigenvalues of T. Then the sum $\mathrm{E}(\lambda_1, T) + \dots + \mathrm{E}(\lambda_m, T)$ is a direct sum.*

If V is finite dimensional, then

$$\dim(\mathrm{E}(\lambda_1, T)) + \dots + \dim(\mathrm{E}(\lambda_m, T)) \leq \dim(V).$$

Moreover, in this case the equality $\dim(\mathrm{E}(\lambda_1, T)) + \dots + \dim(\mathrm{E}(\lambda_m, T)) = \dim(V)$ holds if and only if V is the direct sum of the $\mathrm{E}(\lambda_i, T)$.

Proof. Let $0 = u_1 + \dots + u_m$ with $u_i \in \mathrm{E}(\lambda_i, T)$. By linear independence, (Theorem 5.1.16), all the u_i are 0. We hence conclude that the sum of the $\mathrm{E}(\lambda_i, T)$ is direct.

Since the $\mathrm{E}(\lambda_i, T)$ are in direct sum we have

$$\dim\left(\sum_{i=1}^{m} \dim(\mathrm{E}(\lambda_i, T))\right) = \dim\left(\bigoplus_{i=1}^{m} \mathrm{E}(\lambda_i, T)\right) \leq \dim(V).$$

Since V is finite dimensional, we have equality in the above inequation if and only if $\bigoplus_{i=1}^{m} \mathrm{E}(\lambda_i, T) = V$. □

Corollary 5.1.18. *Let V be a finite dimensional vector space and let $T \in \mathcal{L}(V)$ be an operator. Then T has at most $\dim(V)$ distinct eigenvalues.*

Proof. Let $\lambda_1, \dots, \lambda_m$ be a complete list of eigenvalues for T. Then for all $i \in \{1, \dots, m\}$ we have $\dim \mathrm{E}(\lambda_i, T) \geq 1$. Therefore,

$$m \leq \sum_{i=1}^{m} \dim \mathrm{E}(\lambda_i, T) \leq \dim(V).$$ □

Example 5.1.19. Let S_θ be the reflection across the line L from Example 5.0.1. Then v_1 is an eigenvector of eigenvalue 1, while v_2 is an eigenvector with eigenvalue $-1 \neq 1$. They are linearly independent as announced by Lemma 5.1.15. Moreover, we have $\dim(\mathbf{R}^2) = 2 = 1 + 1 = \dim(\mathrm{E}(1, S_\theta)) + \dim(\mathrm{E}(-1, S_\theta))$ and $\mathbf{R}^2 = \mathrm{E}(1, S_\theta) \oplus \mathrm{E}(-1, S_\theta)$ as predicted by Corollary 5.1.17. Finally, 1 and -1 are the only eigenvalues of S_θ by Corollary 5.1.18, but this can also be directly checked by hand.

Infinite dimensional spaces

In infinite dimensional vector spaces, the formula $\dim(\mathrm{E}(\lambda_1, T)) + \cdots + \dim(\mathrm{E}(\lambda_m, T)) \leq \dim(V)$ of Corollary 5.1.17 remains true. Moreover, we still have the implication $[V = \bigoplus_{i=1}^m \mathrm{E}(\lambda_i, T)] \implies [\sum_{i=1}^n \dim(\mathrm{E}(\lambda_i, T) = \dim(V)]$. However, the converse implication is not true anymore, as V might have a proper subspace $U \subsetneq V$ with $\dim(U) = \dim(V)$.
The statement of Corollary 5.1.18 still holds true for infinite dimensional vector spaces. However, in order to prove it we need versions of Theorem 5.1.16 and Corollary 5.1.17 that are true for infinite lists. We would also need to make sense of what is an infinite sum, which goes beyond this course.

5.1.3 Diagonalizability

Standing assumption

In this Subsection, V will always be a finite dimensional vector space.

One of the very nice feature of Example 5.0.1 is the existence of a basis $\mathcal{B} = (v_1, v_2)$ such that $[S_\theta]_\mathcal{B}^\mathcal{B}$ is diagonal. If we write $\mathcal{E}$ for the standard basis and compare

$$[S_\theta]_\mathcal{B}^\mathcal{B} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \qquad [S_\theta]_\mathcal{E}^\mathcal{E} = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix},$$

it is clear that the left expression is easier to use for computations than the right one.

Definition 5.1.20.

Let V be a finite dimensional vector space, of dimension n. An operator T is **diagonalizable** if there exists a basis $\mathcal{B} = (v_1, \dots, v_n)$ of V such that

$$[T]_\mathcal{B}^\mathcal{B} = \underbrace{\begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}}_{\text{diagonal matrix}}$$

for some $\lambda_1, \dots, \lambda_n \in \mathbf{F}$.

Remark 5.1.21. It follows from the definition of $[T]_\mathcal{B}^\mathcal{B}$ that the operator $T \in \mathcal{L}(V)$ is diagonalisable if and only if there exists a basis $\mathcal{B} = (v_1, \dots, v_n)$ and scalars $\lambda_1, \dots, \lambda_n \in \mathbf{F}$ with $Tv_i = \lambda_i v_i$. That is, if and only if there exists a basis $\mathcal{B} = (v_1, \dots, v_n)$ of eigenvectors, with eigenvalues $\lambda_1, \dots, \lambda_n \in \mathbf{F}$.

As we just discussed, diagonalizability of an operator is linked to its eigenvalues. But it can also be characterized by its eigenspaces.

Theorem 5.1.22. *Let V be an n-dimensional vector space and let $T \in \mathcal{L}(V)$ be an operator. Let $\lambda_1, \dots, \lambda_m$ be the eigenvalues of T (so $m \leq n$ by Corollary 5.1.18). Then the following are equivalent.*

1. *T is diagonalizable;*
2. *There exists a basis of V consisting of eigenvectors of T;*
3. *There exists 1-dimensional T-invariant subspaces $U_1, \dots, U_n$ such that $V = U_1 \oplus \cdots \oplus U_n$;*
4. *$V = \mathrm{E}(\lambda_1, T) \oplus \cdots \oplus \mathrm{E}(\lambda_m, T)$;*
5. *$\dim(V) = \sum_{i=1}^{m} \dim(\mathrm{E}(\lambda_i, T))$.*

Proof. "1 $\iff$ 2 and 4 $\iff$ 5" These are Remark 5.1.21 and Corollary 5.1.17.

"2$\Rightarrow$3" Suppose $\mathcal{B} = (v_1, \dots, v_n)$ is a basis of eigenvectors and let $U_i := \operatorname{span}(v_i)$ for $i \in \{1, \dots, n\}$. Each of the U_i is a T-invariant subspace of dimension 1. Since $\mathcal{B}$ is a basis, it is also a linearly independent list. Therefore, the sum $U_1 + \cdots + U_n \leq V$ is direct. Finally, $n \leq \dim(U_1 \oplus \cdots \oplus U_n) \leq \dim(V) = n$, which implies $V = U_1 \oplus \cdots \oplus U_n$.

"3$\Rightarrow$4" Each U_i is a T-invariant subspace of dimension 1, and hence contained in a eigensubspace. Up to reordering, we can suppose that $U_1 \oplus \cdots \oplus U_{r_1} \subseteq \mathrm{E}(\lambda_1, T)$, $U_{r_1+1} \oplus \cdots \oplus U_{r_2} \subseteq \mathrm{E}(\lambda_2, T)$ etc. for some $1 \leq r_1 < r_2 < \cdots < r_m = n$. By a dimension argument, we conclude that $U_1 \oplus \cdots \oplus U_{r_1} = \mathrm{E}(\lambda_1, T)$ and so on.

"4$\Rightarrow$2" For each $i \in \{1, \dots, m\}$, choose a basis $\mathcal{B}_i$ for $\mathrm{E}(\lambda_i, T)$. Then $\mathcal{B} := \mathcal{B}_1 \cup \cdots \cup \mathcal{B}_m$ is a basis of V consisting of eigenvectors of T. $\square$

Let us work through some of these equivalences on a concrete case.

Example 5.1.23. Let $V = \mathbf{R}^3$ and let $\mathcal{E} = (e_1, e_2, e_3)$ be the standard basis. Let T the operator represented in the standard basis by

$$[T]_{\mathcal{E}}^{\mathcal{E}} = \begin{bmatrix} 4 & & \\ & 5 & \\ & & 4 \end{bmatrix}.$$

Then T is diagonalizable (1), with eigenvalues 4 and 5. Moreover, e_1, e_2 and e_3 are all eigenvectors (2), corresponding respectively to eigenvalues 4, 5 and 4.

If we put $U_i := \operatorname{span}(e_i)$, then $U_1 = \{[x, 0, 0]^T \mid x \in \mathbf{R}\}$ is the x-axis, U_2 is the y-axis and U_3 is the z-axis. All the U_i are 1 dimensional invariant subspaces and we indeed have $\mathbf{R}^3 = U_1 \oplus U_2 \oplus U_3$ (3).

Now, since both e_1 and e_3 have the same eigenvalue 4, we write (4)

$$\mathbf{R}^3 = (U_1 \oplus U_3) \oplus U_2 = \mathrm{E}(4, T) \oplus \mathrm{E}(5, T).$$

Finally, $\mathrm{E}(4,T) = U_1 \oplus U_3 = \{[x,0,z]^T \mid x,z \in \mathbf{R}\}$ is of dimension 2 while $\mathrm{E}(5,T) = U_3$ is of dimension 1. So we indeed have $\dim(\mathbf{R}^3) = 3 = 2+1 = \dim(\mathrm{E}(4,T)) + \dim(\mathrm{E}(5,T))$ (5).

Using Theorem 5.1.22, one can exhibit operators that are not diagonalizable, even for $\mathbf{F} = \mathbf{C}$.

Example 5.1.24. Let $T \in \mathcal{L}(\mathbf{C}^2)$ be defined by $T\left[\begin{smallmatrix} w \\ z \end{smallmatrix}\right] := \left[\begin{smallmatrix} z \\ 0 \end{smallmatrix}\right]$. Is T diagonalizable?

Solution. First, we find all eigenvalues of T. Let $\lambda\left[\begin{smallmatrix} w \\ z \end{smallmatrix}\right] = T\left[\begin{smallmatrix} w \\ z \end{smallmatrix}\right] = \left[\begin{smallmatrix} z \\ 0 \end{smallmatrix}\right]$. Then we have

$$\begin{cases} \lambda w = z, \\ \lambda z = 0. \end{cases}$$

This implies that $\lambda^2 w = 0$ and therefore that $\lambda = 0$, as otherwise $w = z = 0$. Therefore, 0 is the only eigenvalue of T. Moreover, $\mathrm{E}(0,T) = \mathrm{null}(T - 0\,\mathrm{Id}) = \mathrm{null}(T) = \{\left[\begin{smallmatrix} w \\ 0 \end{smallmatrix}\right] \mid w \in \mathbf{C}\}$. We hence have $\dim(\mathbf{C}^2) = 2 > 1 = \dim(\mathrm{E}(0,T))$. We conclude by Theorem 5.1.22 that T is not diagonalizable. □

Theorem 5.1.22 says that T is diagonalizable if, and only if, it has enough linearly independent eigenvectors. Using this, one can show that having enough distinct eigenvalues is a sufficient condition for T being diagonalizable.

Corollary 5.1.25. *Let V be a finite dimensional vector space and let $T \in \mathcal{L}(V)$ be an operator. If T has $\dim(V)$ distinct eigenvalues, then it is diagonalizable.*

Proof. Let $n = \dim(V)$ and let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of T. Then all the $\mathrm{E}(\lambda_i, T)$ have dimension at leat 1 by Proposition 5.1.11 and therefore V is the direct sum of the $\mathrm{E}(\lambda_i, T)$ by Corollary 5.1.17. We conclude by Theorem 5.1.22. □

The converse of the above proposition is false, as demonstrated by $T = \mathrm{Id}_V$ which has only one eigenvalue (1) despite being diagonalizable.

Until now, we have seen conditions that imply (or are equivalent to) diagonalizability. One important consequence of diagonalizability is the following more precise version of the fundamental theorem of linear maps (Theorem 3.0.3).

Lemma 5.1.26. *Let V be a finite dimensional vector space and let $T \in \mathcal{L}(V)$ be an operator. If T is diagonalizable, then $V = \mathrm{range}(T) \oplus \mathrm{null}(T)$.*

Proof. Let $\mathcal{B} = (v_1, \dots, v_n)$ be a basis for which

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}.$$

Up to reordering the v_i, one can suppose that $\lambda_1 = \dots = \lambda_k = 0$ while $\lambda_{r+1}, \dots, \lambda_n$ are non-zero. Then $\mathrm{null}(T) = \mathrm{span}(v_1, \dots, v_k)$ and $\mathrm{range}(T) = \mathrm{span}(v_{k+1}, \dots, v_n)$. □

Can you find an operator T such that $\mathrm{range}(T) \oplus \mathrm{null}(T) \subsetneq V$? Hint: T cannot be diagonalizable.

Infinite dimensional spaces

If T is an operator on an infinite dimensional vector space V, then the equivalences 2 $\iff$ 3 $\iff$ 4 of Theorem 5.1.22 are still true. Some people use them to define the diagonalizability of T. That is, $T \in \mathcal{L}(V)$ is said to be **diagonalizable** if there exists a basis of V consisting of eigenvectors of T. While the infinite dimensional situation is much more complex than the finite dimensional one, some results remain true, as for example Lemma 5.1.26.

5.1.4 Nilpotent operators

In this subsection, we introduce a new kind of operators that have a special relationship with the eigenvalue 0. This definition will be useful to show some of the forthcoming results.

Definition 5.1.27.

An operator $T \in \mathcal{L}(V)$ is **nilpotent** if there exists $k \in \mathbf{N}$ such that $T^k = 0_{\mathcal{L}(V)}$.

The 0 operator is of course nilpotent, but it is easy to find other examples.

Example 5.1.28. The differentiation operator $D \in \mathcal{L}(\mathcal{P}_n(\mathbf{R}))$ defined by $Dp := p'$ is nilpotent. Indeed, $D^n p = p^{(n)} = 0$ for every polynomial of degree at most n.

Example 5.1.29. For $\lambda \in \mathbf{F}$, define an operator $T \in \mathcal{L}(\mathbf{F}^2)$ by

$$Tv = \begin{bmatrix} 0 & \lambda \\ 0 & 0 \end{bmatrix} v.$$

If $\lambda = 1$, this is the operator from Example 5.1.24. Using the matrix, one can check that $T^2 = 0$, and hence that T is nilpotent.

More generally, if $T \in \mathcal{L}(\mathbf{F}^n)$ is represented by a matrix with zeros on and below the diagonal, it is nilpotent. In fact, we will soon show that all nilpotent operators appear in this form. In order to do that, we will need a few technical lemmas about nullspaces.

Lemma 5.1.30. *Let V be a vector space and let $T \in \mathcal{L}(V)$ be an operator.*

1. $\{0\} = \operatorname{null}(T^0) \subseteq \operatorname{null}(T^1) \subseteq \cdots \subseteq \operatorname{null}(T^k) \subseteq \operatorname{null}(T^{k+1}) \subseteq \cdots$;
2. *If there exists m with $\operatorname{null}(T^m) = \operatorname{null}(T^{m+1})$, then $\operatorname{null}(T^m) = \operatorname{null}(T^n)$ for all $n \geq m$.*

Proof. For the first assertion, suppose that $v \in \operatorname{null}(T^k)$ for some non-negative integer k. Then $T^{k+1}v = T(T^k v) = T0 = 0$ and therefore $v \in \operatorname{null}(T^{k+1})$ as desired.

For the second assertion, we want to prove that $\operatorname{null}(T^{m+k}) = \operatorname{null}(T^{m+k+1})$ for every non-negative integer k. We already know that the left to right inclusion is true. To prove the other inclusion, let v be in $\operatorname{null}(T^{m+k+1})$. Then $T^{m+1}(T^k v) = T^{m+k+1}v = 0$ and thus $T^k v$ is in $\operatorname{null}(T^{m+1}) = \operatorname{null}(T^m)$. Therefore, $T^{m+k}v = T^m(T^k v) = 0$, which shows that $v \in \operatorname{null}(T^{m+k})$. □

The above result is particularly useful for finite dimensional spaces. Indeed, in this case we have the following.

Lemma 5.1.31. *Let V be a finite dimensional vector space. Then* $\mathrm{null}(T^{\dim(V)}) = \mathrm{null}(T^{\dim(V)+1}) = \mathrm{null}(T^{\dim(V)+2}) = \cdots$.

Proof. By the first assertion of the last lemma, we have

$$0 \leq \dim(\mathrm{null}(T)) \leq \cdots \leq \dim(\mathrm{null}(T^{\dim(V)})) \leq \dim(\mathrm{null}(T^{\dim(V)+1})) \leq \dim(V).$$

By the pigeonhole principle, at least one of these inequalities is an equality. Therefore, there exists $m \leq \dim(V)$ with $\dim(\mathrm{null}(T^m)) = \dim(\mathrm{null}(T^{m+1}))$. We conclude using the second assertion of Lemma 5.1.30. ☐

To go further

Results similar to Lemmas 5.1.30 and 5.1.31 hold for $\mathrm{range}(T^k)$, but with the inclusions reversed. More precisely, for any vector space T and any operator, one always have

$$V = \mathrm{range}(T^0) \supseteq \mathrm{range}(T) \supseteq \mathrm{range}(T^2) \supseteq \cdots \supseteq \mathrm{range}(T^k) \subseteq \mathrm{range}(T^{k+1}) \supseteq \cdots$$

Moreover, f there exists m with $\mathrm{range}(T^m) = \mathrm{range}(T^{m+1})$, then $\mathrm{range}(T^m) = \mathrm{range}(T^n)$ for all $n \geq m$. Finally, if V is finite dimensional, then $\mathrm{range}(T^{\dim(V)}) = \mathrm{range}(T^m)$ for all $m \geq \dim(V)$. The proofs of these statements are similar to the proofs of Lemmas 5.1.30 and 5.1.31.

Using Lemma 5.1.31, one obtain a variations of Lemma 5.1.26.

Proposition 5.1.32. *Let V be a finite dimensional vector space and let T be an operator on V. Then*

$$V = \mathrm{null}(T^{\dim(V)}) \oplus \mathrm{range}(T^{\dim(V)}).$$

Proof. Let $n = \dim(V)$. We first prove that the sum is direct, that is:

$$\mathrm{null}(T^n) \cap \mathrm{range}(T^n) = \{0\}. \tag{5.4}$$

Suppose v is in the intersection $\mathrm{null}(T^n) \cap \mathrm{range}(T^n)$. Then $T^n v = 0$ and there exists $u \in V$ with $T^n u = v$. If we apply T^n to both sides of the last equation, we obtain $T^{2n}u = T^n v = 0$. Therefore u is in $\mathrm{null}(T^{2n}) = \mathrm{null}(T^n)$ and $v = T^n u = 0$.

Equation (5.4) implies that $\mathrm{null}(T^n) + \mathrm{range}(T^n)$ is a direct sum. By Theorem 3.0.3 we have

$$\dim\big(\mathrm{null}(T^n) \oplus \mathrm{range}(T^n)\big) = \dim\big(\mathrm{null}(T^n)\big) + \dim\big(\mathrm{range}(T^n)\big) = \dim(V)$$

and we hence conclude that $V = \mathrm{null}(T^{\dim(V)}) \oplus \mathrm{range}(T^{\dim(V)})$ as desired. ☐

Proposition 5.1.33. *Let V be a finite dimensional vector space and let $T \in \mathcal{L}(V)$ be a nilpotent operator. Then $T^{\dim(V)} = 0$.*

Proof. Since T is nilpotent, there exists $k \in \mathbf{N}$ with $T^k = 0$, and thus $\operatorname{null}(T^k) = V$. If $k \leq \dim(V)$, then $\operatorname{null}(T^{\dim(V)}) = 0$ by Lemma 5.1.30, while if $k \geq \dim(V)$, then $\operatorname{null}(T^{\dim(V)})$ is still 0, but this time by Lemma 5.1.31. □

As announced after Example 5.1.29, nilpotent operators have a special matrix representation.

Proposition 5.1.34. *Let V be a finite dimensional vector space and let $T \in \mathcal{L}(V)$ be an operator. Then T is nilpotent if and only if there exists a basis $\mathcal{B}$ of V such that*

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{bmatrix} 0 & * & \cdots & * \\ \vdots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & * \\ 0 & \cdots & \cdots & 0 \end{bmatrix}.$$

Proof. "$\Rightarrow$" We have $\operatorname{null}(T) \subseteq \operatorname{null}(T^2) \subseteq \cdots \subseteq \operatorname{null}(T^n) = V$. Moreover, since T is nilpotent it is not injective. In particular, $\operatorname{null}(T) \neq \{0\}$. So we can start with a basis $\mathcal{B}_1$ of $\operatorname{null}(T)$ and extend it to a basis $\mathcal{B}_2$ of $\operatorname{null}(T^2)$ and continue step by step to obtain a basis $\mathcal{B} = (v_1, \dots, v_n)$ of V. By the above, v_1 is in $\operatorname{null}(T)$. Therefore, $Tv_1 = 0$ and hence the first column of $[T]_{\mathcal{B}}^{\mathcal{B}}$ contains only 0. More generally, v_k is in $\operatorname{null}(T^k)$. Therefore, Tv_k is in $\operatorname{null}(T^{k-1}) \subseteq \operatorname{span} v_1, \dots, v_{k-1}$. In other words, $Tv_k = a_{1,k}v_1 + \cdots + a_{k,k-1}v_{k-1}$ for some $a_{j,k} \in \mathbf{F}$. That is, the k^{th} column of $[T]_{\mathcal{B}}^{\mathcal{B}}$ has zeroes on the diagonal and below.

"$\Leftarrow$" Suppose that $A = [T]_{\mathcal{B}}^{\mathcal{B}}$ is an upper triangular $n \times n$ matrix with 0 on the diagonal. Then A^n is the zero matrix, and therefore T^n is the zero operator. □

When $\mathbf{F} = \mathbf{C}$, nilpotent operators can be characterized by their eigenvalues.

Proposition 5.1.35. *Let V be a vector space and let $T \in \mathcal{L}(V)$ be an operator.*

1. *If T is nilpotent, then 0 is an eigenvalue of T, and T has no other eigenvalues;*
2. *If $\mathbf{F} = \mathbf{C}$, V is finite dimensional and 0 is the only eigenvalue of T, then T is nilpotent.*

Proof. 1. If T is nilpotent, then $T^k = 0$ for some $k \in \mathbf{N}$, so T is not injective and hence 0 is an eigenvalue. Now, suppose that λ is an eigenvalue of T with eigenvector $v \neq 0$. Then $0 = T^k v = \lambda^k v$. We conclude that $\lambda^k = 0$ and hence that $\lambda = 0$.

2. The proof is postponed to Proposition 5.1.47. □

5.1.5 Generalized eigenvectors

At the end of the chapter on linear maps, we asked

Major Question 3.1.26.

Given V and $T \in \mathcal{L}(V)$, find a basis $\mathcal{B}$ such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is "as simple as possible" (for example: diagonal, upper triangular, with many zeroes, ...).

A first answer to this question was given by Theorem 5.1.22 which asserts that T is diagonalizable if and only if there exists a basis $\mathcal{B}$ of V consisting of eigenvectors of T, if and only if $\dim(V) = \sum_{i=1}^{m} \dim(\mathrm{E}(\lambda_i, T))$, where the sum is taken over all the eigenvalues of T. If T does not have enough eigenvectors, then it is not diagonalizable. But there is still hope we can find a basis $\mathcal{B}$ consisting "almost eigenvectors" such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is "almost diagonal". See Proposition 5.1.50 for an intermediate result in this direction.

Definition 5.1.36.

Let V be a vector space, let $T \in \mathcal{L}(V)$ be an operator and let $\lambda \in \mathbf{F}$ be an eigenvalue of T. A non-zero vector v is a **generalized eigenvector** of T corresponding to λ if there exists a positive integer j such that $(T - \lambda \operatorname{Id}_V)^j v = 0$.
For $\lambda \in \mathbf{F}$ (not necessarily an eigenvalue), its **generalized eigenspace** is the set

$$\mathrm{G}(\lambda, T) := \{v \in V \mid \exists j, (T - \lambda \operatorname{Id}_V)^j v = 0\}.$$

Observe that given a generalized eigenvector v, one can always obtain an eigenvector w by applying $T - \lambda \operatorname{Id}_V$ to v enough time.

Lemma 5.1.37. *Let V be a vector space and let $T \in \mathcal{L}(V)$ be an operator. Let v be a generalized eigenvector corresponding to the eigenvalue $\lambda \in \mathbf{F}$. Then there exists $j \in \mathbf{N}$ such that $w := (T - \lambda \operatorname{Id}_V)^j v$ is an eigenvector of T corresponding to λ.*

Proof. Let m be the smallest non-negative integer such that $(T - \lambda \operatorname{Id}_V)^m v \neq 0$. Since $v \neq 0$, such an m always exists. Then $w := (T - \lambda \operatorname{Id}_V)^m v$ is non-zero and $(T - \lambda \operatorname{Id}_V) w = 0$ by minimality of m. □

The following lemma is trivially true when λ is an eigenvalue, but still hold for general $\lambda \in \mathbf{F}$.

Lemma 5.1.38. *For V a vector space, $T \in \mathcal{L}(V)$ and operator and $\lambda \in \mathbf{F}$ (not necessarily an eigenvalue), we have*

$$\mathrm{G}(\lambda, T) = \{0\} \sqcup \{v \in V \mid v \text{ is a generalized eigenvector of } T \text{ corresponding to } \lambda\}.$$

Proof. If λ is an eigenvalue, the statement is clear.
If λ is not an eigenvalue, then $\mathrm{G}(\lambda, T)$ always contains $\{0\}$. So we only need to prove that if λ is not an eigenvalue then T has no generalized eigenvector. We do the proof by contraposition. Suppose that T has at least one generalized eigenvector $v \neq 0$ corresponding to λ. Then by the above remark, there exists m such that $w := (T - \lambda \operatorname{Id}_V)^m v$ is an eigenvector, and hence λ is an eigenvalue. □

In the last section, we proved some results about nullspaces. As an interesting consequence of Lemma 5.1.31, we show that generalized eigenspaces are indeed subspaces of V.

Proposition 5.1.39. *Let V be a finite dimensional vector space and let $T \in \mathcal{L}(V)$ be an operator. Then for all $\lambda \in \mathbf{F}$*

$$\mathrm{G}(\lambda, T) = \operatorname{null}(T - \lambda \operatorname{Id}_V)^{\dim(V)}.$$

In particular, $\mathrm{G}(\lambda, T)$ *is a subspace of V.*

Proof. It follows from the definition that we have $\mathrm{G}(\lambda, T) = \bigcup_{j \geq 1} \operatorname{null}(T - \lambda \operatorname{Id}_V)^j$. Using Lemmas 5.1.30 and 5.1.31 we have $\bigcup_{j \geq 1} \operatorname{null}(T - \lambda \operatorname{Id}_V)^j = \operatorname{null}(T - \lambda \operatorname{Id}_V)^{\dim(V)}$ which concludes the proof. □

As we just saw, $\mathrm{G}(\lambda, T)$ is a subspace of V and we have $\mathrm{E}(\lambda, T) \subseteq \mathrm{G}(\lambda, T) \subseteq V$. This explains the name of generalized eigenspace. If v is an eigenvector, then the corresponding eigenvalue λ is uniquely determined by the equation $Tv = \lambda v$. For generalized eigenvectors it is priori not obvious that the corresponding eigenvalue is unique. Fortunately, this is the case.

Lemma 5.1.40. *Let V be a finite dimensional vector space and let $T \in \mathcal{L}(V)$ be an operator. Then each generalized eigenvector corresponds to a unique eigenvalue.*

Proof. Suppose that v is a generalized eigenvector of T corresponding to eigenvalues λ and μ. We need to show that $\lambda = \mu$. Let m be the smallest positive integer such that $(T - \mu \operatorname{Id}_V)^m v = 0$, and let $n = \dim(V)$. Then

$$\begin{aligned} 0 &= (T - \lambda \operatorname{Id}_V)^n v \\ &= ((T - \mu \operatorname{Id}_V) + (\mu - \lambda) \operatorname{Id}_V) v \\ &= \sum_{k=0}^{n} b_k (\mu - \lambda)^{n-k} (T - \mu \operatorname{Id}_V)^k v, \end{aligned}$$

where $b_0 = 1$ and the values of the other binomial coefficients b_k do not matter. If we apply the operator $(T - \mu \operatorname{Id}_V)^{m-1}$ to both sides of the above equation we obtain

$$0 = (\mu - \lambda)^n (T - \mu \operatorname{Id}_V)^{m-1} v.$$

By minimality of m, $(T - \mu \operatorname{Id}_V)^{m-1} v \neq 0$ and hence $\mu = \lambda$. □

Example 5.1.41. Let $T \in \mathcal{L}(\mathbf{C}^3)$ be the operator defined by $T \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} := \begin{bmatrix} 4z_2 \\ 0 \\ 5z_3 \end{bmatrix}$. Let $\mathcal{E} = (e_1, e_2, e_3)$ be the standard basis of $\mathbf{C}^3$. Then

$$A := [T]_{\mathcal{E}}^{\mathcal{E}} = \begin{bmatrix} 0 & 4 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

Then the eigenvalues of T are 0 and 5 (see Section 5.2 for how to easily compute them). Let us compute the eigenspaces and the generalized eigenspaces of the eigenvalues.

Eigenspace and generalized eigenspace for $\lambda = 0$. We have

$$A - 0\,\mathrm{Id} = A, \qquad (A - 0\,\mathrm{Id})^3 = A^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5^3 \end{bmatrix}.$$

Therefore,

$$\mathrm{E}(0,T) = \operatorname{null} T = \left\{ \begin{bmatrix} z_1 \\ 0 \\ 0 \end{bmatrix} \middle| \, z_1 \in \mathbf{C} \right\}, \quad \mathrm{G}(0,T) = \operatorname{null} T^3 = \left\{ \begin{bmatrix} z_1 \\ z_2 \\ 0 \end{bmatrix} \middle| \, z_1, z_2 \in \mathbf{C} \right\}.$$

So $\mathrm{E}(0,T)$ is of dimension 1 and hence a strict subspace of $\mathrm{G}(0,T)$ which is of dimension 2.

Eigenspace and generalized eigenspace for $\lambda = 5$. We have

$$A - 5\,\mathrm{Id} = \begin{bmatrix} -5 & 4 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad (A - 5\,\mathrm{Id})^3 = A^3 = \begin{bmatrix} -5^3 & 300 & 0 \\ 0 & -5^3 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Therefore,

$$\mathrm{E}(5,T) = \operatorname{null} T = \left\{ \begin{bmatrix} 0 \\ 0 \\ z_3 \end{bmatrix} \middle| \, z_3 \in \mathbf{C} \right\}, \qquad \mathrm{G}(0,T) = \operatorname{null} T = \left\{ \begin{bmatrix} 0 \\ 0 \\ z_3 \end{bmatrix} \middle| \, z_3 \in \mathbf{C} \right\}.$$

So $\mathrm{E}(5,T) = \mathrm{G}(5,T)$ is of dimension 1.

Altogether, we have

$$\mathbf{C}^3 = \mathrm{G}(0,T) \oplus \mathrm{G}(5,T) \supsetneq \mathrm{E}(0,T) \oplus \mathrm{E}(5,T).$$

Some of the results of Section 5.1.2 for eigenvectors still hold for generalized eigenvectors.

Lemma 5.1.42. *Let V be a finite dimensional vector space and let $T \in \mathcal{L}(V)$ be an operator. Suppose that v_1 and v_2 are two generalized eigenvectors, corresponding to eigenvalues λ_1 and λ_2 respectively. If $\lambda_1 \neq \lambda_2$ the vectors v_1 and v_2 are linearly independent.*

Proof. Let μ_1 and μ_2 be two scalars such that

$$0 = \mu_1 v_1 + \mu_2 v_2. \tag{5.5}$$

We want to show that $\mu_1 = \mu_2 = 0$. Let $n = \dim(V)$, let $k := \max\{l | v_1 \notin \operatorname{null}(T - \lambda_1 \mathrm{Id}_V)^l\}$ and let $w := (T - \lambda_1 \mathrm{Id}_V)^k v_1 \neq 0$. By maximality of k we have $(T - \lambda_1 \mathrm{Id}_V) w = 0$. Let $f(t) = (t - \lambda_1)^k$ and let $g(t) = (t - \lambda_2)^n$. Both are polynomials in $\mathcal{P}(\mathbf{F})$. Now, we apply $f(T)g(T) = g(T)f(T)$ to Equation (5.5) and obtain

$$\begin{aligned} 0 &= \mu_1 (T - \lambda_2 \mathrm{Id}_V)^n (T - \lambda_1 \mathrm{Id}_V)^k v_1 + \mu_2 (T - \lambda_1 \mathrm{Id}_V)^k (T - \lambda_2 \mathrm{Id}_V)^n v_2 \\ &= \mu_1 (T - \lambda_2 \mathrm{Id}_V)^n w + 0. \end{aligned}$$

Since $w \neq 0$ is an eigenvector corresponding to λ_1, and since $\lambda_1 \neq \lambda_2$, w is not a generalized eigenvector corresponding to λ_2. We conclude that $(T - \lambda_2 \operatorname{Id}_V)^n w \neq 0$ and hence that $\mu_1 = 0$. A similar argument shows that $\mu_2 = 0$ and therefore that v_1 and v_2 are linearly independent. $\square$

The above Lemma easily generalize to the case of finite families of vectors. The proof is let to the reader.

Theorem 5.1.43.

Let V be a finite dimensional vector space and let $T \in \mathcal{L}(V)$ be an operator. Suppose that $v_1, \dots, v_n$ are n eigenvectors, corresponding to eigenvalues $\lambda_1, \dots, \lambda_n$ respectively. If the λ_i are pairwise distincts, then $v_1, \dots, v_n$ is a linearly independent list.

We have seen that if T is an operator on V, then both range(T) and null(T) are T-invariant. We can generalize this.

Lemma 5.1.44. *Let V be a vector space and let $T \in \mathcal{L}(V)$ be an operator. Let $p(t) \in \mathcal{P}(\mathbf{F})$ be any polynomial. Then both* range$(p(T))$ *and* null$(p(T))$ *are T-invariant.*

Proof. Suppose u belongs to null$(p(T))$. Then $p(T)u = 0$. Thus

$$(p(T))(Tu) = (p(T)T)u = (Tp(T))u = T(p(T)u) = T0 = 0.$$

Hence Tu is in null$(p(T))$. So we have just proved that null$(p(T))$ is invariant under T as desired.

Suppose u belongs to range$(p(T))$. Then there exists $v \in V$ such that $u = p(T)v$. Thus

$$Tu = T(p(T)v) = p(T)(Tv),$$

where we used the fact that for polynomials $p(t)$ and $q(t)$ we always have $p(t)q(t) = q(t)p(t)$. By the above equation, Tu is still in range$(p(T))$, which shows that range$(p(T))$ is T-invariant. $\square$

In Corollary 5.1.17, we showed that $\sum_{i=1}^{m} \operatorname{E}(\lambda, T)$ is always a direct sum, and a subspace of V. In Theorem 5.1.22, we then showed that $\bigoplus_{i=1}^{m} \operatorname{E}(\lambda, T) = V$ if and only T is diagonalizable, if and only if there exists a basis consisting of eigevectors. Theorems 5.1.45 and 5.1.46 will show that even if T is not diagonalizable a similar formula holds for generalized eigenspaces.

Theorem 5.1.45.

Let V be a finite dimensional vector space over $\mathbf{C}$ and let $T \in \mathcal{L}(V)$ be an operator. Then there exists a basis of V consisting of generalized eigenvectors of T.

Proof. We will do an induction on $n = \dim(V)$. Observe that the desired result holds if $n = 0$ as in this (degenerated) case we have no generalized eigenvectors but the empty list is a basis. If $n = 1$, then any non-zero vector is both a basis of V and an eigenvector for T.

Now, suppose that $n \geq 2$ and that the desired result holds for all complex vector spaces of dimension strictly less than n. Let λ be an eigenvalue of T, whose existence is guarantee by Theorem 5.1.13. Applying Proposition 5.1.32 to $T - \lambda \operatorname{Id}_V$ we obtain

$$V = \operatorname{null}(T - \lambda \operatorname{Id}_V)^n \oplus \operatorname{range}(T - \lambda \operatorname{Id}_V)^n.$$

On one hand, every non-zero vector in $\operatorname{null}(T - \lambda \operatorname{Id}_V)^n \neq \{0\}$ is a generalized eigenvector of T. Hence, any basis $\mathcal{B}$ of $\operatorname{null}(T - \lambda \operatorname{Id}_V)^n \neq \{0\}$ consists of generalized eigenvectors of T. On the other hand, $\operatorname{null}(T - \lambda \operatorname{Id}_V)^n \neq \{0\}$ as λ is an eigenvalue of T. Therefore, $\operatorname{range}(T - \lambda \operatorname{Id}_V)^n \neq V$ is of dimension strictly less than n, and is invariant under T (by Lemma 5.1.44 for the polynomial $p(t) = (t - \lambda)^n$). Let $S \in \mathcal{L}(\operatorname{range}(T - \lambda \operatorname{Id}_V)^n)$ be the restriction of T to $\operatorname{range}(T - \lambda \operatorname{Id}_V)^n$. By the induction hypothesis applied to the operator S, there exists a basis D of $\operatorname{range}(T - \lambda \operatorname{Id}_V)^n$ consisting of generalized eigenvectors of S. Since S is the restriction of T, every generalized eigenvectors of S is also a generalized eigenvector of T. Altogether, $\mathcal{B} \cup \mathcal{D}$ is a basis of V consisting of generalized eigenvectors of T. $\square$

The following theorem makes the link between nilpotent operators and generalized eigenspaces. It also shows why generalized eigenspaces are the right enlargement of eigenspaces.

Theorem 5.1.46.

Let V be a finite dimensional vector space over $\mathbf{C}$, and let $T \in \mathcal{L}(V)$ be an operator. Let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues of T. Then

1. *Each of the $\mathrm{G}(\lambda_j, T)$ is a T-invariant subspace;*
2. $V = \mathrm{G}(\lambda_1, T) \oplus \cdots \oplus \mathrm{G}(\lambda_m, T)$;
3. *The operator $T_j := (T - \lambda_j \operatorname{Id}_V)\big|_{\mathrm{G}(\lambda_j, T)}$ is a nilpotent operator on $\mathrm{G}(\lambda_j, T)$.*

Proof. 1. We have $\mathrm{G}(\lambda_i, T) = \operatorname{null}(T - \lambda_i \operatorname{Id}_V)^n$. We conclude by Lemma 5.1.44.

2. Firstly, the sum $\sum_{j=1}^m \mathrm{G}(\lambda_j, T)$ is a direct sum since generalized eigenvectors corresponding to distinct eigenvalues are linearly independent the sum. See the proof of Corollary 5.1.17 for more details. Secondly, the existence of a basis of generalized eigenvectors (Theorem 5.1.45) implies that $V = \mathrm{G}(\lambda_1, T) + \cdots + \mathrm{G}(\lambda_m, T)$.

3. By 1, $T_j := T - \lambda_j \operatorname{Id}$ is an operator on $\mathrm{G}(\lambda_j, T)$. It then follows from Proposition 5.1.39 that $T_j^{\dim(\mathrm{G}(\lambda_j, T))}$ is the zero operator on $\mathrm{G}(\lambda_j, T)$, and hence that T_j is nilpotent. $\square$

As an easy corollary, one can finally shows that an operator whose eigenvalues are all 0 is nilpotent. This finishes the proof of Proposition 5.1.35.

Proposition 5.1.47. *Let V be a finite dimensional vector space over $\mathbf{C}$ and let $T \in \mathcal{L}(V)$ be an operator. Suppose that 0 is the only eigenvalue of T. Then T is nilpotent.*

Proof. Since 0 is the only eigenvalue, $T = T - 0\,\mathrm{Id}_v$ is nilpotent on $V = \mathrm{G}(0,T)$. □

Can you find an example of an operator T on $\mathbf{R}^3$ such that 0 is the only eigenvalue of T, but T is not nilpotent? Hint, by the above proposition, this means that 0 is the only real eigenvalue of T, but that T has non-zero complex eigenvalues.

As another corollary of Theorem 5.1.46 we have the following.

Proposition 5.1.48. *Let V be a finite dimensional vector space over $\mathbf{C}$, and let $T \in \mathcal{L}(V)$ be an operator. Then T is diagonalizable if and only if $\mathrm{E}(\lambda,T) = \mathrm{G}(\lambda,T)$ for all $\lambda \in \mathbf{C}$.*

To go further

Theorems 5.1.45 and 5.1.46 and Propositions 5.1.47 and 5.1.48 are true for all algebraically closed fields $\mathbf{F}$.

Definition 5.1.49.

Let V be a vector space, let $T \in \mathcal{L}(V)$ be an operator and let $\lambda \in \mathbf{F}$ be an eigenvalue of T. The **algebraic multiplicity** of λ in T is defined as

$$m_\lambda = m_\lambda(T) := \dim(\mathrm{G}(\lambda,T)).$$

The **geometric multiplicity** of λ in T is defined as

$$d_\lambda = d_\lambda(T) := \dim(\mathrm{E}(\lambda,T)).$$

Clearly, we always have $d_\lambda \leq m_\lambda$. It also follows from Proposition 5.1.48 that T is diagonalisable if and only if $d_\lambda(T) = m_\lambda(T)$ for all $\lambda \in \mathbf{F}$. The name "geometric multiplicity" is because it is related to $\mathrm{E}(\lambda,T)$, which we think of as some geometric invariant of T. The name "algebraic multiplicity" is because we can compute it directly from a polynomial associated to T. We will talk more about this polynomial in the next section.

Before ending this section, let us mention one more use of Theorem 5.1.46. Namely the fact that for operators on complex vector spaces, one can always find a basis $\mathcal{B}$ for which $[T]_{\mathcal{B}}^{\mathcal{B}}$ is particularly nice.

Proposition 5.1.50. *Let $V \neq \{0\}$ be a finite dimensional vector space over $\mathbf{C}$, and let $T \in \mathcal{L}(V)$ be an operator. Then there exists a basis $\mathcal{B}$ for which $[T]_{\mathcal{B}}^{\mathcal{B}}$ is* diagonal by [triangular blocks]. *More precisely, $[T]_{\mathcal{B}}^{\mathcal{B}}$ is of the form (all matrix entries not shown are* *zero)*

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{bmatrix} \boxed{\begin{matrix} \lambda_1 & & * \\ & \ddots & \\ & & \lambda_1 \end{matrix}} & & & \\ & \boxed{\begin{matrix} \lambda_2 & & * \\ & \ddots & \\ & & \lambda_2 \end{matrix}} & & \\ & & \ddots & \\ & & & \boxed{\begin{matrix} \lambda_m & & * \\ & \ddots & \\ & & \lambda_m \end{matrix}} \end{bmatrix},$$

where the j^{th} diagonal block is an $m_{\lambda_j} \times m_{\lambda_j}$ triangular matrix with λ_j on the diagonal and 0 below the diagonal.

Proof. Let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues of T and let us write $U_j := \mathrm{G}(\lambda_j, T)$. For each $j \in \{1, \dots, m\}$, the restriction $T_j := (T - \lambda_j \operatorname{Id}_V)\big|_{U_j}$ is a nilpotent operator on U_j. So by Proposition 5.1.34 there exists a basis $\mathcal{B}_j$ of U_j such that $[T_j]_{\mathcal{B}_j}^{\mathcal{B}_j}$ is a $m_{\lambda_j} \times m_{\lambda_j}$ matrix with zeros on the diagonal and below the diagonal. This implies that $[T|_{U_j}]_{\mathcal{B}_j}^{\mathcal{B}_j}$ is a $m_{\lambda_j} \times m_{\lambda_j}$ matrix with λ_j on the diagonals and 0 below the diagonal.

$$\Big[(T - \lambda_j \operatorname{Id}_V)\big|_{\mathrm{G}(\lambda_j, T)}\Big]_{\mathcal{B}_j}^{\mathcal{B}_j} = \begin{bmatrix} 0 & * & \cdots & * \\ & \ddots & \ddots & \vdots \\ & & \ddots & * \\ & & & 0 \end{bmatrix}, \qquad \Big[T\big|_{\mathrm{G}(\lambda_j, T)}\Big]_{\mathcal{B}_j}^{\mathcal{B}_j} = \begin{bmatrix} \lambda_j & * & \cdots & * \\ & \ddots & \ddots & \vdots \\ & & \ddots & * \\ & & & \lambda_j \end{bmatrix}.$$

Let $\mathcal{B} := \mathcal{B}_1 \cup \cdots \cup \mathcal{B}_m$. Since V is a direct sum of the generalized eigenspaces, $\mathcal{B}$ is a basis of V. It then follows from the T-invariance of the U_j that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is diagonal by block, with the j^{th} diagonal block being equal to $[T|_{U_j}]_{\mathcal{B}_j}^{\mathcal{B}_j}$. □

In this subsection, we defined generalized eigenvectors and we have seen that V is a direct sum of generalized eigenspaces. Moreover, we also proved that (if $\mathbf{F} = \mathbf{C}$, there exists a basis $\mathcal{B}$ for which $[T]_{\mathcal{B}}^{\mathcal{B}}$ is diagonal by [triangular blocks]. But is is possible to find an even “nicer” (with more zeroes) matrix representation of T. In order to do that, we need to introduce some new tools. The first of them, the characteristic polynomial, will help us to compute eigenvalues.

5.2 Characteristic and minimal polynomials

Standing assumption

In this Section, $V \neq \{0\}$ will always be a finite dimensional non-zero vector space.

The aim of this section is to define powerful tools, called the characteristic polynomial and the minimal polynomials, that will help us to compute eigenvalues of an operator. The definition of the characteristic polynomial of T will use $[T]_{\mathcal{B}}^{\mathcal{B}}$. We hence start by developing an eigen-theory for matrices similar to the one for operator.

5.2.1 Characteristic polynomial for matrices

The eigen-theory for matrices mirrors to the one for operators studied in Section 5.1.2. Let us make that explicit.

Definition 5.2.1.

Let $A \in \mathbf{F}^{n,n}$ be a matrix. We say that $\lambda \in F$ is an **eigenvalue** of A if there exists a non-zero $v \in \mathbf{F}^n$ such that $Av = \lambda v$.
A non-zero vector v satisfying $Av = \lambda v$ is called an **eigenvector** of A, corresponding to eigenvalue λ.
The **eigenspace** of T corresponding to $\lambda \in \mathbf{F}$ is the set

$$\mathrm{E}(\lambda, A) := \{v \in \mathbf{F}^n \mid Av = \lambda v\}.$$

Recall that given a matrix $A \in \mathbf{F}^{n,n}$ we have an operator $\mathrm{L}_A \in \mathcal{L}(\mathbf{F}^n)$

$$\begin{aligned} \mathrm{L}_A \colon \mathbf{F}^n &\longrightarrow \mathbf{F}^n \\ x &\longmapsto Ax. \end{aligned}$$

It directly follows from the definitions that λ is an eigenvalue for A if and only if it is an eigenvalue for L_A. Similar statements hold for eigenvectors and eigenspaces. This implies that results similar to the ones in Section 5.1.2 hold for matrices.

As for operators, it is important to clarify the base field $\mathbf{F}$. The following example is simply Example 5.1.12 rewritten in term of matrices.

Example 5.2.2. Let $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. Then L_A is the operator T from Example 5.1.12, so we know that A has no eigenvalues if $\mathbf{F} = \mathbf{R}$, and has eigenvalues i and $-i$ if $\mathbf{F} = \mathbf{C}$. One can also observe that whenever $\mathbf{F} = \mathbf{R}$, L_A is the rotation by angle $\pi/2$, which explains why it has no eigenvalues.

Using matrices instead of operators allow us to use the determinant. We can hence define

Definition 5.2.3.

Let $A \in \mathbf{F}^{n,n}$ be a square matrix. The **characteristic polynomial** of A is

$$\chi_A(t) := \det(A - t\,\mathrm{Id}_n).$$

If A is a $n \times n$ matrix, then its characteristic polynomial, is a polynomial in t of degree n, with coefficients in $\mathbf{F}$.

The characteristic polynomial is a powerful tool that allows to easily compute the eigenvalues.

Theorem 5.2.4.

Let $A \in \mathbf{F}^{n,n}$ be a square matrix and let $\lambda \in \mathbf{F}$. Then λ is an eigenvalue for A if and only if it is a root of χ_A (that is, if and only if $\chi_A(\lambda) = 0$).

Proof. Analogously to Proposition 5.1.11, λ is an eigenvalue of A if and only if $A - \lambda\,\mathrm{Id}_n$ is not invertible, that is if and only if $\det(A - \lambda\,\mathrm{Id}_n) = 0$. □

Example 5.2.5. Let $A \in \mathbf{F}^{n,n}$ be a square matrix. If A is a diagonal matrix or a triangular matrix (upper triangular, or lower triangular), then the eigenvalues of A are exactly its diagonal coefficients.

Let us revisite Examples 5.2.2 and 5.1.12.

Example 5.2.6. Let $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. Then

$$\chi_A(t) = \det \begin{bmatrix} -t & 1 \\ -1 & -t \end{bmatrix} = t^2 + 1.$$

Therefore, it is clear that A has no eigenvalues if $\mathbf{F} = \mathbf{R}$, and eigenvalues i and $-i$ if $\mathbf{F} = \mathbf{C}$.

The characteristic polynomial is of great help to prove fundamental properties of eigenvalues.

Proposition 5.2.7. *Let $A \in \mathbf{R}^{n,n}$ be a matrix with real entries. If $\lambda \in \mathbf{C}$ is a complex eigenvalue of A, then its conjugate $\overline{\lambda}$ is also an eigenvalue of A.*

Proof. In general, λ is an eigenvalue of A if and only if $\chi_A(\lambda) = 0$, if and only if $\overline{\chi_A(\lambda)} = 0$. But since A has real entries, its characteristic polynomial $\chi_A(t) = a_0 + a_1 t + \dots + a_n t^n$ has real coefficients. Therefore, $\overline{\chi_A(t)} = a_0 + a_1 \overline{t} + \dots + a_n \overline{t}^n$. We conclude that $\overline{\chi_A(\lambda)} = 0$ if and only if $\chi_A(\overline{\lambda}) = 0$, if and only if $\overline{\lambda}$ is an eigenvalue of A. □

Theorem 5.2.8.

Let $A \in \mathbf{C}^{n,n}$. Then

1. $\chi_A(t) = \det(A) + a_1 t + a_2 t^2 + \dots + a_{n-1} t^{n-1} + (-1)^n t^n$, *with* $a_1, \dots, a_{n-1} \in \mathbf{C}$;

2. *$\chi_A(t) = (\lambda_1 - t)(\lambda_2 - t)\cdots(\lambda_n - t)$, where $\lambda_1, \dots, \lambda_n \in \mathbf{C}$ are the $\mathbf{C}$-eigenvalues of A, possibly with repetition.*

Proof. The first assertion comes from the product formula for the determinant. The second assertion, is the fundamental theorem of algebra: an n-degree polynomial with complex coefficients has n roots when counted with multiplicity. □

The characteristic polynomial also helps us to show that two similar matrices have the same eigenvalues. Recall that two matrices $A, B \in \mathbf{F}^{n,n}$ and B are similar if there exists an invertible matrix $P \in \mathbf{F}^{n,n}$ with $A = P^{-1}BP$.

Proposition 5.2.9. *If $A, B \in \mathbf{F}^{n,n}$ are similar matrices, then*

1. $\det(A) = \det(B)$*;*
2. $\chi_A(t) = \chi_B(t)$*.*

Proof. Let P be an invertible matrix such that $A = P^{-1}BP$. Then $\det(A) = \det(P^{-1}BP) = \det(P^{-1})\det(B)\det(P) = \det(B)$.

For the second assertion,

$$\begin{aligned} \chi_A(t) &= \det(A - t\,\mathrm{Id}_n) = \det(P^{-1}BP - t\,\mathrm{Id}_n) \\ &= \det(P^{-1}(B - t\,\mathrm{Id}_n)P) \\ &= \det(B - t\,\mathrm{Id}_n) = \chi_B(t). \end{aligned}$$ □

5.2.2 Characteristic polynomial for operators

We now have everything we need to define the characteristic polynomial of an operator. Indeed, if $T \in \mathcal{L}(V)$ is an operator on a finite dimensional vector space, and $\mathcal{B}$ and C are two basis of V, it follows from Proposition 3.1.25 that the matrices $[T]_{\mathcal{B}}^{\mathcal{B}}$ and $[T]_{\mathrm{C}}^{\mathrm{C}}$ are similar. Therefore, by Proposition 5.2.9, $\det([T]_{\mathcal{B}}^{\mathcal{B}} - t\,\mathrm{Id}_n) = \det([T]_{\mathrm{C}}^{\mathrm{C}} - t\,\mathrm{Id}_n)$.

Definition 5.2.10.

Let $V \neq \{0\}$ be a finite dimensional vector space and let $T \in \mathcal{L}(V)$ be an operator. The **characteristic polynomial** of T is

$$\chi_T(t) := \det([T]_{\mathcal{B}}^{\mathcal{B}} - t\,\mathrm{Id}_n),$$

where B is any basis of V.

Similarly, if $T \in \mathcal{L}(V)$ is an operator on a finite dimensional vector space, using Proposition 5.2.9 one can define its **determinant** by $\det(T) := \det([T]_{\mathcal{B}}^{\mathcal{B}})$ for any basis $\mathcal{B}$ of V.

Using our work on matrices, one can show the analogous of Theorem 5.2.4.

Theorem 5.2.11.

Let $V \neq \{0\}$ be a finite dimensional space, let $T \in \mathcal{L}(V)$ be an operator and let $\lambda \in \mathbf{F}$. Then λ is an eigenvalue for A if and only if it is a root of χ_A (that is, if and only if $\chi_A(\lambda) = 0$).

Proof. Let B be a basis for V. Then λ is an eigenvalue of T if and only if $T - \lambda \operatorname{Id}_V$ is not invertible, if and only if

$$[T_\lambda \operatorname{Id}_V]_{\mathcal{B}}^{\mathcal{B}} = [T]_{\mathcal{B}}^{\mathcal{B}} - \lambda \operatorname{Id}_n$$

is not invertible, if and only if λ is an eigenvalue of $[T]_{\mathcal{B}}^{\mathcal{B}}$. We conclude by Theorem 5.2.4. □

Corollary 5.2.12. *Let V be a finite dimensional vector space over $\mathbf{C}$ and let $T \in \mathcal{L}(V)$ be an operator. Then T is nilpotent if and only if $\chi_T(t) = (-t)^n$.*

Proof. By the previous theorem, $\chi_T(t) = (-t)^n$ if and only if 0 is the only eigenvalue of T. By Proposition 5.1.35 this is equivalent to T being nilpotent. □

So the eigenvalues of T are exactly the eigenvalues of $[T]_{\mathcal{B}}^{\mathcal{B}}$, where $\mathcal{B}$ is any basis of V. How about the eigenvectors of T? Recall that given a basis $\mathcal{B}$ of a n dimensional vector space V, we have an isomorphism

$$\begin{aligned} [\,\cdot\,]_{\mathcal{B}} \colon V &\longrightarrow \mathbf{F}^n \\ v &\longmapsto [v]_{\mathcal{B}}. \end{aligned}$$

Theorem 5.2.13.

Let $V \neq \{0\}$ be a n dimensional vector space, let $T \in \mathcal{L}(V)$ and let $\mathcal{B}$ be a basis of V. Then for any $v \in V$, the vector $v \in V$ is an eigenvector of $T \in \mathcal{L}(V)$ (corresponding to eigenvalue λ) if and only if $[v]_{\mathcal{B}} \in \mathbf{F}^n$ is an eigenvector of $[T]_{\mathcal{B}}^{\mathcal{B}} \in \mathbf{F}^{n,n}$ (corresponding to eigenvalue λ).

Proof. Recall we have a commutative diagram

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[d, "{[\,\cdot\,]_{\mathcal{B}}}"'] & V \arrow[d, "{[\,\cdot\,]_{\mathcal{B}}}"] \\ \mathbf{F}^n \arrow[r, "{[T]_{\mathcal{B}}^{\mathcal{B}}}"'] & \mathbf{F}^n \end{tikzcd}$$

where $[\,\cdot\,]_{\mathcal{B}} \colon V \to \mathbf{F}^n$ is an isomorphism. For every $v \in V$ we have $[Tv]_{\mathcal{B}} = [T]_{\mathcal{B}}^{\mathcal{B}}[v]_{\mathcal{B}}$. Therefore, $Tv = \lambda v$ if and only if $[Tv]_{\mathcal{B}} = [\lambda v]_{\mathcal{B}}$ if and only if $[T]_{\mathcal{B}}^{\mathcal{B}}[v]_{\mathcal{B}} = \lambda[v]_{\mathcal{B}}$, which finishes the proof. □

Example 5.2.14. Let $T \in \mathcal{L}(\mathcal{P}_2(\mathbf{R}))$ be defined by $T(f(x)) := f(x) + (x+1)f'(x)$. Find the eigenvalues and eigenvectors of T.

Solution. Let $\mathcal{B} = \{1, x, x^2\}$ be the standard basis of $\mathcal{P}_2(\mathbf{R})$. We compute:

$$\begin{aligned} T(1) &= 1, \\ T(x) &= x + (x+1) \cdot 1 = 1 + 2x, \\ T(x^2) &= x^2 + (x+1) \cdot 2x = 2x + 3x^2. \end{aligned}$$

We hence have

$$A := [T]_{\mathcal{B}}^{\mathcal{B}} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}.$$

The characteristic polynomial of T (and hence of A) is given by the following easy computation

$$\chi_T(t) = \det(A - t\,\mathrm{Id}_3) = \det \begin{bmatrix} 1-t & 1 & 0 \\ 0 & 2-t & 2 \\ 0 & 0 & 3-t \end{bmatrix} = (1-t)(2-t)(3-t).$$

We conclude that the eigenvalues of T (and of A) are 1, 2 and 3. Finally, the eigenvectors of T are given the eigenvectors of A. Therefore, to find it we need to solve system of linear equations.

Eigenspace for $\lambda = 1$. We have $\mathrm{E}(1, [T]_{\mathcal{B}}^{\mathcal{B}}) = \mathrm{E}(1, A) = \mathrm{null}(A - \mathrm{Id}_3)$ and hence

$$\begin{aligned} \mathrm{E}(1, T) &= \{z \in \mathbf{R}^3 \mid (A - \mathrm{Id}_3)z = 0\} \\ &= \left\{ \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} \;\middle|\; \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\} \\ &= \left\{ \begin{bmatrix} a \\ 0 \\ 0 \end{bmatrix} \;\middle|\; a \in \mathbf{R} \right\}. \end{aligned}$$

Therefore, $\mathrm{E}(1, T) = \{a \mid a \in \mathbf{R}\}$. In other words, the eigenvectors of T corresponding to the eigenvalue 1 are exactly the non-zero constant polynomials.

Eigenspace for $\lambda = 2$. This time we solve

$$\begin{aligned} \mathrm{E}(2, T) &= \{z \in \mathbf{R}^3 \mid (A - 2\,\mathrm{Id}_3)z = 0\} \\ &= \left\{ \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} \;\middle|\; \begin{bmatrix} -1 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\} \\ &= \left\{ \begin{bmatrix} a \\ a \\ 0 \end{bmatrix} \;\middle|\; a \in \mathbf{R} \right\}. \end{aligned}$$

Therefore, $\mathrm{E}(2, T) = \{a + ax \mid a \in \mathbf{R}\}$ and the eigenvectors of T corresponding to the eigenvalue 2 are $\{a + ax \mid a \in \mathbf{R}, a \neq 0\}$.

Eigenspace for $\lambda = 3$. Finally we solve

$$\begin{aligned}
\mathrm{E}(3, T) &= \{z \in \mathbf{R}^3 \mid (A - 3\,\mathrm{Id}_3)z = 0\} \\
&= \left\{ \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} \,\middle|\, \begin{bmatrix} -2 & 1 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\} \\
&= \left\{ \begin{bmatrix} a \\ 2a \\ a \end{bmatrix} \,\middle|\, a \in \mathbf{R} \right\}.
\end{aligned}$$

Therefore, $\mathrm{E}(2, T) = \{a + 2ax + ax^2 \mid a \in \mathbf{R}\}$ and the eigenvectors of T corresponding to the eigenvalue 3 are $\{a + 2ax + ax^2 \mid a \in \mathbf{R}, a \neq 0\}$. □

The characteristic polynomial behaves well with respect to restriction to invariant subspaces.

Lemma 5.2.15. *Let V be a finite dimensional vector space and let $T \in \mathcal{L}(V)$ be an operator. If $W \subseteq V$ is a T-invariant subspace, then $\chi_{T|_W}(t)$ divides $\chi_T(t)$.*

Proof. Let $\mathcal{B} = (w_1, \dots, w_m)$ be a basis of W and extend it to $\mathcal{D} = (w_1, \dots, w_m, v_1, \dots, v_{n-m})$ a basis of V, where $m = \dim(W)$ and $n = \dim(V) \geq m$. Write $A = [T]_{\mathcal{D}}^{\mathcal{D}}$ and $B = [T|_W]_{\mathcal{B}}^{\mathcal{B}}$. Then it follows from the definition of matrix representations that

$$A = \begin{bmatrix} B & C \\ 0 & D \end{bmatrix}$$

for some $C \in \mathbf{F}^{m,n-m}$ and $D \in \mathbf{F}^{n-m,n-m}$. Then $\chi_A(t) = \det(A - t\,\mathrm{Id}_n) = \det(B - t\,\mathrm{Id}_m) \cdot \det(D - t\,\mathrm{Id}_{n-m}) = \chi_B(t)\chi_D(t)$. □

While we were able to define the characteristic polynomial for a finite dimensional space over $\mathbf{F}$, for the following result we will need that $\mathbf{F} = \mathbf{C}$. This result explains the name "algebraic mutliplicity" for $m_\lambda(T)$.

Proposition 5.2.16. *Let V be a finite dimensional vector space over $\mathbf{C}$ and let $T \in \mathcal{L}(V)$ be an operator. Let $\lambda_1, \dots, \lambda_k$ be all the distinct eigenvalues of T. Then $\chi_T(t) = (\lambda_1 - t)^{m_{\lambda_1}} \cdots (\lambda_k - t)^{m_{\lambda_k}}$.*

Proof. This directly follows from the matrix representation of T given in Proposition 5.1.50. □

Theorem 5.2.17 (Cayley-Hamilton theorem)**.**

Let V be a finite dimensional vector space. Then $\chi_T(T) = 0_{\mathcal{L}(V)}$.

Proof. We first prove the theorem for $\mathbf{F} = \mathbf{C}$. Let $\lambda_1, \dots, \lambda_k$ be the eigenvalue of T. Then $V = \bigoplus_{j=1}^{k} \mathrm{G}(\lambda_j, T)$ by Theorem 5.1.46. Let v be any vector in V. Then $v = v_1 + \dots + v_k$ for some $v_j \in \mathrm{G}(\lambda_j, T)$. Using that $(T - \lambda_j \operatorname{Id}_V)|_{\mathrm{G}(\lambda_j,T)}$ is nilpotent on $\mathrm{G}(\lambda_j, T)$ we have $(T - \lambda_j \operatorname{Id}_V)^{m_{\lambda_j}(T)} v_j = 0$ for all $j \in \{1, \dots, k\}$. It follows that

$$\begin{aligned}\chi_T(T)v_1 &= \big((T - \lambda_2 \operatorname{Id}_V)^{m_{\lambda_2}} \cdots (T - \lambda_k \operatorname{Id}_V)^{m_{\lambda_k}}\big)(T - \lambda_1 \operatorname{Id}_V)^{m_{\lambda_1}} v_1 \\ &= \big((T - \lambda_2 \operatorname{Id}_V)^{m_{\lambda_2}} \cdots (T - \lambda_k \operatorname{Id}_V)^{m_{\lambda_k}}\big)0 = 0.\end{aligned}$$

And similarly, $\chi_T(T)v_j = 0$ for all $j \in \{1, \dots, k\}$. Therefore,

$$\chi_T(T)v = \big(\chi_T(T)\big)(v_1 + \dots + v_k) = 0 + \dots + 0 = 0.$$

We now prove the case $\mathbf{F} = \mathbf{R}$. Let T be an operator on a real vector space V of dimension n. Fix a basis $\mathcal{B}$ of V and let $A = [T]_{\mathcal{B}}^{\mathcal{B}} \in \mathbf{R}^{n,n}$. The trick is that a $n \times n$ matrix with real coefficients is also a $n \times n$ matrix with complex coefficients, so we can apply the Cayley-Hamilton theorem to the complex matrix A. Here are the details. One can define an operator S on $\mathbf{C}^n$ such that the matrix representation of S, with respect to the standard basis of $\mathbf{C}$, is also A. That is, $S = \mathrm{L}_A \colon \mathbf{C}^n \to \mathbf{C}^n$, $v \mapsto Av$. Then $\chi_T(t) = \det(A - t \operatorname{Id}_n) = \chi_S(t)$ and $\chi_T(T) = \chi_T(A) = \chi_S(A) = \chi_S(S) = 0$. $\square$

To go further

Corollary 5.2.12 and Proposition 5.2.16 are true for all algebraically closed fields (see the remark after Theorem 5.1.13).
The Cayley-Hamilton theorem is true for all fields $\mathbf{F}$, and even for $\mathbf{Z}$ (or even more generally: for *commutative rings* $R, +, -, \cdot, 0, 1$, which are "fields without division").
The proof of the Cayley-Hamilton for $\mathbf{C}$ works for all algebraically closed fields. Now, if $\mathbf{F}$ is any field, there always exist an algebraic closed field $\overline{\mathbf{F}} \subseteq \mathbf{F}$ and we can apply the matrix-trick from the proof to the pair $\mathbf{F} \subseteq \overline{\mathbf{F}}$. Finally, if R is a commutative ring (think $\mathbf{Z}$), we still have the existence of and algebraic closed field $\overline{\mathbf{F}} \supseteq R$ (think $\mathbf{C} \supseteq \mathbf{Z}$) and the matrix-trick still works.

5.2.3 Minimal polynomial

The characteristic polynomial is a powerful tool that encode informations about eigenvalues. But it is sometimes not precise enough. For example, the following two matrices have the same characteristic polynomial $(1-t)^2$ despite being quite different:

$$\operatorname{Id}_2 = \begin{bmatrix} 1 & \\ & 1 \end{bmatrix}, \qquad A = \begin{bmatrix} 1 & 1 \\ & 1 \end{bmatrix}.$$

Indeed, we have $\mathbf{C}^2 = \mathrm{E}(1, \operatorname{Id}) \supsetneq \mathrm{E}(1, A) = \{ \left[\begin{smallmatrix} z \\ 0 \end{smallmatrix}\right] \mid z \in \mathbf{C} \}$.

In this subsection, we will introduce another polynomial naturally associated to an operator, which will be able to distinguish between Id_2 and A.

Definition 5.2.18.

A non-zero polynomial $p(t) = a_n t^n + \dots + a_1 t + a_0$ ($a_n \neq 0$) is **monic** if the leading coefficient is 1, that is if $a_n = 1$.

Example 5.2.19. If $T \in \mathcal{L}(V)$ is an operator on finite dimensional vector space $V \neq 0$, then $(-1)^{\dim(V)} \chi_T(t)$ is monic.

Proposition 5.2.20. *Let $V \neq \{0\}$ be a finite dimensional non-zero vector space and let $T \in \mathcal{L}(V)$ be an operator. Then there exists a unique monic polynomial $M_T(t)$ such that:*

1. $M_T(T) = 0_{\mathcal{L}(V)}$*;*
2. $M_T(t)$ *has the smallest degree among non-zero polynomials satisfying 1.*

Proof. Let $n = \dim(V)$. Then $\dim(\mathcal{L}(V)) = n^2$ and hence the family $\mathrm{Id}_V, T \dots, T^{n^2}$ is linearly dependent in $\mathcal{L}(V)$. Let $m := \min\{k \mid T^k \in \operatorname{span}(\mathrm{Id}_V, T, \dots, T^{k-1})\}$, so $m \leq n^2$. By definition of m, there exists $\lambda_0, \lambda_1, \dots, \lambda_{m-1}$ in $\mathbf{F}$ such that $0 = \lambda_0 \mathrm{Id}_V + \lambda_1 T + \dots + \lambda_{m-1} T^{m-1} + T^m$. Define $M_T(t) := t^m + \lambda_{m-1} t^{m-1} + \dots + \lambda_1 t + \lambda_0$. Then $M_T(t)$ satisfies 1, but also 2 by minimality of m.

Suppose now that $q(t)$ is also a monic polynomial satisfying 1 and 2. Then, $(M_T - q)(T) = 0 - 0 = 0$ and $\deg(M_T - q) \leq m - 1$. We conclude that $M_T - q$ is the zero polynomial and thus $M_T = q$ is unique. □

Definition 5.2.21.

Let $V \neq \{0\}$ be a finite dimensional non-zero vector space and let $T \in \mathcal{L}(V)$ be an operator. The **minimal polynomial** of T is the unique monic polynomial $M_T(t) \in \mathcal{P}(\mathbf{F})$ given by Proposition 5.2.20.

Lemma 5.2.22. *Let $V \neq \{0\}$ be a finite dimensional vector space and let $T \in \mathcal{L}(V)$ be an operator. Then $\deg(M_T(t)) \geq 1$. Moreover, $\deg(M_T(t)) = 1$ if and only if there exists $\lambda \in \mathbf{F}$ such that $T = \lambda \mathrm{Id}_V$.*

Proof. Since $M_T(t)$ is monic, it is not the zero polynomial. But since $M_T(T) = 0$, $M_T(t)$ cannot be a non-zero constant polynomial. We conclude that $M_T(t)$ has degree at least 1.

Since the minimal polynomial has degree at least one, we have $M_T(t) = t - \lambda$ if and only if $T - \lambda \mathrm{Id}_V = 0$, if and only if $T = \lambda \mathrm{Id}_V$. □

The above lemma implies that the minimal polynomial distinguishes the matrices from the introduction of this subsection.

Example 5.2.23. Let $\mathcal{E}$ be the standard basis of $\mathbf{C}^2$ and let T be the operator represented by $[T]^{\mathcal{E}}_{\mathcal{E}} = \begin{bmatrix} 1 & 1 \\ & 1 \end{bmatrix}$. Then $\chi_T(t) = (1-t)^2 = \chi_{\mathrm{Id}_{\mathbf{C}^2}}(t)$. However, $M_T(t)$ has degree at least 2, while $M_{\mathrm{Id}_V}(t) = t - 1$. Actually, it follows from the forthcoming Corollary 5.2.25 that $M_T(t)$ has degree at most 2, and hence exactly 2.

Not only the minimal polynomial is the unique polynomial of smallest degree annulling T, it is also the smallest one for divisibility.

Proposition 5.2.24. *Let $V \neq \{0\}$ be a finite dimensional non-zero vector space and let $T \in \mathcal{L}(V)$ be an operator. Then for any $q(t) \in \mathcal{P}(\mathbf{F})$, $q(T) = 0$ if and only if $M_T(t)$ divides $q(t)$.*

Proof. "$\Leftarrow$" Suppose $M_T(t)$ divides $q(t)$. Then there exists a polynomial $p(t) \in \mathcal{P}(\mathbf{F})$ such that $q(t) = M_T(t)p(t)$. Therefore, $q(T) = p(T)M_T(t) = 0$.

"$\Rightarrow$" By the Euclidean division of polynomials, there exists $r(t), s(t) \in \mathcal{P}(\mathbf{F})$ such that $q(t) = s(t)M_T(t) + r(t)$ and $\deg r(t) < \deg M_T(t)$. Now, if $q(T) = 0$ we also have $r(T) = 0$, which by minimality of $M_T(t)$ implies that $r(t)$ is the zero polynomial. That is, $q(t) = s(t)M_T(t)$. □

Corollary 5.2.25. *Let $V \neq \{0\}$ be a finite dimensional vector space and let $T \in \mathcal{L}(V)$ be an operator. Then the minimal polynomial $M_T(t)$ divides the characteristic polynomial $\chi_T(t)$.*

Proof. This directly follows from the proposition and the Cayley-Hamilton theorem (Theorem 5.2.17). □

Theorem 5.2.26.

Let $V \neq \{0\}$ be a finite dimensional vector space, let $T \in \mathcal{L}(V)$ be an operator and let $\lambda \in \mathbf{F}$. Then λ is an eigenvalue of T if and only if it is a root of the minimal polynomial.

Proof. "$\Leftarrow$" If $M_T(\lambda) = 0$, then $\chi_T(\lambda) = 0$ and thus λ is an eigenvalue of T by Theorem 5.2.11.

"$\Rightarrow$" Suppose that λ is an eigenvalue of T. Then there exists a non-zero vector v with $Tv = \lambda v$. Therefore, for all positive integer j we have $T^j v = \lambda^j v$. Let $M_T(t) = t^m + a_{m-1}t^{m-1} + \dots + a_1 t + a_0$. Then $0 = M_T(t)v = (\lambda^m + a_{m-1}\lambda^{m-1} + \dots + a_1\lambda + a_0)v = M_T(\lambda)v$. Since $v \neq 0$, we conclude that $M_T(\lambda) = 0$. □

Corollary 5.2.27. *Let $V \neq \{0\}$ be a finite dimensional vector space over $\mathbf{C}$ and let $T \in \mathcal{L}(V)$ be an operator. Let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues of T. Then there exists integers $k_1, \dots, k_m$ with $1 \leq k_i \leq \dim(\mathrm{G}(\lambda_i, T))$ such that $M_T(t) = (t-\lambda_1)^{k_1} \cdots (t-\lambda_m)^{k_m}$.*

Example 5.2.28. Let $\mathcal{E}$ be the standard basis of $\mathbf{C}^3$ and let $T \in \mathcal{L}(\mathbf{C}^3)$ be the operator given by

$$[T]_{\mathcal{E}}^{\mathcal{E}} = \begin{bmatrix} 6 & 3 & 4 \\ 0 & 6 & 2 \\ 0 & 0 & 7 \end{bmatrix} =: A.$$

Then $\chi_T(t) = (6-t)^2(7-t)$ and the eigenvalues of T are 6 (with geometric multiplicity 2) and 7 (with geometric multiplicity 1). Then we have two possibilities for the minimal polynomial: either $M_T(t) = (t-6)^2(t-7)$, or $M_T(t) = (t-6)(t-7)$. The second possibility occurs if and only if $(A - 6\,\mathrm{Id}_3)(A - 7\,\mathrm{Id}_3) = 0$. But

$$(A - 6\,\mathrm{Id}_3)(A - 7\,\mathrm{Id}_3) = \begin{bmatrix} 0 & 3 & 4 \\ 0 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 3 & 4 \\ 0 & -1 & 2 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -3 & * \\ * & * & * \\ * & * & * \end{bmatrix} \neq 0,$$

where the $*$ are entries that might or might not be 0. We conclude that $M_T(t) = (t-6)^2(t-7)$. Therefore, for all eigenvalues, the geometric and the algebraic multiplicities are the same, which implies that T is diagonalisable.

The minimal polynomial has the advantage to be able to distinguish operators that the characteristic polynomial cannot. However, the minimal polynomial has the disadvantage to be more complicated to compute in general. Hopefully, for nilpotent operators, one can easily compute it.

Proposition 5.2.29. *Let $V \neq \{0\}$ be a finite dimensional vector space and let $T \in \mathcal{L}(V)$ be a nilpotent operator. Let m be the smallest integer such that $T^m = 0$. Then the minimal polynomial of T is equal to t^m.*

Proof. The minimal polynomial $M_T(t)$ is a monic polynomial that divides (by Corollary 5.2.25) $\chi_T(t) = (-t)^{\dim V}$. Therefore, $M_T(t) = t^k$ for some $1 \leq m \leq \dim(V)$. But, by definition m is the smallest integer such that $T^m = 0$. □

Corollary 5.2.30. *Let $V \neq \{0\}$ be a finite dimensional vector space and let $T \in \mathcal{L}(V)$ be an operator such that $T - \lambda\,\mathrm{Id}_V$ is nilpotent for some $\lambda \in \mathbf{F}$ Let m be the smallest integer such that $(T - \lambda\,\mathrm{Id}_V)^m = 0$. Then the minimal polynomial of T is equal to $(t-\lambda)^m$.*

5.2.4 A word about matrices

All the work we did on nilpotent operators, the characteristic polynomial and the minimal polynomial naturally translates to the realm of matrices. For example, we say that a matrix $A \in \mathbf{F}^{n,n}$ is **nilpotent** if there exists an integer k such that $A^k = 0$ is the zero matrix. One easily shows that an operator $T \in \mathcal{L}(V)$ on a finite dimensional vector space is nilpotent if and only if the matrix $[T]^{\mathcal{B}}_{\mathcal{B}}$ is nilpotent for any basis $\mathcal{B}$ of V. One can also define the generalized eigenspace of A as $\mathrm{G}(\lambda, A) := \mathrm{G}(\lambda, \mathrm{L}_A) = \mathrm{null}(A - \lambda\,\mathrm{Id}_n)^n$, where L_A is the operator defined by $\mathrm{L}_A\, x = Ax$. Finally, the minimal polynomial of A is defined in a natural way, and one have $M_T(t) = M_{[T]^{\mathcal{B}}_{\mathcal{B}}}(t)$.

The Cayley-Hamilton theorem Theorem 5.2.17 remains true for matrices. More precisely, it says that for any square matrix A we have $\chi_A(A) = 0 \in \mathbf{F}^{n,n}$.

Remark 5.2.31. When seeing the Cayley-Hamilton theorem for the first time, some people would like to write the following incorrect "proof" of the matrix version of the theorem. "*We have* $\chi_A(A) = \det(A - A\,\mathrm{Id}) = \det(0) = 0$". While this seems appealing, this does not work. First of all, the matrix version of the theorem says that $\chi_A(A) = 0_{\mathbf{F}^{n,n}}$ is the zero matrix, not the scalar $0_{\mathbf{F}} = \det(0_{\mathbf{F}^{n,n}})$. Second, in the expression $\det(A - t\,\mathrm{Id})$, the variable t appears as the diagonal entries of the matrix $t\,\mathrm{Id}_n$. To illustrate this, let us consider the simple example $A = \left[\begin{smallmatrix}1&0\\0&0\end{smallmatrix}\right]$ (this is the orthogonal projection onto the x-axis). This matrix has eigenvalues 0 and 1, and hence characteristic polynomial $t(1-t)$. We can easily verify that the matrix A indeed satisfy $A(1-A)=0$. However, we have $A - t\,\mathrm{Id} = \left[\begin{smallmatrix}1-t&0\\0&-t\end{smallmatrix}\right]$. If we substitute the matrix A for t in this expression, we obtain

$$\begin{bmatrix} 1-\left[\begin{smallmatrix}1&0\\0&0\end{smallmatrix}\right] & 0 \\ 0 & -\left[\begin{smallmatrix}1&0\\0&0\end{smallmatrix}\right]\end{bmatrix},$$

which is not even a valid matrix expression.

When solving concrete problems about $T \in \mathcal{L}(V)$, it is often easier to take a basis $\mathcal{B}$ of V, translate the problem into a problem for $[T]^{\mathcal{B}}_{\mathcal{B}}$, solve the matrix version and then translate the solution back to T. Let us exemplify that by computing the minimal polynomial of an operator.

Example 5.2.32. Let $T \in \mathcal{L}(V)$ be an operator on a finite dimensional vector space. Suppose we have a basis $\mathcal{B}$ such that

$$[T]^{\mathcal{B}}_{\mathcal{B}} = \begin{bmatrix} \boxed{\begin{matrix}\lambda_1 & & * \\ & \ddots & \\ & & \lambda_1\end{matrix}} & & & \\ & \boxed{\begin{matrix}\lambda_2 & & * \\ & \ddots & \\ & & \lambda_2\end{matrix}} & & \\ & & \ddots & \\ & & & \boxed{\begin{matrix}\lambda_m & & * \\ & \ddots & \\ & & \lambda_m\end{matrix}} \end{bmatrix}.$$

If V is a complex vector space, such a basis always exists by Proposition 5.1.50. The minimal polynomial of T is hence of the form $M_T(t) = (t-\lambda_1)^{d_1}\cdots(t-\lambda_m)^{d_m}$ for some $d_j \in \{1,\dots,m_{\lambda_j}\}$. Write $A_1,\dots,A_m$ for the diagonal blocs ocurring in $[T]^{\mathcal{B}}_{\mathcal{B}}$. To compute d_1, observe that $N_1 := A_1 - \lambda_1\,\mathrm{Id}_{m_{\lambda_1}}$ is a triangular matrix with 0s on the diagonal, and hence nilpotent. Therefore, $M_{B_1}(t) = t^{d_1}$, where d_1 is the smallest integer such that $(B_1)^{d_1} = 0$. In other words, to find d_1 it is enough to compute $B_1, B_1^2, \dots, B_1^{m_{\lambda_1}}$ and to stop as soon one of those matrix is the 0 matrix. Finally, $M_{A_1}(t) = M_{B_1}(t-\lambda_1) = (t-\lambda_1)^{d_1}$. The other exponents $d_2,\dots,d_m$ are computed in a similar way.

Let us apply this procedure on a concrete example. Suppose that

$$[T]^{\mathcal{B}}_{\mathcal{B}} = A = \begin{bmatrix} 1 & 2 & & \\ & 1 & & \\ & & 3 & \\ & & & 3 \end{bmatrix}.$$

Then $\chi_T(t) = (1-t)^2(3-t)^2$ and $M_T(t) = (t-1)^{d_1}(t-3)^{d_2}$ with $1 \leq d_1, d_2 \leq 2$. Using the above notation we have $A_1 = \begin{bmatrix} 1 & 2 \\ & 1 \end{bmatrix}$ and $B_1 = \begin{bmatrix} 1 & 2 \\ & 1 \end{bmatrix}$. So $B_1 \neq 0$ but $B_1^2 = 0$ and we conclude that $d_1 = 2$. For the eigenvalue 3 we have $A_2 = \begin{bmatrix} 3 & \\ & 3 \end{bmatrix}$ and $B_2 = 0$, so $d_2 = 1$. Altogether, $M_T(t) = (t-1)^2(t-3)$.

5.3 Jordan normal form

In this section, we will finally give the definitive answer to our main question.

Major Question 3.1.26.

Given V and $T \in \mathcal{L}(V)$, find a basis $\mathcal{B}$ such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is "as simple as possible" (for example: diagonal, upper triangular, with many zeroes, ...).

5.3.1 Existence of the Jordan form

We already saw in Proposition 5.1.50 that if T is an operator on a complex finite dimensional vector space, there always exists a basis such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is a diagonal by blocks matrix. Moreover, if $\lambda_1, \dots, \lambda_m$ are the eigenvalues of T, then $[T]_{\mathcal{B}}^{\mathcal{B}}$ has m diagonal block $A_1, \dots, A_m$, and each A_j is an $m_{\lambda_j} \times m_{\lambda_j}$ upper triangular matrix with λ_j on the diagonal. In this section, we will see that one can always choose a basis such that the A_j itself are made of particularly nice blocks.

Definition 5.3.1.

A square matrix $J \in \mathbf{F}^{k,k}$ is called a **Jordan block** of eigenvalue $\lambda \in \mathbf{F}$ if it is of the form

$$J = \begin{bmatrix} \lambda & 1 & & & \\ & \lambda & 1 & & \\ & & \ddots & \ddots & \\ & & & \ddots & 1 \\ & & & & \lambda \end{bmatrix}.$$

In particular, if $k = 1$, then a Jordan block of eigenvalue λ is simply the following 1×1 matrix $\begin{bmatrix} \lambda \end{bmatrix}$.

It directly follows from the definition that the eigenvalues of a $k \times k$ "Jordan block of eigenvalue λ" is indeed λ, with algebraic multiplicity k. We have $\chi_J(t) = (\lambda - t)^k$ and the matrix $J - \lambda \operatorname{Id}_k$ is nilpotent. Moreover, by direct matrix computations we have

$$(J - \lambda \operatorname{Id}_k)^{k-1} = \begin{bmatrix} 0 & \cdots & 0 & 1 \\ & \ddots & & 0 \\ & & \ddots & \vdots \\ & & & 0 \end{bmatrix}.$$

Therefore, $(J - \lambda \operatorname{Id}_k)^k = 0$ and $M_J(t) = (t - \lambda)^k$ by Corollary 5.2.30.

Proposition 5.3.2. *Let*

$$J = \begin{bmatrix} J_1 & & & \\ & J_2 & & \\ & & \ddots & \\ & & & J_m \end{bmatrix} \in \mathbf{F}^{n,n}$$

be a matrix where each $J_j \in \mathbf{F}^{k_j,k_j}$ is a Jordan block of eigenvalue λ and size k_j. (So $n = \sum_{j=1}^m k_j$.) Then $\chi_J(t) = (\lambda - t)^n$ and $M_J(t) = (t-\lambda)^k$ where $k = \max\{k_1, \dots, k_m\}$.

Proof. This is basically Example 5.2.32. But here is a detailed proof. The matrix J is upper triangular, of size n with λ on the diagonal. This directly implies the formula for the characteristic polynomial. For the minimal polynomial, since J is diagonal by block, we have

$$(J - \lambda \operatorname{Id}_n)^l = \begin{bmatrix} (J_1 - \lambda \operatorname{Id}_{k_1})^l & & & \\ & (J_2 - \lambda \operatorname{Id}_{k_2})^l & & \\ & & \ddots & \\ & & & (J_m - \lambda \operatorname{Id}_{k_m})^l \end{bmatrix}$$

for all $l \in \mathbf{N}$. Therefore, $M_J(t) = (t-\lambda)^k$ for

$$k = \min\{l \mid \forall j \in \{1, \dots, m\}, (J_j - \lambda \operatorname{Id}_{k_j}) = 0\} = \max\{k_1, \dots, k_m\}. \qquad \square$$

Example 5.3.3. Let $n = 4$ and $\lambda = 2$. Up to permutation of the blocks, there is 5 matrices that are diagonal by [Jordan blocks]:

$$J = \begin{bmatrix} 2 & 1 & & \\ & 2 & 1 & \\ & & 2 & 1 \\ & & & 2 \end{bmatrix}, \quad K = \begin{bmatrix} 2 & 1 & & \\ & 2 & 1 & \\ & & 2 & \\ & & & 2 \end{bmatrix}, \quad L = \begin{bmatrix} 2 & 1 & & \\ & 2 & & \\ & & 2 & 1 \\ & & & 2 \end{bmatrix},$$

$$M = \begin{bmatrix} 2 & 1 & & \\ & 2 & & \\ & & 2 & \\ & & & 2 \end{bmatrix}, \quad N = \begin{bmatrix} 2 & & & \\ & 2 & & \\ & & 2 & \\ & & & 2 \end{bmatrix}.$$

All of these matrices have characteristic polynomial $\chi(t) = (2-t)^4$. For the minimal polynomials, we have $M_J(t) = (2-t)^4$, $M_K(t) = (2-t)^3$, $M_L(t) = M_M(t) = (2-t)^2$ and $M_N(t) = 2 - t$.

The case of Jordan blocks of distinct eigenvalues is similar to what happen in Proposition 5.3.2. Simply treat together all the block sharing one common eigenvalue.

Example 5.3.4. Let

$$A = \begin{bmatrix} 4 & 1 & & \\ & 4 & & \\ & & 4 & \\ & & & 5 \end{bmatrix}.$$

Then we have $M_A(t) = (t-4)^2(t-5)$, where the exponent 2 in $(t-4)^2$ is the size of the largest Jordan block of eigenvalue 4, while the exponent 1 in $(t-5)^1$ is the size of the largest Jordan block of eigenvalue 1.

In Question 3.1.26 we asked if it was always possible to have a “nice” matrix representation of an operator. By nice, we mean: a product of Jordan blocks.

Definition 5.3.5.

Let $V \neq \{0\}$ be a finite dimensional vector space and let $T \in \mathcal{L}(V)$ be an operator. A **Jordan form** (also called a **Jordan normal form**) for T is a matrix representation of the form

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{bmatrix} J_1 & & & \\ & J_2 & & \\ & & \ddots & \\ & & & J_k \end{bmatrix}, \tag{5.6}$$

where J_i is a Jordan block of an eigenvalue of T (different blocks might have the same eigenvalue).

A basis $\mathcal{B}$ for which $[T]_{\mathcal{B}}^{\mathcal{B}}$ is a Jordan form is called a **Jordan basis** for T.

A square matrix is a **Jordan matrix** if it is diagonal by block, with each block a Jordan block. That is, if it is as the matrix appearing in the right-hand side of Equation (5.6).

Let $A \in \mathbf{F}^{n,n}$ be a square matrix. The **Jordan form** of A is the Jordan form the operator $\mathrm{L}_A \in \mathcal{L}(\mathbf{F}^n)$, $\mathrm{L}_A : v \mapsto Av$.

As a direct corollary of Proposition 5.3.2 we obtain.

Corollary 5.3.6. *Let $V \neq \{0\}$ be a finite dimensional vector space and let $T \in \mathcal{L}(V)$ be an operator. Let $\lambda_1, \ldots, \lambda_m$ be the eigenvalue of T and let $M_T(t) = (t-\lambda_1)^{c_{\lambda_1}} \cdots (t-\lambda_m)^{c_{\lambda_m}}$ be its minimal polynomial. If T admits a Jordan form J, then c_{λ_j} is the maximal size of a Jordan block of eigenvalue λ_j ocuring in J.*

Before proving our next result on Jordan form, we need a technical lemma about linear independence and nullspaces.

Lemma 5.3.7. *Let $V \neq \{0\}$ be a finite dimensional vector space and let $T \in \mathcal{L}(V)$ be a nilpotent operator. Let $v_1, \ldots, v_k$ be vectors in $\operatorname{null}(T^m)$ but not in $\operatorname{null}(T^{m-1})$ for some $m \geq 2$. If $v_1, \ldots, v_k$ are linearly independent, then $(v_1, \ldots, v_k, Tv_1, \ldots, Tv_k)$ is linearly independent.*

Proof. Suppose we can write $0 = \lambda_1 v_1 + \cdots + \lambda_k v_k + \mu_1 T v_1 + \cdots + \mu_k T v_k$. We want to show that all the λ_j and the μ_j are 0. □

Every nilpotent operator admits a Jordan form (here we don't need that $\mathbf{F} = \mathbf{C}$).

Lemma 5.3.8. *Let $V \neq \{0\}$ be a finite dimensional vector space and let $T \in \mathcal{L}(V)$ be a nilpotent operator. Then there exists a Jordan basis $\mathcal{B}$ for T. Moreover, the Jordan form for T is unique up to a permutation of the Jordan blocks.*

Proof. The proof is done by induction on $n = \dim(V)$. If $n = 1$, the only nilpotent operator is the zero operator, which has matrix representation $[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{bmatrix}0\end{bmatrix}$ in any basis, so we are done.

Now, suppose that $n \geq 2$ and that the theorem has been proven for every vector space of dimension strictly smaller than n. If T is the zero operator, then its matrix representation is a 0 matrix in any basis, and hence a Jordan form. So one can suppose that $T \neq 0$. Our aim is to find a direct sum decomposition $V = U \oplus W$ such that both U and W are of dimension strictly less than V and T-invariant. Indeed, in this case one can apply the induction hypothesis to U and obtain a Jordan basis $\mathcal{B}_U$ for $T|_U$. One also obtain a Jordan basis $\mathcal{B}_W$ for $T|_W$. Altogether, $\mathcal{B} = \mathcal{B}_U \cup \mathcal{B}_W$ is a basis of V in which $[T]_{\mathcal{B}}^{\mathcal{B}}$ is diagonal by blocks (this follows from the T-invariance), with blocks $[T|_U]_{\mathcal{B}_U}^{\mathcal{B}_U}$ and $[T|_W]_{\mathcal{B}_W}^{\mathcal{B}_W}$.

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{bmatrix} [T|_U]_{\mathcal{B}_U}^{\mathcal{B}_U} & 0 \\ 0 & [T|_W]_{\mathcal{B}_W}^{\mathcal{B}_W} \end{bmatrix}.$$

In particular, $[T]_{\mathcal{B}}^{\mathcal{B}}$ is a Jordan form for T and B is a Jordan basis.

We first describe U. Let m be the smallest integer such that $T^m = 0$. Therefore, we have

$$\{0\} \subsetneq \operatorname{null}(T) \subsetneq \operatorname{null}(T^2) \subsetneq \cdots \subsetneq \operatorname{null}(T^{m-1}) \subsetneq \operatorname{null}(T^m) = V.$$

Then there exists $u_m \in V$ but not in $\operatorname{null}(T^{m-1})$. For such an u_m, $u_{m-1} := Tu_m$ is in $\operatorname{null}(T^{m-1})$ but not in $\operatorname{null}(T^{m-2})$ and so on, until we have $u_1 := T^{m-1}u \in \operatorname{null}(T)$ but $T^{m-1}u \neq 0$. In particular, the list $u_m, u_{m-1}, \dots, u_1$ is linearly independent. Let $U := \operatorname{span}(u_m, u_{m-1}, \dots, u_1)$. This is a m-dimensional subspace of V with basis $\mathcal{B}_U := (u_1, \dots, u_m)$. We have $Tu_j = u_{j-1}$ for $j \in \{2, \dots, m\}$ and $Tu_1 = 0$. This implies both that U is T-invariant and that $[T|_U]_{\mathcal{B}_U}^{\mathcal{B}_U}$ is a single Jordan block of eigenvalue 0 and size m. If $U = V$, then $\mathcal{B}_U$ is a Jordan basis of V and we are done. Thus we can assume that $U \neq V$ is of dimension strictly less than n. Moreover, $U \neq 0$, so any direct sum complement W will have dimension strictly less than n.

All that remind to do is to find a direct sum complement $W \subseteq V$ of U, which is also T-invariant. Let $X = \operatorname{span}(u_1)$ and let Y be any direct sum complement: $X \oplus Y = V$. (For example, one can take $Y = X^\perp$ if V is an inner product space). So we have a map $P\colon V \to \mathbf{F}$ defined by $P(\lambda u_1, y) = \lambda$. ($P$ is the projection onto U followed by the isomorphism $U \cong \mathbf{F}$.) One can hence defined a map $S\colon V \to \mathbf{F}^m$ by

$$Sv := \big(P(v), P(Tv), \dots, P(T^{m-1}v)\big).$$

Finally, we let $W := \operatorname{null}(S) \subseteq V$. We want to prove that W is T-invariant and a direct sum complement of U. So let $v \in W$ and look at Tv. Then

$$S(Tv) := \big(P(Tv), P(T^2v), \dots, P(T^{m-1}v), P(T^m v)\big).$$

Since v is in $\text{null}(S)$ we have $P(Tv) = P(T^2v) = \dots = P(T^{m-1}v = 0)$, while $T^m = 0$ implies $T^m v = 0$. Altogether, $S(Tv) = (0, \dots, 0)$, which proves the T-invariance of W. Now, let $u \in U$. Then $u = \lambda_1 u_1 + \cdots \lambda_m u_m$. Therefore, $Su = (\lambda_1, \lambda_2, \dots, \lambda_m)$. We conclude $U \cap W = \{0\}$. Therefore, the sum $U + W$ is direct. Finally, we have$\dim(W) = \dim(\text{null}(S)) = \dim(V) - \dim(\text{range}(S)) \geq n - m$. But then, $\dim(U \oplus W) = \dim(U) + \dim(W) \geq m + n - m = n$, which implies $U \oplus W = V$.

In order to finish the proof, with still need to show the unicity of the Jordan from. But every Jordan block of size m in a Jordan form of T correspond to a vector in $\text{null}(T^m) \setminus \text{null}(T^{m-1})$. Therefore, the number of Jordan blocks of size m is equal to $\dim(\text{null}(T^m)) - \dim(\text{null}(T^{m-1}))$, which depends only on T. Similarly, one can show that for every $j \in \{1, \dots, m\}$, the number of Jordan blocks of size j depends only on T. This proves that the Jordan form is unique, up to permutation of the Jordan blocks. □

To go further

It is possible to write a more algorithmic proof of Lemma 5.3.8. While the idea is similar to the above proof, this time we do not use induction, but instead directly construct the Jordan blocks, inspired by the unicity part of the above proof.

Alternative proof of Lemma 5.3.8. If $T = 0$ is the zero operator, then in any basis the matrix representing T is the 0 matrix and hence a Jordan matrix. We can hence suppose that $T \neq 0$. Let m be the smallest integer such that $T^m = 0$. By the above, $m \geq 2$.

$$\{0\} = \text{null}(T^0) \subsetneq \text{null}(T) \subsetneq \text{null}(T^m) \subsetneq \cdots \subsetneq \text{null}(T^{m-1}) \subsetneq \text{null}(T^m) = V.$$

In Proposition 5.1.34, we started with a basis of $\text{null}(T)$ an extended it step by step to obtain a basis $\mathcal{D}$ of $\text{null}(T^m) = V$ such that $[T]_{\mathcal{D}}^{\mathcal{D}}$ is upper triangular (with 0 on the diagonal). Here we will go in the opposite direction: start with a maximal independent family of $\text{null}(T^m) \setminus \text{null}(T^{m-1})$ and extend it carefully to a basis $\mathcal{B}$ of V such that $[T]_{\mathcal{B}}^{\mathcal{B}}$ is a Jordan matrix.
For $j \in \{1, \dots, m\}$, let us write $d_j := \dim \text{null}(T^j)$. So $1 \leq d_1 < d_2 < \cdots < d_m$ and $d_1 = \dim(\text{E}(0, T))$ is the geometric multiplicity of 0 while $d_m = \dim(\text{E}(0, T)) = \dim(V)$ is the algebraic multiplicity of 0. Write $c_m := d_m - d_{m-1} \geq 1$. We will show that J has c_m blocks of size m. We have $d_m = d_{m-1} + c_m$. Therefore, it is possible to find c_m linearly independent vectors $v_{m,1}, v_{m,2}, \dots, v_{m,c_m}$ in $\text{null}(T^m)$ such that $\text{span}(\text{null}(T^{m-1}), v_{m,1}, v_{m,2}, \dots, v_{m,c_m}) = \text{null}(T^m)$. (For example, take any basis of $\text{null}(T^{m-1})$ and extend it to a basis of $\text{null}(T^m)$). This implies in particular that the $v_{m,1}, v_{m,2}, \dots, v_{m,c_m}$ are linearly independent and moreover that a non-trivial linear combination of the $v_{m,1}, v_{m,2}, \dots, v_{m,c_m}$ never belongs to $\text{null}(T^{m-1})$.
For $k \in \{1, \dots, c_m\}$ and $j \in \{1, m-1, \dots,\}$, let $v_{j,k} := T^{m-j} v_{m,k}$. So for example, $v_{m-2,3} = T^2 v_{m,3}$. Since the $v_{m,k}$ are in in $\text{null}(T^m)$ but not in $\text{null}(T^{m-1})$, the $v_{j,k}$ are in in $\text{null}(T^j)$ but not in $\text{null}(T^{j-1})$. Moreover, it follows from the fact

that non-trivial linear combination of the $v_{m,1}, v_{m,2}, \dots, v_{m,c_m}$ never belongs to $\mathrm{null}(T^{m-1})$ that the list

$$\mathcal{B}_m := (v_{1,1}, v_{2,1}, \dots, v_{m,1}, \dots, v_{1,c_m}, \dots, v_{m,c_m})$$

is linearly independent.

Observe that $T(v_{j,k}) = v_{j-1,k}$ if $j \geq 2$ by definition, and $T(v_{1,k}) = 0$ as $v_{1,k}$ is in $\mathrm{null}(T)$. Before going further, let us look a little more at $\mathcal{B}_m$. As we have just seen, $U_m := \mathrm{span}(\mathcal{B}_m)$ is T-invariant and $[T|_{U_m}]_{\mathcal{B}_m}^{\mathcal{B}_m}$ is a diagonal product of c_m Jordan blocks of size m.

$$[T|_{U_m}]_{\mathcal{B}_m}^{\mathcal{B}_m} = \begin{bmatrix} J_1 & & & \\ & J_2 & & \\ & & \ddots & \\ & & & J_{c_m} \end{bmatrix}, \qquad J_j = \begin{array}{c c} & \begin{array}{cccc} 1 & 2 & \cdots & m \end{array} \\ \begin{array}{c} 1 \\ 2 \\ \vdots \\ m \end{array} & \begin{bmatrix} 0 & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & 0 \end{bmatrix} \end{array}.$$

We will now take care of the Jordan blocks of size $m-1$ (if any). Let $c_{m-1} := d_{m-1} - d_{m-2} - c_m$. By the above, $c_{m-1} \geq 0$. If $c_{m-1} = 0$, then there will be no Jordan blocks of size $m-1$ and we can directly go to the next step. If $c_{m-1} \geq 1$, take any basis $\mathcal{D}_{m-2}$ of $\mathrm{null}(T^{m-2})$. It is possible to complete $\mathcal{D}_{m-2}, v_{m-1,1}, \dots, v_{m-1,c_m}$ to a basis $\mathcal{D}_{m-2}, v_{m-1,1}, \dots, v_{m-1,c_m}, v_{m-1,c_m+1}, \dots, v_{m-1,c_m+c_{m-1}}$. For $k \in \{c_m + 1, \dots, c_m + c_{m-1}\}$ and $j \in \{1, m-2, \dots,\}$, let $v_{j,k} := T^{m-j} v_{m,k}$. Then both the list

$$\mathcal{B}_{m-1} := (v_{1,c_m+1}, \dots, v_{m-1,c_m+1}, \dots, v_{1,c_m+c_{m-1}}, \dots, v_{m-1,c_m+c_{m-1}})$$

and $\mathcal{B}_m \cup \mathcal{B}_{m-1}$ are linearly independent. Moreover, for any $v_{j,k} \in \mathcal{B}_m \cup \mathcal{B}_{m-1}$, $T(v_{j,k}) = v_{j-1,k}$ if $j \geq 2$ while $T(v_{1,k}) = 0$. Finally, $U_{m-1}\, \mathrm{span}(\mathcal{B}_{m-1})$ is T-invariant and $[T|_{U_{m-1}}]_{\mathcal{B}_{m-1}}^{\mathcal{B}_{m-1}}$ is a diagonal product of c_{m-1} Jordan blocks of size $m-1$. We conclude that $U_m + U_{m-1}$ is a direct sum, that $U_m \oplus U_{m-1}$ is T-invariant and that $[T|_{U_m \oplus U_{m-1}}]_{\mathcal{B}_m \cup \mathcal{B}_{m-1}}^{\mathcal{B}_m \cup \mathcal{B}_{m-1}}$ is a diagonal product of c_m Jordan blocks of size m

followed by c_{m-1} Jordan blocks of size $m-1$.

$$[T|_{U_m \oplus U_{m-1}}]^{\mathcal{B}_m \cup \mathcal{B}_{m-1}}_{\mathcal{B}_m \cup \mathcal{B}_{m-1}} = \begin{bmatrix} J_1 & & & & & \\ & \ddots & & & & \\ & & J_{c_m} & & & \\ & & & J_{c_m+1} & & \\ & & & & \ddots & \\ & & & & & J_{c_m+c_{m-1}} \end{bmatrix},$$

$$J_j = \begin{cases} \begin{array}{cc} & \begin{array}{ccccc} 1 & 2 & \cdots & m-1 & m \end{array} \\ \begin{array}{c} 1 \\ 2 \\ \vdots \\ m-1 \\ m \end{array} & \begin{bmatrix} 0 & 1 & & & \\ & \ddots & \ddots & & \\ & & \ddots & 1 & \\ & & & 0 & 1 \\ & & & & 0 \end{bmatrix} \end{array} & \text{if } j \in \{1, \dots, c_m\}, \\ \begin{array}{cc} & \begin{array}{cccc} 1 & 2 & \cdots & m-1 \end{array} \\ \begin{array}{c} 1 \\ 2 \\ \vdots \\ m-1 \end{array} & \begin{bmatrix} 0 & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & 0 \end{bmatrix} \end{array} & \text{if } j \in \{c_m+1, \dots, c_m+c_{m-1}\}. \end{cases}$$

Iterating this procedure, we obtain integers $c_{m-2} = d_{m-2} - d_{m-3} - c_m - c_{m-1}$ up to $c_1 = d_1 - 0 - c_m - \cdots - c_2$ and $\mathcal{B} := \mathcal{B}_m \cup \mathcal{B}_{m-1} \cup \cdots \cup \mathcal{B}_1$ is a basis for V such that $[T]^{\mathcal{B}}_{\mathcal{B}}$ is a diagonal product of c_m Jordan blocks of size m, c_{m-1} Jordan blocks of size $m-1$, …, and c_1 Jordan blocks of size 1. □

We finally answer Question 3.1.26: every operator on a complex vector space admits a nice matrix representation.

Theorem 5.3.9.

Let $V \neq \{0\}$ be a finite dimensional vector space over $\mathbf{C}$ and let $T \in \mathcal{L}(V)$ be an operator. Then there exists a basis $\mathcal{B}$ of V such that $[T]^{\mathcal{B}}_{\mathcal{B}}$ is a Jordan form for T. Moreover, the Jordan form for T is unique up to a permutation of the Jordan blocks.

Proof. Let $\lambda_1, \dots, \lambda_m$ be the eigenvalues of T. Then, by Theorem 5.1.46, $V = \mathrm{G}(\lambda_1, T) \oplus \cdots \oplus \mathrm{G}(\lambda_m, T)$, each of the $U_j := \mathrm{G}(\lambda_j, T)$ is invariant by $T - \lambda_j \operatorname{Id}_V$ and $S_j := T - \lambda_j \operatorname{Id}_V|_{U_j}$ is a nilpotent operator on U_j. Therefore, by Lemma 5.3.8 there exists a basis $\mathcal{B}_j$ be a basis of U_j for which $[S_j]^{\mathcal{B}_j}_{\mathcal{B}_j}$ is a product of Jordan blocks of eigenvalue 0. We conclude that $[T|_{U_j}]^{\mathcal{B}_j}_{\mathcal{B}_j}$ is a product of Jordan blocks of eigenvalues λ_j. Finally, $\mathcal{B} = \mathcal{B}_1 \cup \cdots \mathcal{B}_m$ is a basis for V (by direct sum) and $[T]_{\mathcal{B}}^{\mathcal{B}}$ is diagonal by block, with blocks $[T|_{U_1}]_{\mathcal{B}_1}^{\mathcal{B}_1}, \dots, [T|_{U_m}]_{\mathcal{B}_m}^{\mathcal{B}_m}$ (by invariance of the U_j and construction of $\mathcal{B}$). □

Corollary 5.3.10. *Let $V \neq \{0\}$ be a finite dimensional vector space and let $T \in \mathcal{L}(V)$ be an operator. Let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues of T. Then T is diagonalizable if and only if $M_T(t) = (t - \lambda_1) \cdots (t - \lambda_m)$.*

Proof. A matrix T is diagonalisable if and only if it admits a Jordan form where all Jordan block have size 1. The left-to-right implication then follows from Proposition 5.3.2.

For right-to-left implication, if V is a complex vector space, then T admits a Jordan form. But then, all Jordan blocks have size 1 by Proposition 5.3.2. More generally, if V is a vector space over $\mathbf{F}$, the fact that $M_T(t) = (t - \lambda_1) \cdots (t - \lambda_m)$ split as a product of monomial is enough to guarantee the existence of a Jordan form. □

In term of matrices, we obtain the following results.

Lemma 5.3.11. *For every $A \in \mathbf{C}^{n,n}$ there exists an invertible matrix $P \in \mathbf{C}^{n,n}$ such that $P^{-1}AP$ is the Jordan form of A.*

Proof. Let us write $\mathrm{L}_A \in \mathcal{L}(\mathbf{C}^n)$ for the operator $\mathrm{L}_A(v) = Av$. Then we have $[\mathrm{L}_A]_{\mathcal{E}}^{\mathcal{E}} = A$ for the standard basis $\mathcal{E}$, while by Theorem 5.3.9 there exists a Jordan basis $\mathcal{B}$ such that $[\mathrm{L}_A]_{\mathcal{B}}^{\mathcal{B}} =: J$ is a Jordan matrix. But then the matrices A and J are similar. □

Corollary 5.3.12. *Two complex square matrices $A, B \in \mathbf{C}^{n,n}$ are similar if and only if they have the same Jordan form.*

Proof. "$\Rightarrow$" If the matrices A and B are similar, they represent the same operator T and hence have the same Jordan form (up to permutation of the Jordan block).

"$\Leftarrow$" Let J be the Jordan form of A and B. Since the Jordan form is unique (up to permutation of the Jordan block), $J = P^{-1}AP = Q^{-1}BQ$ for invertible matrices $P, Q \in \mathbf{C}^{n,n}$. This gives $A = (QP^{-1})^{-1}B(QP^{-1})$. (Observe that the "$\Rightarrow$" implication is true for any $\mathbf{F}$.) □

To go further

Theorem 5.3.9, Corollaries 5.3.10 and 5.3.12, and Lemma 5.3.11 are true for all algebraically closed fields $\mathbf{F}$. More generally, they remain true for every operator T on a finite dimensional $\mathbf{F}$ vector space if T has all its eigenvalues in $\mathbf{F}$. That is, if the characteristic polynomial can be written as a product of degree 1 polynomials: $\chi_T(t) = (\lambda_1 - t)^{m_{\lambda_1}} \cdots (\lambda_m - t)^{m_{\lambda_m}}$.

The following summarizes the important relations between eigen-theory and the Jordan form.

Summary 5.3.13.

Let $V \neq \{0\}$ be a n dimensional vector space over $\mathbf{C}$ and let $T \in \mathcal{L}(V)$ be an operator. To such a T, one can associate:

- *Its distinct eigenvalues $\lambda_1, \dots, \lambda_k$ (we necessarily have $1 \leq k \leq n$);*
- *For each eigenvalue, its algebraic multiplicity $m_{\lambda_j} = \dim(\mathrm{G}(\lambda_j, T))$;*
- *For each eigenvalue, its geometric multiplicity $d_{\lambda_j} = \dim(\mathrm{E}(\lambda_j, T))$;*
- *The characteristic polynomial $\chi_T(t) = (\lambda_1 - t)^{m_{\lambda_1}} \cdots (\lambda_k - t)^{m_{\lambda_k}}$;*
- *The minimal polynomial $M_T(t) = (t - \lambda_1)^{c_{\lambda_1}} \cdots (t - \lambda_k)^{c_{\lambda_k}}$;*
- *The Jordan form J of T.*

Then we have the following:

- *$m_{\lambda_j} \in \{1, \dots, n\}$ and $\sum_{j=1}^{k} m_{\lambda_j} = n$ (Theorem 5.1.46);*
- *$J = \begin{bmatrix} J_1 & & & \\ & J_2 & & \\ & & \ddots & \\ & & & J_k \end{bmatrix}$, where J_j is the Jordan form of $T|_{\mathrm{G}(\lambda_j, T)}$ (Theorem 5.3.9);*
- *$d_{\lambda_j} \in \{1, \dots, m_{\lambda_j}\}$ is the number of Jordan blocks of eigenvalue λ_j in J_j, in particular J_j is diagonal if and only if and only if $d_{\lambda_j} = m_{\lambda_j}$ (proof of Lemma 5.3.8);*
- *$c_{\lambda_j} \in \{1, \dots, m_{\lambda_j}\}$ is the size of the biggest Jordan block of eigenvalue λ_j in J_j, in particular J_j is diagonal if and only if and only if $c_{\lambda_j} = 1$ (Corollary 5.3.6);*
- *The sum of the size of all Jordan blocks of eigenvalue λ_j is equal to m_{λ_j} (follows from the definition of m_{λ_j}).*

5.3.2 Computation of the Jordan form

Knowing that the Jordan form exists is good. Being able to compute it is better. This is the subject of this subsection. As we have seen in the last subsection, in order to find a Jordan basis for a matrix A or for an operator T, it is enough to be able to find a Jordan basis $\mathcal{B}_i$ it for each generalized eigenspace $\mathrm{G}(\lambda_i, A)$, which is done by looking at $\mathrm{null}(A - \lambda_i \mathrm{Id}) \subsetneq \cdots \subsetneq \mathrm{null}(A - \lambda_i \mathrm{Id})^{m_i} = \mathrm{G}(\lambda_i, A)$, where $1 \leq m_i \leq m_{\lambda_i}(A)$ is the smallest integer such that $(A - \lambda_i \mathrm{Id})^{m_i} = 0$. Alternatively, m_i is the size of the biggest Jordan block of eigenvalue λ_i in the Jordan form of A.

Let us work on a concrete example.

Example 5.3.14. Let A be the complex square matrix

$$A = \begin{bmatrix} 2 & -4 & 2 & 2 \\ -2 & 0 & 1 & 3 \\ -2 & -2 & 3 & 3 \\ -2 & -6 & 3 & 7 \end{bmatrix}, \qquad \text{so } A - t\,\mathrm{Id}_4 = \begin{bmatrix} 2-t & -4 & 2 & 2 \\ -2 & -t & 1 & 3 \\ -2 & -2 & 3-t & 3 \\ -2 & -6 & 3 & 7-t \end{bmatrix}.$$

We start by computing the characteristic polynomial of A. We will prove that $\chi_A(t) = (2-t)^2(4-t)^2$. It is of course possible and easier to ask a computer to do it. We will do it by hand to show how it can be done. First do rows operations (that do not change the determinant) on $A - t\,\mathrm{Id}_4$: $r_1 \mapsto r_1 + (1 - t/2)r_2$, $r_3 \mapsto r_3 - r_2$ and $r_4 \mapsto r_4 - r_2$ to obtain

$$\begin{aligned} \det(A - t\,\mathrm{Id}_4) &= \det \begin{bmatrix} 0 & -4-t+\frac{t^2}{2} & 3-\frac{t}{2} & 5-\frac{3t}{2} \\ -2 & -t & 1 & 3 \\ 0 & -2+t & 2-t & 0 \\ 0 & -6+t & 2 & 4-t \end{bmatrix} \\ &= 2\det \begin{bmatrix} -4-t+\frac{t^2}{2} & 3-\frac{t}{2} & 5-\frac{3t}{2} \\ -2+t & 2-t & 0 \\ -6+t & 2 & 4-t \end{bmatrix}, \end{aligned}$$

which we can develop along the third column to obtain

$$\begin{aligned} \det(A - t\,\mathrm{Id}_4) &= (10-3t)\det \begin{bmatrix} -2+t & 2-t \\ -6+t & 2 \end{bmatrix} + (8-2t)\det \begin{bmatrix} -4-t+\frac{t^2}{2} & 3-\frac{t}{2} \\ -2+t & 2-t \end{bmatrix} \\ &= (10-3t)\big(-4+2t-(-12+8t-t^2)\big) \\ &\quad + (8-2t)\big(-8+2t+2t^2-\tfrac{t^3}{2}-(-6+4t-\tfrac{t^2}{2})\big) \\ &= (10-3t)(8-6t+t^2) + (4-t)(-4-4t+5t^2-t^3) \\ &= (80-84t+28t^2-3t^3) + (-16-12t+24t^2-9t^3+t^4) \\ &= t^4-12t^3+52t^2-96t+64. \end{aligned}$$

So we have $\chi_A(t) = t^4 - 12t^3 + 52t^2 - 96t + 64$ and we still need to find the roots. Since all the coefficients are integers, if an rational root exist it is of the form p/q, where p is a divisor of 64 (constant coefficient) and q a divisor of 1 (leading coefficient). So the possible rational roots are in $\{\pm 1, \pm 2, \pm 4, \pm 8, \pm 16, \pm 32, \pm 64\}$. By testing, we find that both 2 and 4 are roots. Therefore, $(2-t)(4-t) = t^2 - 6t + 8$ divides $\chi_A(t)$. A polynomial division (similar to a long division) gives us $t^4 - 12t^3 + 52t^2 - 96t + 64 / t^2 - 6t + 8 = t^2 - 6t + 8 = (2-t)(4-t)$. We have just showed $\chi_A(t) = (2-t)^2(4-t)^2$. Therefore, the eigenvalues of A are $\lambda_1 = 2$ and $\lambda_2 = 4$, both with algebraic multiplicity 2.

For $\lambda_1 = 2$,

$$A - 2\,\mathrm{Id}_4 = \begin{bmatrix} 0 & -4 & 2 & 2 \\ -2 & -2 & 1 & 3 \\ -2 & -2 & 1 & 3 \\ -2 & -6 & 3 & 5 \end{bmatrix}.$$

To compute $\mathrm{null}(A - 2\,\mathrm{Id}_4)$, we solve the system

$$\begin{bmatrix} 0 & -4 & 2 & 2 \\ -2 & -2 & 1 & 3 \\ -2 & -2 & 1 & 3 \\ -2 & -6 & 3 & 5 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \iff \begin{bmatrix} 0 & -4 & 2 & 2 \\ -2 & -2 & 1 & 3 \\ -2 & -6 & 3 & 5 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Moreover, in the last matrix we have $r_1 + r_2 = r_3$. Therefore, $\mathrm{null}(A - 2\,\mathrm{Id}_4)$ is of dimension at least $4 - 2 = 2$. Since $\mathrm{null}(A - 2\,\mathrm{Id}_4) \subseteq \mathrm{G}(2, A)$, we conclude $\mathrm{null}(A - 2\,\mathrm{Id}_4) = \mathrm{G}(2, A)$. Therefore, we can choose any basis of $\mathrm{null}(A - 2\,\mathrm{Id}_4)$ as a basis of $\mathrm{G}(2, A)$. For example, one can take $v_1 = [2, 1, 0, 2]^\mathsf{T}$ and $v_2 = [0, 1, 2, 0]^\mathsf{T}$.

We now take care of $\lambda_2 = 4$. We have

$$A - 4\,\mathrm{Id}_4 = \begin{bmatrix} -2 & -4 & 2 & 2 \\ -2 & -4 & 1 & 3 \\ -2 & -2 & -1 & 3 \\ -2 & -6 & 3 & 3 \end{bmatrix}.$$

To compute $\mathrm{null}(A - 4\,\mathrm{Id}_4)$, we solve the system (where we do $r_2 \mapsto r_2 - r_1$, $r_3 \mapsto r_3 - r_1$)

$$\begin{bmatrix} -2 & -4 & 2 & 2 \\ -2 & -4 & 1 & 3 \\ -2 & -2 & -1 & 3 \\ -2 & -6 & 3 & 3 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \implies \begin{bmatrix} -2 & -4 & 2 & 2 \\ 0 & 0 & -1 & 1 \\ 0 & 2 & -3 & 1 \\ -2 & -6 & 3 & 3 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

We hence have $y = z = w$ and $x = 0$. Since $\mathrm{null}(A - 4\,\mathrm{Id}_4)$ is of dimension at least 1, we conclude $\{0\} \subsetneq \mathrm{null}(A - 4\,\mathrm{Id}_4) = \{[0, y, y, y]^\mathsf{T} \mid y \in \mathbf{C}\} \subsetneq \mathrm{null}(A - 4\,\mathrm{Id}_4)^2 = \mathrm{G}(4, A)$. We have

$$(A - 4\,\mathrm{Id}_4)^2 = 4 \begin{bmatrix} 1 & 2 & -1 & -1 \\ 1 & 1 & 0 & -1 \\ 1 & 0 & 1 & -1 \\ 1 & 2 & -1 & -1 \end{bmatrix}.$$

It is not necessary to have a full description of $\mathrm{null}(A - 4\,\mathrm{Id}_4)^2$, only to find a vector v_4 that is inside $\mathrm{null}(A - 4\,\mathrm{Id}_4)^2$, but not inside $\mathrm{null}(A - 4\,\mathrm{Id}_4)$. One can solve a system to find v_4 or simply guess $v_4 = [1, -1, -1, 0]^\mathsf{T}$. Finally, taking $v_3 = (A - 4\,\mathrm{Id}_4) v_4 = [0, 1, 1, 1]^\mathsf{T}$ gives us a basis v_3, v_4 of $\mathrm{null}(A - 4\,\mathrm{Id}_4)^2 = \mathrm{G}(4, A)$ with $v_3 \in \mathrm{null}(A - 4\,\mathrm{Id}_4)$.

Let us take $\mathcal{B} = (v_1, v_2, v_3, v_4)$ for a basis. We have $Av_1 = 2v_1$, $Av_2 = 2v_2$, $Av_3 = 4v_3$ and $Av_4 = 4v_4$. So, for $P := [v_1\ v_2\ v_3\ v_4]$ we have

$$AP = [Av_1\ Av_2\ Av_3\ Av_4] = [v_1\ v_2\ v_3\ v_4] \begin{bmatrix} 2 & & & \\ & 2 & & \\ & & 4 & 1 \\ & & & 4 \end{bmatrix}.$$

That is:

$$P^{-1} A P = \begin{bmatrix} 2 & & & \\ & 2 & & \\ & & 4 & 1 \\ & & & 4 \end{bmatrix} =: J$$

is the Jordan form of A.

Example 5.3.15. For the following four matrices, decide which ones are similar. (They all have characteristic polynomial $(-t)^4$ and are hence nilpotent.)

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

$$C = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \qquad D = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Solution. The matrices A, B and C are already in Jordan form. These Jordan form are distinct (even up to permutation of the blocks), which implies that these matrices are mutually non-similar. For D, we have $\operatorname{rank}(D) = 2$ so $\dim(\operatorname{null}(D)) = 2$. But $D^2 = 0$ and thus $\dim(\operatorname{null}(D^2)) = 4$. So we choose $2 = \dim(\operatorname{null}(D^2)) - \dim(\operatorname{null}(D))$ vectors u_2, v_2 in $\operatorname{null}(D^2)$ but not in $\operatorname{null}(D)$. Then we write $u_1 := Du_2$ and $v_1 := Dv_2$ to obtain a basis (u_1, u_2, v_1, v_2) of $\mathbf{C}^4$. Since we have two sequences u_1, u_2 and v_1, v_2 of length 2 we have two Jordan blocks of size 2×2. In other words, the Jordan form of D is

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = B.$$

Therefore, D and B are similar. □

While the proof of Lemma 5.3.8 (or more precisely its algorithmic version) gives us a general procedure to find a Jordan basis, it might be difficult to use in general. Hopefully, for small dimensions there is an easier procedure, which works well as soon as for each eigenvalue there is at most one Jordan block of size bigger than 1. To find the Jordan form of a small matrix it is usually enough to do the following steps:

1. Find the eigenvalues of A by solving $\det(A - \lambda \operatorname{Id}) = 0$.
2. For each eigenvalue λ, find the corresponding eigenvectors by finding a basis $v_1, \dots, v_k$ of $\{v \mid (A - \lambda \operatorname{Id})v = 0\} = \operatorname{E}(\lambda, A)$.
3. For each eigenvector v_i try to find a generalized eigenvector $v_{i,2}$ by solving, if possible, $(A - \lambda \operatorname{Id})w = v_i$ for w.
4. If you still miss generalized eigenvectors, iterate: solve $(A - \lambda \operatorname{Id})u = v_{i,2}$ for u to find a new generalized eigenvector $v_{i,3}$, and so on.

Example 5.3.16. Let $A = \begin{bmatrix} 2 & & \\ 1 & 2 & \\ & & 2 \end{bmatrix}$. Then

1. Since A is triangular, its only eigenvalue is 2 (with algebraic multiplicity 3).
2. We solve $\begin{bmatrix} 0 & & \\ 1 & 0 & \\ & & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ and find $\mathrm{E}(2, A) = \{ \begin{bmatrix} 0 \\ y \\ z \end{bmatrix} \mid y, z \in \mathbf{R} \}$. So one can take $v_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ as a basis $\mathrm{E}(2, A)$.
3. We try to solve $\begin{bmatrix} 0 & & \\ 1 & 0 & \\ & & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ and find it to be impossible. So v_1 will correspond to a Jordan block of size 1. We then try to solve $\begin{bmatrix} 0 & & \\ 1 & 0 & \\ & & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ and find $v_{2,2} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ as a solution (we could have taken any other solution for $v_{2,2}$, as for example $v_{2,2} = [1, 1, 1]^\mathsf{T}$). So $(v_2, v_{2,2})$ will give us a Jordan block of size at least 2.

   Since A is a 3×3 matrix and we already have Jordan blocks of size 1 and 2, we can stop here.

Altogether, we have the following Jordan form for A: $J = \begin{bmatrix} 2 & & \\ & 2 & 1 \\ & & 2 \end{bmatrix}$, and $v_1, v_2, v_{2,2}$ is the corresponding Jordan basis.

While the minimal polynomial and the characteristic polynomial are useful tools to compute the Jordan form, they are not always sufficient to recover the Jordan form as demonstrated by the following example.

Example 5.3.17. The following two matrices have the same characteristic polynomial $(2-t)^4$ and the same minimal polynomial $(t-2)^2$, but not the same Jordan form.

$$A = \begin{bmatrix} 2 & 1 & & \\ & 2 & & \\ & & 2 & 1 \\ & & & 2 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & 1 & & \\ & 2 & & \\ & & 2 & \\ & & & 2 \end{bmatrix}.$$

Observe that the geometric multiplicity of 2 in A is 2, while it is 3 in B, so we can still distinguish A and B using eigen-theory. (In both cases, the algebraic multiplicity of 2 is 4.)

For following two matrices, the only eigenvalue 2 has geometric multiplicity 2 and algebraic multiplicity 4 in both cases, but the Jordan form are not the same.

$$C = \begin{bmatrix} 2 & 1 & & \\ & 2 & 1 & \\ & & 2 & \\ & & & 2 \end{bmatrix}, \qquad D = \begin{bmatrix} 2 & 1 & & \\ & 2 & & \\ & & 2 & 1 \\ & & & 2 \end{bmatrix}.$$

Observe that the above two matrices do not have the same minimal polynomial and so can still be distinguished by eigen-theory. Indeed $M_C(t) = (t-2)^3$ while $M_D(t) = (t-2)^2$.

It is possible to construct bigger matrices such that eigen-theory is not able to distinguish between them. The following two matrices have the same characteristic polynomial $(2-t)^7$ and the same minimal polynomial $(t-2)^3$. Moreover, they have the same eigenvalues 2, with the same algebraic multiplicity 7 and the same geometric multiplicity 3. Nevertheless, they do not have the same Jordan form.

$$E = \begin{bmatrix} 2 & 1 & & & & & \\ & 2 & 1 & & & & \\ & & 2 & & & & \\ & & & 2 & 1 & & \\ & & & & 2 & 1 & \\ & & & & & 2 & \\ & & & & & & 2 \end{bmatrix}, \qquad F = \begin{bmatrix} 2 & 1 & & & & & \\ & 2 & 1 & & & & \\ & & 2 & & & & \\ & & & 2 & 1 & & \\ & & & & 2 & & \\ & & & & & 2 & 1 \\ & & & & & & 2 \end{bmatrix}.$$

The matrices from Example 5.3.17 were not randomly chosen. Indeed, one can prove:

Lemma 5.3.18. *Let V be a finite dimensional vector space over $\mathbf{C}$ and let T be an operator with only one eigenvalue, so $\chi_T(t) = (\lambda - t)^{m_\lambda}$ and $M_T(t) = (t-\lambda)^{c_\lambda}$. Then (recall that $d_{\lambda_j} = \dim(\mathrm{E}(\lambda_j, T))$ is the geometric multiplicity)*

1. *If $m_\lambda \leq 3$, then (m_λ, c_λ) determines the Jordan form J of T;*
2. *If $m_\lambda \leq 3$, then (m_λ, d_λ) determines the Jordan form J of T;*
3. *If $m_\lambda \leq 6$, then $(m_\lambda, c_\lambda, d_\lambda)$ determines the Jordan form J of T.*

Proof. By Theorem 5.3.13, J is a matrix of size m_λ with d_λ blocks of maximal size c_λ. To determine J, it is enough to know the number b_k of blocks of size k for each $1 \leq k \leq m_\lambda$. But the total size of J is the sum of the size of the blocks:

$$m_\lambda = \underbrace{1 + \dots + 1}_{b_1} + \underbrace{2 + \dots + 2}_{b_2} + \dots + \underbrace{c_\lambda + \dots + c_\lambda}_{b_{c_\lambda}}, \tag{5.7}$$

where $b_k \geq 0$ for $k \in \{1, \dots, c_\lambda - 1\}$, $b_{c_\lambda} \geq 1$ and $b_1 + b_2 + \dots + b_{c_\lambda} = d_\lambda$ is the number of blocks.

For a given $m_\lambda \leq 6$, one can list all the decompositions according to Equation (5.7) and check that no two such decompositions share the same triple $(m_\lambda, c_\lambda, d_\lambda)$. Alternatively, one can try to find the smallest counterexample possible. This will both prove the statement and explain how to obtain the matrices Example 5.3.17.

First of all, in order two find two distinct decompositions of a given m_λ, one should have $c_\lambda \geq 2$, as there exists a unique decomposition of m_λ using only 1s. It is also trivial that such a counterexample need to use at least two summands, that is $d_\lambda \geq 2$.

If $c_\lambda \geq 3$, then $m_\lambda \geq 4$. If $c_\lambda = 2$, then the only possibility is decompose one 2 into two 1s. But in $2 = 1 + 1$, the left hand side has $c_\lambda = 2$ while the right hand side has $c_\lambda = 1$. Therefore, the smallest counterexample to 1 is $2+2 = 2+1+1$, which is matrices A and B from Example 5.3.17.

If one uses only 1s and 2s, one cannot write m_λ using the same number of summands. In other words, a counterexample to 2 necessarily has $c_\lambda \geq 3$. Which, using $d_\lambda \geq 2$, gives us $m_\lambda \geq 4$. For an explicit counterexample, one can hence take $3+1=2+2$, which is matrices C and D from Example 5.3.17.

For 3, one already knows that a counterexample has $c_\lambda \geq 3$. Moreover, since both decomposition have the same c_λ the smallest counterexample possible is $3+3+1 = 3+2+2$, which is matrices E and F from Example 5.3.17. □

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Rethink the logical structure. Add intro § to every (sub)section

Add at the beginning of each section/in the section title the standing assumptions (finite dimensional, $\mathbf{F} = \mathbf{R}/\mathbf{C}$, ...)

Add a symbol list.

Say that for graphics in $\mathbf{R}^2$ and $\mathbf{R}^3$, we will freely use [point (x, y) (blacknode)]<—>[vector $\begin{bmatrix} x \\ y \end{bmatrix}$ (arrow)]

Say that for matrices, empty cells represent 0

(Re)write the whole section on matrix representation.

Totally rewrite the section on eigenvectors and co. In particular, define the characteristic polynomial to be $\det(t \operatorname{Id}_n - A)$ in order to have a monic polynomial.

Provide shortcaptions for figures. (As well as a listoffigures?)

Bibliography

Better index.

Write semantic commands for color in graphics: vectors 1,2, exists; range+image, black+gray dots, plots, tcolorbox,...

Better enumerate environment

koma headings? look at headers/footers of pages starting with a float

Rethink the subdefi (and addindex (and defi)) commands.

Choose better notation for algebraic multiplicity (and geometric multiplicity) versus minimal polynomial

Better matrices, solve the problem with $T|_U$.

Todo list